

SIR KARL RAIMUND POPPER, C.H., F.B.A.

28 July 1902–17 September 1994

Elected F.R.S. 1976

BY DAVID MILLER

Department of Philosophy, University of Warwick, Coventry CV4 7AL UK

Karl Raimund Popper, the philosopher and methodologist of science, died on 17 September 1994 at the age of 92 years. Locke was a Fellow of the Royal Society, of course, and some other early Fellows, Boyle in particular, have a secure place in the history of philosophy. Whewell, an intellectual forerunner of Popper's (Medawar 1967), was elected in 1820 when the Royal Society was, according to a contemporary philosopher, 'in a state of unstable equilibrium'; he was a mere 26, a lecturer in mathematics, and was to occupy the chair of mineralogy at Cambridge for some years before writing his exceptional studies in the history and philosophy of science. Whitehead and Russell were elected as Fellows for their mathematical work, as have been one or two more recently elected logicians. Popper is the only philosopher in modern times to have been elected a Fellow of the Royal Society primarily in recognition of his philosophical achievement.

Popper's intellectual life spanned more than 75 years, with little interruption. He worked, and thought, and wrote, and rethought, and rewrote, almost unceasingly. He made outstanding (in some cases revolutionary) advances in the philosophy of science, the theory of probability, the theory of knowledge, metaphysics, social and political philosophy, and the philosophy of history; he contributed importantly (usually controversially) to our understanding of logic, the history of philosophy (especially the pre-Socratics, Plato and Marx), classical mechanics, classical thermodynamics, quantum physics, evolutionary biology, psychology and music. Magee (1997, p. 211) writes: 'No other thinker of the twentieth century has come anywhere near matching ... [Popper's] range of effectiveness as a destroyer of the dominant myths of the age ... To the end of his life he was astonishingly fertile in new ideas.' Popper's list of publications contains over 1200 entries, including reprints and translations, and his archive (up to 1985) occupies 463 manuscript boxes in the library of the Hoover Institution at Stanford University (the University of Klagenfurt in Austria has a complete photocopy; a further copy is held on 473 rolls of microfilm at the London School of Economics). What follows here does no more than identify the crucial events in Popper's intellectual and scientific life, outline and



Karl Popper

evaluate his leading contributions to science and to the understanding of science—indicating some of the countless criticisms that they have attracted—, and in passing reveal some aspects of his lovable though formidable character.

IN VIENNA, 1902–36

Popper was born on 28 July 1902 at Himmelhof in Ober St Veit on the western outskirts of Vienna, the only son of Dr Simon Siegmund Carl Popper and Jenny Popper (*née* Schiff). His sister Dora (Emilie Dorothea) was then eight years old, his sister Annie (Anna Lydia) four. His parents were Jewish, but had converted to the Protestant faith. His mother, who was born in Vienna in 1864, came from a richly musical family; her parents were founder-members of the *Gesellschaft der Musikfreunde*, and she and her two sisters were talented pianists. Popper's father, whose family came from Kolin, was born in 1856 in Roudnice nad Labem (Raudnitz) in what is now the Czech Republic. He was brought up separately from his two brothers, but the three followed the same career, becoming doctors of law of the University of Vienna. He was a cultivated man, historian, poet, social theorist, 'really more of a scholar than a lawyer' according to his son. The Popper family lived in the centre of Vienna in the handsome 18th-century house at the corner of Freisingergasse and Bauernmarkt where Popper's father maintained his successful law practice (Geier 1994, p. 39). In §§2–6 of his intellectual autobiography *Unended quest* (23),* Popper has written movingly about one or two episodes in his childhood, and no more will be said about it here.

Popper's father died in 1932, the year in which Dora Popper (who had been a nurse and also a civil servant) committed suicide during a bout of illness. Popper and his wife Hennie (Josefine Anna Henninger, born 1906) left Austria in 1937. His mother died in 1938, after a long illness. Some 16 members of the family died as a result of the Nazi terror. A month after her mother's death Annie Popper escaped to Paris and eventually to Switzerland, where she lived until her death in the 1970s. She wrote sentimental stories and novels, one of which, for example, relates the exploits of a female Tarzan.

Between the ages of six and eleven Popper was educated in a private school; thereafter he attended the Franz Josef Gymnasium [366.3]. But he reports that, desperately bored in most of his lessons, he effectively left school at the age of 16. His first attempt at the *Matura*—the entrance examination for the University—was unsuccessful, as he failed in two subjects (one failure could be condoned): logic and Latin. In logic he was examined by Edgar Zilsel, later a member of the Vienna Circle, with whom he had a disagreement about Aristotelian logic (probably about the existential commitment of universal quantification). The Latin examination required the oral translation of an unseen Horace ode. In 1922, the second attempt at the *Matura* was successful; there was no conflict with his logic examiner, and the Horace ode was perhaps less unseen than the earlier one had been.

Popper remained at the University until 1928. He studied mathematics formally with Hahn, Helly, Furtwängler and Menger; physics with Ehrenhaft, Lecher and Thirring; psychology with Bühler; and philosophy with Heinrich Gomperz and Schlick. But he did many things besides. He tried his hand as a manual worker. This was not a success. From about 1917 he

* References in this form refer to the bibliography at the end of the text. References to the Hoover archive are in the form [x.z], where x is the number of the box, z the number of the folder. For the sake of brevity I have omitted most references to *Unended quest*, which I have drawn on freely.

had been actively concerned in progressive politics, even for a short time thinking of himself as a communist. But after the callous shooting of unarmed demonstrators by the police in Hörlgasse in 1919, he turned against the cynical readiness of Marxists to sacrifice others' lives for indeterminate ideals. Despite misgivings, he remained sympathetic to the social democratic movement; and, although never a pacifist, even to pacifism (Kiesewetter 1995, p. 278). For some years after 1920 he was apprenticed to a cabinetmaker. This too was not an unqualified success. (Years later, in New Zealand, he designed and had made some handsome furniture in a somewhat functional style.) He worked in Adler's child guidance clinic, and later did social work with neglected children. In 1923 he qualified as a primary school teacher. After sampling, and not much liking, the music performed in Schoenberg's *Verein für musikalische Privataufführungen*, he sought admission in 1922 or 1923 to the Vienna Konservatorium, writing for this purpose an organ fugue in F sharp minor. He later said of this fugue, which was first publicly performed in 1992 by Dame Gillian Weir, that it was an attempt to honour Bach through its second subject, 'a quotation from Bach's *Passion Music According to St Matthew*' (39). The pianist Julien Musafia, who has arranged the fugue as a piano work for four hands, describes it as 'a composition of romantic impulse vested in baroque form—it would have been a most creditable piece of work to have submitted at the end of his studies at the Konservatorium, let alone at the beginning' (1996, personal communication). But Popper, convinced that he was not talented enough to become a professional musician, stayed only one year.

In a statement, written a decade later, of the details of his education Popper wrote [366.3]: '[In 1925] I understood that circumstances in Austria would not permit me to obtain an official position at the university.' This doubtless alludes not only to the prevalent anti-semitic sentiment, but also to Popper's relative remoteness from philosophers of influence; the academic establishment in those days was for the most part conservative, clerical and catholic, and not natural quarters for an independent-minded social democrat such as Popper. 'I therefore decided to prepare myself for a post at the public school. I attended lectures on education and began to deal with paedagogic problems' The City of Vienna had recently opened its new Pedagogic Institute, in which Popper enrolled. It was there that he met his wife, Hennie, who was training as a physical education teacher. He attended Bühler's university lectures on psychology, and discovered the doctrines of the Würzburg school, especially those of Külpe and Selz. But although attracted by many of these doctrines, particularly the doctrine that thinking develops not through the passive association of ideas but under the active impulse of problem-solving, Popper was becoming persuaded that logic is the key to psychology, not the other way round (22, chapter 1, §10); that the growth of knowledge can be understood only if its logic is understood. His Ph.D. dissertation of 1928, 'Zur Methodenfrage der Denkpsychologie' ('On the problem of method in the psychology of thinking'), examined by Bühler (who praised it very highly [406.1]) and Schlick, the founder of the Vienna Circle, marked a turning point; thereafter he never wavered from the view that scientific knowledge, though humanly created, is objective knowledge, and must be analysed in objectivist terms. The thesis "Gewohnheit" und "Gesetzerlebnis" in der Erziehung' ("Habit" and 'experience of laws' in education') submitted to the Pedagogic Institute in 1927 or 1928 was never properly finished.

Popper's interest in the logic and methodology of science had started long before his attendance at Bühler's lectures. He has related on more than one occasion (16, chapter 1; 23, §8) the story of his realization, in 1919, that what is distinctive about the best scientific

theories, those that contribute most to our understanding and our knowledge, is not that experience invariably confirms them, but that it invariably challenges them, by exposing them to the risk of falsification. He contrasted general relativity, whose daring prediction that light rays bend in gravitational fields had recently been tested by Eddington's eclipse observations at Sobral and Principe, with pseudoscientific doctrines—he cited vulgar Marxism and psychoanalysis—whose principal claim to attention seems to be their ability to explain anything and everything that might happen. The predictions of some astrologers and fortune-tellers are similar, cautiously or ambiguously phrased so as to fit any future contingency. Popper noticed that such plasticity, such ability to explain anything, whatever it is, is not a virtue in a theory; on the contrary, it reveals that the theory excludes nothing, that it has no empirical content. To be sure, a successful theory will explain a great deal, perhaps everything, that does happen within its domain; what it must not do is explain everything that might happen. A theory, Popper's criterion of demarcation says, must be falsifiable in principle if it is to belong to empirical science, if it is to say anything about the world that we experience. But it is of course desirable that, despite multifarious attempts at falsification, it should be unfalsified in practice. If a theory is to be informative, it must take risks; if it is to be correct, it must survive. Popper has attributed his appreciation of the virtue of daring speculation in scientific hypotheses to the intrepidity of the polar explorer Nansen, one of his boyhood heroes (43, pp. 321f.; Kieseewetter 1995, pp. 278f.). His appreciation of the fact that a theory may pass any number of tests and still be false he attributed to the supersession of Newtonian gravitational theory by general relativity. Indeed, he often declared that without Einstein's revolution he would have remained a Kantian, convinced that classical mechanics was the final truth.

The most long-lasting and sympathetic philosophical influence during these student years was Julius Kraft, a philosopher of law from Göttingen, who had studied with Leonard Nelson. Kraft and Popper had many informal conversations during 1924–25, their arguments revolving round profound difficulties that each saw in Kant's synthetic *a priori*, the doctrine that we can know what we do know about the world of experience only if some of the most fundamental things we know (for instance that space is three-dimensional and Euclidean) are available to us independently of experience. Like Kant himself, Kraft and Popper were anxious to avoid the sceptical abrogation of rationality into which Hume had toppled, but they were in stark disagreement about how to avoid it (15, pp. 4–6). Popper for one already knew that some of Kant's synthetic *a priori* propositions had been refuted. But it was only at the end of his student days that it became clear to him that if scientific theories boldly anticipate experience they cannot be staidly grounded in experience. The traditional creed, proclaimed by Bacon and institutionalized by J.S. Mill, that science advances from facts to generalizations by a logically defensible process of induction, must be rejected both as psychology and as logic. Our theories are not born out of experience, nor are they always borne out by it. Psychologically theories are conjectures, guesses whose role—as the Würzburg school emphasized—is to solve problems, to probe the unknown, not merely to organize what is already known (itself, admittedly, sometimes also a problem). The same holds in logic. No universal theory or prediction follows from the known facts by any logical process (Hume had shown this 150 years before); *there is no inductive logic*. Facts, observations, experiments, these serve in science not as a foundation, or even a support, for the theoretical superstructure, but only as tests of its correctness. Although no accumulation of experiences can verify a universal theory, or even provide grounds for supposing it to be true, one counterexample, if upheld, will falsify it.

This, in brief, is the method of conjectures and refutations, Popper's solution to the

problem of induction, his resolution of the Kantian problem of how the intellectual (*a priori*) and the experiential (*a posteriori*) components of human knowledge are integrated. Much criticized, endlessly dissected, in the last 70 years, it has gladdened scientists with its insistence that their work is inventive and imaginative; comforted those brought up to believe that only the maladroit make mistakes; and cheered those troubled by humane scepticism about induction. *There is no induction. But there need not be scepticism either.* Our knowledge can grow, provided we forego the traditional demand for theories that are verified, or proved, or justified, and ask no more than that they be correct. A correct guess is not beyond us, if we are lucky, though we can never be sure that we have made one. Scientific discovery differs from commonsense (and even animal) learning not, as Bacon thought, in its avoidance of *parti pris* and prejudice, for scientists too jump to conclusions. The difference is that a conclusion in science marks the beginning of the investigation, not the end of it; it is through its self-conscious use of the critical method to identify and eliminate errors that scientific thinking (more generally, rational thinking) is distinguished from animal thinking; 'from the amoeba to Einstein is just one step' (22, chapter 6, §xx).

In 1929 Popper submitted to the Pedagogic Institute a third thesis, entitled 'Axiome, Definitionen und Postulate der Geometrie' ('Axioms, definitions and postulates of geometry')[4.6], to qualify as a science teacher in lower secondary schools. In 1930, the year of his marriage, he was appointed to his first official teaching position. He continued to think and write without respite. At about this time he met Herbert Feigl, one of the younger members of the Vienna Circle. This group, sometimes known as 'logical positivists' or 'logical empiricists', combined an enthusiasm for contemporary mathematical logic with traditional positivistic emphases on certainty and on the epistemological priority of sense experience. Although Popper's views on demarcation and induction had been developed, partly in discussions with Kraft, as an extension and correction of the philosophies of Kant, Fries, and Nelson, and not deliberately in opposition to the ideas of the Vienna Circle, he had long known that they conflicted on numerous points with the Circle's views, pre-eminently its doctrine (derived from Wittgenstein, not himself a member of the Circle) that empirical verifiability provides a criterion of meaningfulness (and the consequence that metaphysics is meaningless). Feigl persuaded him that his criticisms of the Circle were so significant that they should be published as a book. In response, Popper, who had so far published only a few short articles on educational topics, planned a huge work, *Die beiden Grundprobleme der Erkenntnistheorie* (*The two fundamental problems of the theory of knowledge*), in two volumes, entitled *Das Induktionsproblem* (*The problem of induction*) and *Das Abgrenzungsproblem* (*The problem of demarcation*). Never quite finished, it was abridged, then expanded, then abridged again. The result was *Logik der Forschung* (1), translated into English as *The logic of scientific discovery* (14). What survives of the original two-volume typescript was published as (28), but is not yet available in English. The fate of the intermediate typescripts is unknown (Hansen 1974).

LOGIK DER FORSCHUNG

The opening chapters of *Logik der Forschung* state with stunning clarity and succinctness all the cardinal theses of falsificationism, the methodology of conjectures and refutations. (a) Induction is not only fallacious but unnecessary. The so-called inductive method is

replaced by ‘*the deductive method of testing*, ... the view that a hypothesis can only be empirically *tested*—and only *after* it has been advanced’ (§1). (b) Hypotheses are to be entertained in science if and only if they are empirically falsifiable. It is admitted that any hypothesis can be rescued from falsification by sufficiently drastic adjustments, for example by postulating *ad hoc* the existence of previously unknown entities or effects; but it is at the heart of the scientific approach, if not its unvarying practice, to exclude evasive tactics of this kind (§6). (c) If we want science to progress, we must not be frightened of mistakes, but we must adopt methodological rules whose overriding objective is to encourage falsification of our theories and the elimination of the mistaken ones among them (§11). (d) Methodology is not to be identified, as positivists identify it, with the empirical discipline of describing how scientists behave, for even the underlying demarcation of scientists from others depends on the adoption of a decision or convention (§4). (e) The basic statements with which scientific theories are confronted in tests are themselves conjectural. Science makes no appeal to incorrigible perceptions or perception statements, and ‘the empirical basis’ is a conventional—though far from arbitrary—stopping point, not an absolute one. No falsification, accordingly, is ever conclusive (but that does not preclude its being a falsification): ‘If you insist on ... strict disproof ... in the empirical sciences, you will never benefit from experience, and never learn from it how wrong you are’ (§9). Succeeding chapters elaborate on these theses, and others.

A major topic of the book is probability. Popper noted in §34 that the word ‘probability’ has many meanings, and he distinguished between the physical probability of an event or type of event, and what he called the logical probability of a statement or theory, which he showed to be inversely related to its empirical content and its degree of falsifiability: the more a theory says, the more it forbids, the more opportunities it offers for falsification, and the less likely it is to be true. He claimed as an immediate consequence that science cannot be interested in obtaining probable theories (§83). In a long chapter devoted to physical probability he dismissed subjectivist apologies that explain it as no more than a compensation for ignorance, and grappled with two major problems in the only extant objectivist theory, that of von Mises, according to which probabilities are the limits to which relative frequencies tend in indefinitely long random reference classes (collectives). *The problem of decidability* asks how statements of probability, which are ‘*in principle impervious to strict falsification*’ (proem to Chapter VIII), can nonetheless belong to empirical science; a crucial challenge to the criterion of demarcation. *The fundamental problem of the theory of chance* (§49) asks for an explanation of why random sequences are statistically stable. (Mises offers little here, as collectives are defined as sequences that are both random and stable.) Popper proposed, but did not fully harmonize, two solutions to the problem of decidability. One, quite in accord with the procedures of classical statistics, involves the adoption of methodological rules to permit the falsification of probabilistic or statistical statements. The other, more original, shows that the attribution of randomness itself is falsifiable if we extend the idea to finite sequences. Although Popper’s account of finite random sequences, one of the earliest, is not wholly satisfactory (Ville 1939; Delahaye 1994, p. 35), it does suggest a solution also to the fundamental problem of the theory of chance; for sequences that are random from the start are, he showed, always statistically stable.

Despite the popularity of the verifiability criterion of meaning, it was usually conceded that no quantity of empirical evidence can turn a scientific theory into a certainty; most philosophers, even in the Vienna Circle, were fallibilists with regard to theories, though not

with regard to statements of evidence. It was claimed, nevertheless, and still is claimed, that certainty can be approached; theories can be made more and more probable (in the sense of logical probability), and in this way the problem of learning from experience, which is identical with the problem of induction, solved. The conditional (or as Popper preferred to call it, relative) probability $p(h \mid e)$ of a hypothesis or theory h on evidence e was treated as a measure of the partial deducibility of h from e (ranging from unity, when h is deducible from e , to zero when h contradicts e); and—on the supposition that induction is generalized deduction—it was therefore thought that $p(h \mid e)$ adequately measures the degree to which h is empirically or inductively supported by e . In the final chapter of the book Popper gave a detailed repudiation of the view, especially in the forms advanced by Reichenbach and by Keynes, that the problem of induction can be tackled by such considerations of probability. There are three items at issue here: the updated probability $p(h \mid e)$ of the theory h ; the degree to which h has withstood the tests whose results are described by e (Popper called this the degree of corroboration of h); and what e tells us about the truth value of h . Popper argued that the first two are distinct, and that, unless the theory is refuted, neither is a measure of the third; for empirical evidence can teach us nothing about a theory except that it is false. In the years to come, criticism of appeals to probability was to be a constant theme in his unrelenting attacks on the philosophy of induction.

Logik der Forschung contains also a substantial chapter criticizing the Copenhagen interpretation of quantum mechanics, a topic apparently distant from methodology. But it is a natural consequence of Popper's insistence on the objectivity of scientific knowledge that he should have sought to purge quantum theory of the subjectivism implanted into it by Heisenberg, in particular the subjectivist interpretation of probabilities as measures of ignorance rather than as statistical frequencies. Popper later changed his mind on the interpretation of quantum-mechanical probabilities (without weakening the demand for objectivity), and it is convenient to postpone the summary of his criticisms of the Copenhagen school until something has been said about the propensity interpretation of probability, which he first proposed in the 1950s.

A teacher in a lower secondary school when this marvellous book was being written, and still a teacher when it was published late in 1934, Popper was becoming restless. His abilities, even in science and mathematics, were not being used. He had begun to meet other members of the Vienna Circle, especially Carnap, who was to be his principal defender among the logical empiricists, and Gödel, as well as distinguished visitors to Vienna, such as Heisenberg in 1935 (30, Preface, §III). But by 1935 Carnap and Gödel rarely took part in meetings of the Circle (Menger 1974), and Popper himself was never invited there. He did speak at some of the peripheral meetings, and a talk he gave on randomness at Menger's 'mathematisches Colloquium' in 1935 first sparked the interest of Wald in the theory of probability. He began to attend international conferences, beginning with the Circle's 'Vorkonferenz' in Prague in 1934. There he met Tarski, who in Vienna shortly afterwards explained to him his celebrated 'semantic conception of truth'. This was a momentous occasion for Popper, who had previously been a realist with an uneasy conscience, unable to make clear to himself how to reconcile the commonsense idea of truth with common sense; that is, how to spell out the correspondence between true statements and facts. Thereafter he championed Tarski's ideas on unnumbered occasions (not always in a way that appealed to all of Tarski's other champions). The rehabilitation, as he called it later, of the commonsense idea of truth is of decisive importance for Popper's philosophy, as it embodies an indispensable distinction

between truth and justified truth. Philosophers since Plato have been much more interested in what can be justified to be true—conclusively or inconclusively—than in what is true; the demand for knowledge, rather than mere opinion, insistently proclaims this preference. In positivism the preference becomes acute, and truth is identified with what can be known. In contrast, a genuine conjecture is plainly a statement that we think may be true even if we cannot know it; not only may it be true-and-not-justified but it may be true-and-not-justifiable.

During the academic year 1935–36 Popper took leave without pay from his teaching position, in order to accept invitations to speak abroad (Hacohen 1996, p. 457). He visited London, Cambridge, Oxford, Brussels, Utrecht, and Copenhagen; he met Russell, Moore, Schrödinger, Hayek, Crawford, Woodger, Haldane, Bernal, Waddington, Oppenheim, Bohr, and others. He continued to be restless. As noted above, he had known for years that he had almost no hope of a university position in Austria; and not much help was offered by Austrian friends who had already obtained posts abroad. At the end of October 1936 he applied (at Woodger's suggestion) to the Universities Bureau of the British Empire for the Professorship of Philosophy in the Department of Education & Philosophy, Canterbury University College, Christchurch, one of four constituent colleges of the federal University of New Zealand, naming Moore and Woodger as referees; and secondarily for a simultaneously advertised lectureship. I.L.G. Sutherland from Wellington, who had obtained his doctorate in Glasgow in 1923 with a thesis in what we now call moral psychology, a man five years Popper's senior, was appointed to the chair. Popper was appointed to the lectureship. Hurriedly leaving Vienna at the beginning of 1937, the Poppers went into exile on the other side of the world ('halfway to the moon' as Hennie described it).

It should not be supposed that *Logik der Forschung* was the first work to reject inductivism and to advocate the hypothetical method. Liebig (1865) and Bernard (1865) had done so, and Whewell (1840), though a believer in the power of empirical evidence to establish the truth of scientific theories, had made plain that hypotheses are essential if science is to have any theories to establish (Medawar 1967). Popper referred to Liebig, but seems to have known nothing of Bernard and Whewell (or of Bachelard, who was saying similar things in France). Yet Popper's originality is not to be doubted. He was the first to stress that the hypothetical character of a theory is never mitigated, however numerous and however breathtaking its empirical successes, and he was the first to confront inductivism with a systematic alternative methodology, in which evidence plays no positive role in the assessment of theories. Indeed, it is not so much inductivism that he demolished, as justificationism; the view that the rationality of science (and other intellectual activities) consists in the pursuit and attainment of (wholly or partially) justified truth, rather than of truth. Critics of falsificationism nearly all stumble at this point, and ascribe to it precisely those justificationist aspirations that it is at such pains to discard. This partly explains why Popper was so dismissive of so many of the objections that purported to attack his views. He encourages us to see science as a reckless human activity, whose extraordinary purchase on the truth about this strange and beautiful world is not less genuine for being quite unsecured. To quote one of his favourite lines from Xenophanes (16, § xv): 'all is but a woven web of guesses'.

IN NEW ZEALAND, 1937–45

In 1937 the Department of Education & Philosophy at Canterbury University College had two professors, one lecturer, one full-time assistant lecturer, two part-time assistants and about 80–100 students [366.3]. Sutherland, the new Professor of Philosophy, was by that time really an anthropologist, an expert in Maori affairs, a man who had been honoured by the Ngati Porou people with the ancestral name Paikea (Beaglehole *et al.* 1952). Popper was the only active philosopher, and did all the teaching in philosophy, fettered to a syllabus that ‘had advanced from Aristotle to Bacon but no further in the logical alphabet’ (Godsiff 1972). Writing to Carnap in 1943 he reported ‘an extensive as well as exhaustive lecturing programme’ of six courses, and complained that his ‘hatred against the empty verbalism and scholasticism of the vast majority of philosophical writings is increasing proportionally with the time I have to devote to the teaching of such matters’ [282.24]. More to his liking was

a course for research workers on Scientific Methods (with discussion of their practical research problems, which have led to considerable practical results; a kind of poli-clinical advisory agency for agricultural chemists etc.). This course ... was very interesting and successful. It is comforting to find that philosophy can be of some practical use!

He contributed to other extramural courses too; for example, to a series of University Extension Lectures in 1940 on religion [366.20], and to a W.E.A. course ‘Science in the making’, for which he produced substantial typewritten notes [366.24]. His enthusiasm and vitality became famous beyond Christchurch: he ‘acted as a kind of intellectual champagne after the dry depression years. ... Staff and students alike crowded his open lectures not for instruction or information, but for enlightenment and the sheer intellectual joy of exploring the unknown with him’ (Gardner *et al.* 1973, pp. 263; quoted in Bartley 1982, p. 253). The short lecture course that Popper was invited to give in Dunedin in 1944 attracted large crowds in the same way (Eccles 1982, p. 221). It had the notable effect also of converting a naive believer in induction (as Eccles described himself) into one of the most vigorous of all scientific advocates of the method of conjectures and refutations.

Although he was promoted to Senior Lecturer as soon as the rank was introduced at the beginning of 1941, Popper felt unappreciated and unrewarded in the Department at Canterbury. There were endless disputes with Sutherland, his superior, blowing up at one point in 1940 into a formal request from Popper to the Rector ‘for an enquiry into my loyalty to the British cause and into my competence to lecture on the present war-issues’, which, he suspected, had been called into question. To this the Rector replied [366.5]; ‘I wish to leave no shadow of doubt regarding my well founded and firm conviction in your absolute loyalty to the British cause in this war. ... I have observed that your influence on your students is of the nature of a strong ethical appeal, backed by personal experience, in support of the British cause.’ In 1944, apparently in the aftermath of some quarrel over the distribution of examiners’ fees, Popper wrote to the Chairman of Council that: ‘For a number of years, co-operation in the department of philosophy between Professor Sutherland and myself has become increasingly difficult, in spite of my earnest endeavour to facilitate such co-operation. The position has now reached such a state that I have no option but to request the Council to review the situation, and to clarify my duties and responsibilities.’ Sutherland, he claimed, insinuated that he was neglecting his academic responsibilities, and described his protests as ‘typical of behaviour that no Britisher or New Zealander would ever adopt’ [366.5]. (This may

be one source of Popper's allusion later to '[t]he British habit of accepting hardship without grumbling' (16, §7.) What lay behind this unreal accusation of dereliction of duty was Popper's attempt, against all odds, to introduce by example a research tradition into a university where such a thing was unknown, a university whose 'governing body ... seemed to regard time spent on research as time filched from the primary job of teaching' (Simkin 1993, p. 183). Throughout his life Popper had the strongest distaste for university politics, and it was probably his most conspicuous political achievement to have founded in the University of New Zealand the idea that research is an indispensable ingredient of university work (5; Gardner *et al.* 1973, pp. 264f.).

Between 1937 and 1943 Popper began some important studies on the axiomatization of boolean algebra and probability theory. But his main research in this period was in social and political philosophy, a field rather less congenial to someone who—despite youthful enthusiasms—was 'neither by inclination nor by training a student of society or politics' [253.3]. Appalled by the tragedy that was engulfing Europe, not accepted for service in the New Zealand armed forces (Gardner *et al.* 1973, p. 262), he set himself as his war work the task of exposing the intellectual origins of totalitarianism, not only fascism but also communism, which he expected to become a major problem if the war against Hitler was won. In certain respects this work was a continuation of the criticism of Marxism that he had begun in 1919. At the centre of the target was a doctrine that he named historicism, the doctrine that the grand sweep of human history, if not its details, is ruled by historical laws or laws of destiny. Historicist myth-making infects most totalitarian thinking from Plato to Hegel and Marx, to Hitler and Stalin, and has been a crippling intellectual burden on mankind, and on man's attempt to make for himself a better life. The more analytical and methodological criticisms of historicism are contained in *The poverty of historicism*, originally published in three parts in *Economica* in 1944–45, later expanded and published in book form as (13); those that interweave more closely with metaphysical, ethical, and political issues, are contained in *The open society and its enemies* (3, 4), which Popper describes in his preface as 'a critical introduction to the philosophy of politics and of history, and an examination of some of the principles of social reconstruction'. Appropriately for a book that was once to be called 'A social philosophy for Everyman', *The open society* is written with transcendent clarity. But Popper wisely warns the reader that '[a]lthough the book presupposes nothing but open-mindedness in the reader, its object is not so much to popularize the questions treated as to solve them' (*loc. cit.*). The text of the book teems with arguments; the abundant footnotes, on a huge range of peripheral topics, only add to the profusion of thoughts.

This is not the place to expound Popper's social and political views (Simkin 1993, Part II; Shearmur 1996), or his sustained critiques of Plato and Marx. But it may be said that these views are by no means independent of his earlier contributions to the theory of knowledge. Popper brought to the study of society the radical anti-authoritarianism that he had brought before to the study of science; in particular the belief that the reasonable demand for control should not be confused with the unreasonable demand for security. There is accordingly an insistence that, since mistakes are unavoidable, we must devote our energies more to their cure than to their prevention, and hence that social reform should be piecemeal and monitored rather than utopian and monolithic. Popper the advocate of revolutionary thinking in science—since bold new theories are easily disposed of, if they are false—is the same thinker as Popper the opponent of revolutionary activity in society. It is to be observed that anti-authoritarianism does not imply, as some recent writers and politicians have imagined, that

state intervention is to be eschewed altogether. Indeed, Popper makes it plain that the state must be ready to intervene, if only to prevent others from intervening. A significant parallel is drawn (4, chapter 24, note 7) between this paradox of freedom (unrestrained freedom subverts freedom) and other paradoxes of sovereignty such as the paradox of majority rule (the people may vote for a tyrant), and the ancient logical paradox of the Liar ('This statement is false'). As noted below, a further significant parallel between the paradoxes of sovereignty and fundamental problems of the theory of knowledge was to be drawn later.

Although the main theses of *The open society* are inevitably of concern to all thinking people, there are one or two matters raised in it that may be of particular interest to scientific men and women. There is, for example, the explanation of scientific objectivity in terms of inter-subjectivity (chapter 23), which acknowledges that individual scientists are not objective, but maintains that in a milieu of free critical discussion, possible only in an open society, subjectivity, where it is dangerous, may be largely neutralized. The social and institutional prerequisites of science are stressed—science too must be an open society—but the deterministic doctrine, proposed by Mannheim, and once again undeservedly popular, that scientific theories are no more than social products, is rejected. There is the intriguing conjecture, added in (9) to note 9 of chapter 6, that the approximate identity $\sqrt{2} + \sqrt{3} \approx \pi$ may have encouraged Plato to think that all irrational numbers are simple functions of $\sqrt{2}$ and $\sqrt{3}$, and to have reinforced his view that the regular platonic solids (in which only these surds appear) hold the key to the understanding of nature. Still of relevance is an important, though brief, criticism of the ideal of a liberal education. Popper wrote (chapter 11, note 6):

...in our day no man should be considered educated if he does not take an interest in science. ... Anybody who does not attempt to acquire an understanding of this movement cuts himself off from the most remarkable development in the history of human affairs. ... [not only does] literary education ... fail to educate the student ... to an understanding of the greatest spiritual movement of his own day, but it also often fails to educate him to intellectual honesty. Only if the student experiences how easy it is to err, and how hard to make even a small advance in the field of knowledge, only then can he obtain a feeling for the standards of intellectual honesty, a respect for truth, and a disregard of authority and bumpiousness.

Quoting T.H. Huxley's view that a liberal education, far from inculcating 'the power of seeing things as they are without regard to authority', does its best to suppress it, Popper went on:

I admit that, unfortunately, this is true also of many courses in science, which by some teachers is still treated as if it was 'a body of knowledge', as the ancient phrase goes. But this idea will one day, I hope, disappear; for science can be taught as a fascinating part of human history—as a quickly developing growth of bold hypotheses, controlled by experiment, and by *criticism*. Taught in this way ... it could become the basis of a new liberal University education; of one whose aim, where it cannot produce experts, will be to produce at least *men who can distinguish between a charlatan and an expert*.

It is a sad fact that these modest aims are even now very far from being achieved. The cult of the expert is flourishing, as is the cult of the charlatan. Few students of the humanities are taught to think of science as a stirring intellectual adventure, calling for both courage and humility. Science students usually fare even worse (Bondi 1992).

The first volume of *The open society* was completed in October 1942, the second volume in February 1943. Each volume had been rewritten many times, and retyped by Hennie many times (Gombrich 1995). Finding a publisher was hardly less arduous. A copy entrusted to a friend in the USA languished on his desk unsubmitted. The view was that the book was too

disrespectful of Aristotle to be worth offering to a publisher. (Aristotle is a relatively minor character in the book. Popper was later to admit that he was unjustly treated.) Finally Popper made contact with Ernst Gombrich, whom he had known in London before leaving for New Zealand. Gombrich accepted the exacting commission to place the book and, with the assistance of F.A. von Hayek, secured the interest of Herbert Read at George Routledge & Sons, though not before it had been rejected by Cambridge University Press for being too disrespectful of Plato. At more or less the same time Hayek, at whose seminar years before Popper had read an early version of *The poverty of historicism*, was manoeuvring to bring him to the London School of Economics (L.S.E.). The previous incumbent of the Chair of Logic and Scientific Method, shared half-time with University College, had retired in 1941 after 35 years' service (Dahrendorf 1995, pp. 205, 369). Hayek and Ginsberg were in agreement that teaching in these subjects must continue, but whereas Ginsberg thought the matter not urgent, Hayek had a candidate up his sleeve; the half-time chair, he proposed, should be converted into a full-time readership, to be held not by a bright young man but by a mature and experienced logician and philosopher [368.1]. Hayek prevailed. In the spring of 1944 Popper received a contract from Routledge for *The open society* (Gombrich 1995, p. 11; Simkin 1993, p. 187); 12 months later he was appointed to the readership at the L.S.E. *The open society* must have been one of Routledge's best investments, for the book has been through half a dozen editions, and has remained in print for 50 years. The L.S.E. did well too. The testimonial that Russell had written in 1936 hardly exaggerated in its description of Popper as 'a man of great ability, whom any university would be fortunate in having on its staff' [406.4]; the passage is quoted by Grattan-Guinness 1992, p. 5).

The Poppers' life in New Zealand seems to have been an austere one, with little diversion from teaching and research. There was some mountaineering and skiing in the Southern Alps, and in 1938 or 1939, he reports in (14), appendix *x, they went digging for the bones of moas in the marshy land of Pyramid Valley, about 50 miles north-west of Christchurch. After the *Anschluss* they organized a committee to help Austrian refugees to obtain New Zealand entry visas. Out of his salary came repayments of a sizable mortgage loan on their wooden house on Cashmere Hill (where Popper seized the opportunity to deploy his woodworking skills), heavy premiums on a life insurance policy, and some help (via the Red Cross) for his sister Annie in Switzerland. Their main luxuries seem to have been ice creams and milkshakes. But austerity and simplicity were what the Social Democrats in Vienna had always prescribed and—departmental hostilities aside—it seems that the Poppers were contented. The writing of *The open society* evidently exhausted them. But even after the L.S.E. appointment had been confirmed, their problems in New Zealand were not quite over. Popper was reluctant to resign his position at Canterbury (in virtue of which they were permitted to live in New Zealand), since they were stateless and the British Government had granted visas for only 12 months. Much as they wanted to return to Europe, they could not throw away their right to live in a free country. But the worry was unnecessary. On 25 October 1945 he was granted leave of absence for the 1946 academic year, and on 1 November he and Hennie became naturalized British subjects [325.27]. (Citizenship had to wait until 1949.) He eventually resigned his New Zealand post on 18 June 1946 [366.4].

AT THE L.S.E., 1946–59

Popper arrived at the L.S.E. in January 1946, again to be in an institution with only one philosopher. (Not for long: J.O. Wisdom joined the Department in 1948.) Still, there were other philosophers nearby, and he was at once made welcome in London and beyond. For the first time for 10 years he was able to attend philosophical meetings and conferences; so much so that he was an invited symposiast at the Joint Session of the Aristotelian Society and the Mind Association (the annual get-together of British philosophers) in each of the years 1946, 1948, and 1953. But despite the great friendliness he was shown, and despite the great respect he had for the abilities of some of the philosophers he came to know [290.6], he found the professional atmosphere in Britain chillingly unrealistic. Here was a country that for six years had been fighting a bloody war for its life, and here were its leading philosophers fribbling over such questions as whether there are other minds, or whether a physical object continues to exist when no one is observing it. Only with Russell and a few others did he feel in sympathy, though he was dismayed by Russell's return in his *Human knowledge* (1948) to something like the inductivism of J.S. Mill. (Russell owned a copy of *Logik der Forschung*, but the pages were barely cut. See Grattan-Guinness 1992, p. 8.) Indeed Popper never came to terms with academic philosophy in Britain (or elsewhere). He wholeheartedly disliked the linguistic philosophy in vogue on his arrival in 1946, which continued to dominate discussions at least until the end of the 1960s, sometimes comparing philosophical preoccupation with the niceties of language to spectacle-cleaning—a necessary task, no doubt, but only incidental to what spectacles are for (21, p. 138). Gellner's shocker *Words and things* (1959) was 'right on almost every point' [386.15]. Problems, Popper maintained, cannot be solved by language analysis, 'problems can only be solved with the help of new ideas' (23, §7). Still less had he any respect for the frivolous reincarnations of Marxism that have constituted much of the philosophical chic of the last 30 years (though he had considerable sympathy for the seriousness of Marxism itself). Always pugnaciously out of fashion, he followed Kraft (1957; see 15, p. 11) in quoting with approval from von Hofmannsthal: 'Philosophy must be a *judge* of her times: things are in a bad way when she becomes an *expression* of the spirit of her times' (17; 43, p. 159).

In New Zealand there had been no other professional philosophers within 200 miles of Christchurch and Popper had made his friends largely outside philosophical circles. Although it would be wrong to suggest that he did not keep on decently cordial terms, at least for some years, with his British colleagues in philosophy, this predilection was to persist for the rest of his life. He had few philosophical friends, apart from his own students. As in Vienna before the war, he had a reputation for being dazzlingly clever, but prickly and difficult to argue with. He was certainly difficult to get the better of, but this is attributable as much to his preparedness—he had usually already considered, and found a way of answering, his critics' objections—as to his combativeness. When something he had not thought of came along, he was all ears (an expression used, in a moment of perhaps unconscious self-mockery, in the fourth speech in chapter 14 of (16)). Among his new acquaintances in Britain, Peter Medawar, whom he met at Woodger's Theoretical Biology Club early in 1946 (33), was of special and almost immediate importance. They soon became friends, and exerted a powerful and positive mutual influence, each excited by and interested in the other's ideas and problems; for example, we find Medawar in June 1947 inviting Popper to come to Oxford to talk to his group about Gödel's theorem, Zeno's paradoxes and quantification theory, quite a strenuous programme for a single evening

[325.25]. They were both members of the Metalogical Society, which met monthly under the auspices of A.J. Ayer between 1949 and 1953 (Crawshaw-Williams 1970, p. 60). Medawar has documented in numerous places (1967, 1986, and elsewhere) his debt to Popper, whose methodological recommendations he embraced as warmly, and put into practice as thoroughly, as had Eccles; and when he was elected to Fellowship of the Royal Society in March 1949 he wrote [325.26] to acknowledge his indebtedness, saying that Popper deserved a 'cut' of the Fellowship. (In October 1960 he again awarded Popper a portion of the credit for his share of the Nobel Prize [325.25].) This cut was eventually paid with what must have been outstanding interest.

Initially Popper had much less teaching to do than he had had in New Zealand—only #390 *Logic* (Sessional) and #391 *Scientific method* (Lent & Summer) are advertised in the 1945–46 L.S.E. Prospectus [389.6]—but by the end of 1947 he was ruefully reporting eight hours of lectures weekly [282.24]. The heady phenomenon of Canterbury was repeated throughout the L.S.E. years. Lecturing with a minimum of notes, Popper attracted large audiences, composed of as many colleagues and visitors as undergraduates and research students. He followed no syllabus, and spoke usually on some topic that held his current interest. Students in *Scientific method* with no training in physics were more than once treated to an attack on the orthodox interpretation of quantum mechanics. His celebrated seminar, in which friends and foes exposed their ideas to his incessant fire, was not an experience for the faint-hearted, and many found it overbearing and oppressive. Fools were suffered, not always gladly, but not upstarts and braggarts. A clear problem, and a provocative thesis stated in simple terms without jargon or technicality, were what every speaker was advised to bring along. But there was no telling from which direction criticism would come. All that was guaranteed was that it would come soon.

In 1947–48 Popper published seven papers on logic, the fruits of intense work done in isolation in New Zealand. In these papers, in which he largely rediscovered the systems of natural deduction of Gentzen, he hoped to solve a problem left open in Tarski's discussion of logical consequence (1936, p. 420), namely that of clarifying the distinction between logical constants (such as **not** and **all**) and extra-logical constants (such as **bat** and **ball**); and thereby to show that the validity of the rules of logic can be reduced to definitions (6). There are errors in the second part of the project, as reviewers were quick to point out; but the papers 'contain a detailed and well-founded proposal' (Schroeder-Heister 1984, p. 80) to characterize the logical constants. It seems that Tarski, whose judgement Popper revered, was not much interested in this work when it was shown to him, and this response, as much as the mistakes he had made (Popper was always very sensitive to mistakes that he had not caught himself), discouraged further efforts.

The open society and other works were successfully presented to the University of London in 1948 as a thesis for the degree of D.Lit. His entitlement to a professorship had by then been occupying Popper's mind for some time (Medawar 1986, pp. 113–115), but the bitterness of its curriculum in logic and scientific method at that time made it hard for the L.S.E. to make a case to the University for the establishment, or re-establishment, of a chair. Had Hennie's mother, who was still in Austria, not become seriously ill, the Poppers might well have been persuaded to return permanently to New Zealand, to the vacant chair at Otago to which Eccles energetically tried to lure him [290.7]. (The only occasion on which they did go back to New Zealand was in 1973, as visitors to the Universities of Otago and Canterbury.) The problem was eventually resolved by Popper's promotion to a personal professorship on 1 January

1949. Later that month there arrived an unexpected letter from Professor Donald Williams, Chairman of the Department of Philosophy at Harvard, asking if Popper 'would entertain an invitation to give the William James Lectures' in the year 1949–50 [304.25]. Popper says that he 'thought a mistake had been made' (23, §28), and that he had been confused with Josef Popper-Lynkeus, a distinguished philosopher and social theorist, also from Kolin, who had belonged to the Monists, the circle of Mach and Ostwald. But this is scarcely believable; on the one hand, Popper-Lynkeus had died nearly 30 years before; on the other, Williams's first letter mentions *The open society* by name. A fee of \$6000 was offered for eight or ten weekly lectures, together with a parallel seminar course. The lectures—under the title 'The study of nature and of society'—were duly given, apparently from notes. The seminars were on 'The structure of experience' (a somewhat uncharacteristic title). Although one of the conditions of the lecture series, not mentioned by Harvard until it was too late, was that the lecturer deposit a manuscript copy of his lectures in the University Library, Popper never complied, and never produced the obligatory work for publication.

Yet he was hardly taking things easy. The 1950s was an extraordinarily productive decade. *Conjectures and refutations* (16) collects a number of striking papers from these years on diverse problems in the philosophy of science, on logic, on the body-mind problem, on metaphysics, on problems of social philosophy, and on the pre-Socratic philosophers. One of these papers (chapter 6) shows to what extent Mach's criticisms of absolute space, time and motion were anticipated by Berkeley in *De motu* (1721). Popper distanced himself from the instrumentalist tenor of these criticisms, and also from the essentialism—in Cotes's explanation of gravity, for example—against which they were aimed; and argued (chapter 3) for an intermediate view of science, as the search for ever deeper, but never ultimate, explanations. Beginning with (10), there were seven notes in five years skirmishing with Carnap and Bar-Hillel over the inductivist identification of probability and degree of corroboration. In their joint paper (Bar-Hillel & Carnap 1954) these authors managed to embrace Popper's doctrine of the inverse relation of probability and content, but separately they resisted the conclusion that empirical investigation is not directed at making our theories more probable. In (12) was initiated a series of notes, mostly in *Nature*, on the arrow of time and the correct interpretation of the second law of thermodynamics, topics on which he had some altercations with Schrödinger. At Hennie's insistence—she had to cry all night before he was persuaded—Popper also undertook (with help from others) an English translation of (1), a project that had been set in motion as far back as 1937 but soon halted for lack of financial support (Godsiff 1972; Woodger completed a translation too in 1946 [12.13], but it was never used). Although the outcome, *The logic of scientific discovery* (14), contained much extra material, packed into new footnotes and new appendices, even more was left out. As the translation proceeded it became clear that the fruits of 20 years' thinking, especially on the interpretation of probability and related issues, would comfortably fill another book; and accordingly another book was conceived and composed, under the title *Postscript: after twenty years*. Projected for 1954, it was more or less in shape by the end of 1957. But for various reasons, including undue perfectionism and serious illness, the correction of the proofs of this companion volume was still unfinished in 1962, when it was set aside for what was to be a second 20 years. It was eventually published as a three-volume work, containing a second bounty of new material in an unwieldy agglutination of prefaces, introductions, and addenda, as (29–31).

Popper travelled widely. Encouraged by Hayek, he began as early as 1948 to attend and

contribute to meetings of The Austrian College Society in Alpbach (Tyrol), a group set up largely by former members of the Resistance to nurture democratic values in post-war Austria. He was invited to spend the 1956–57 academic year at the Institute for Advanced Studies in the Behavioral Sciences at Stanford. This visit was nearly scotched by Popper's guileless application to the US consulate for a visa; having originally written of some of the friends of his youth that they 'later became Communists', he then declared that what he had wanted to say was that they 'called themselves Communists'. He was told at once that this association would have to be referred to Washington, and that meanwhile he could do no better than collect up to 20 declarations by prominent and solid citizens that he was not a communist. His friends obliged, writing letters resonant with disbelief that the US Government could imagine that the author of *The open society* was sympathetic to communism [407.23]. It was only intervention by a retired senior diplomat that put an end to a time-wasting investigation.

It is hardly possible to do justice to the wealth of arrestingly novel ideas that Popper brought to fruition between 1950 and 1959 (he continued to work on many of them for many years more). The principal contributions of *The postscript* will be described in the next section. Before that it is necessary to say something about his remarkable 'autonomous' axiomatizations of the elementary theory of probability.

Popper had published his first (rather clumsy) axiomatization of probability theory as far back as 1938 (2). Nothing more was published until 1955 (11), but in the meantime his thinking on the subject took a most unexpected turn. Originally, knowing nothing of the Kolmogorov axiomatization of 1933, now nearly canonical, he had been concerned only to provide an abstract axiomatization for the singular function $p(y)$. (In terms of this function the conditional probability $p(x \mid z)$ is standardly defined as the quotient $p(xz)/p(z)$, for positive $p(z)$; here concatenation stands for meet or conjunction.) Later, fortuitously misled by the fact that in the propensity interpretation (to be discussed below) probabilities are always relative to the ambient physical conditions, he decided to adopt the binary functor $p(x \mid z)$ as primitive. Call x and z [*probabilistically*] *indistinguishable* if for all y the probabilities $p(x \mid y)$ and $p(z \mid y)$ are equal; in symbols, $x \sim z$. At some stage Popper noticed that, granted $(xz)y \sim (xy)(zy)$, we can easily derive the distribution property $(x \vee z)y \sim xy \vee zy$ (here $\vee$ stands for join or disjunction) from the general addition and multiplication laws

$$\begin{aligned} p(x \mid y) + p(z \mid y) &= p(xz \mid y) + p(x \vee z \mid y), \\ p(xz \mid y) &= p(x \mid zy)p(z \mid y). \end{aligned}$$

(This derivation is the nub of one of the criticisms in (19) of the thesis of Birkhoff and von Neumann that quantum mechanics, which is undoubtedly in some sense a probabilistic theory, requires a non-distributive logic.) Now the identity of the terms $(xz)y$ and $(xy)(zy)$ follows from the familiar lattice laws of idempotence ($yy = y$), commutation ($xz = zx$), and association ($x(yz) = (xy)z$) for meet. Popper had worked in New Zealand on the axiomatization of Boolean algebra (and on Boole [4.12]), and seems to have been carried away with the thought that suitable axioms for the functor p might make it possible to establish the indistinguishability of these terms too, and hence the probabilistic distributive law $p((x \vee z)y \mid w) = p(xy \vee zy \mid w)$. So it proved. One of the many elegant axiom systems that accomplishes this task is this one (40, appendix *V):

- A1 $\exists x \exists z \, p(x \mid x) \neq p(x \mid z)$
 A2 $\forall y [p(y \mid y) \leq p(x \mid z) \leq p(z \mid x)] \implies \forall y [p(y \mid x) \leq p(y \mid z)]$
 A3 $p(x \mid x) \leq p(z \mid z)$
 B1 $p(xz \mid y) \leq p(x \mid y)$
 B2 $p(xz \mid y) = p(x \mid zy) p(z \mid y)$
 C $p(x \mid x) \neq p(y \mid z) \implies p(x \mid x) = p(x \mid z) + p(x' \mid z)$

The accent in axiom C stands for a singular operation that turns out to have all the properties of complementation. The sign $\vee$ for join is introduced by the definition: $x \vee z = (x'z')'$. There are, as in all Popper's axiomatic systems for conditional probability, some unusual features. In the first place, $p(x \mid z)$ is defined for every z , even the contradictory element yy' ; in fact, $p(x \mid yy') = 1$. Secondly, the customary bounds on the function p do not appear explicitly; $0 \leq p(x \mid z) \leq 1$ is a theorem. Most importantly, no assumption is made concerning the domain of elements $x, y, z, \dots$, neither that it is a Boolean algebra, nor even that it is a lattice; none of the defining identities for meet is assumed for the concatenation operation, and no defining identity for the complement is assumed for $'$. It is this freedom from any purely algebraic assumptions that the description 'autonomous' is meant to mark. Nonetheless, the domain is reducible to a Boolean algebra, in the sense that $s = t$ is an identity of Boolean algebra if and only if the probabilistic indistinguishability of s and t is derivable in the system. This establishes a result that previously had been no more than a rumour: namely, that the theory of probability is a genuine generalization of classical propositional logic, not merely an extension of it (in the way that, as here formulated, it is merely an extension of some theory of the real numbers). Indeed the deducibility relation $z \mid x$ is definable in many ways; for instance, by the relation $z \sim xz$, or by the identity $p(x \mid z) = 1$.

An axiomatic system different only in minor respects from the one above was spelt out in detail, and weighed with numerous alternatives, in two of the new appendices, *iv and *v, of (14). Popper ventured there the opinion that it 'can hardly be further simplified' (14, appendix *ii, note *i), but he never stopped working on refinements, improvements and variations to it. Some of the crucial proofs in his original presentation were indeed decisively simplified in (42). Leblanc (1989) is a survey and comparison with other work, including the contemporary axiomatizations published by Rényi (1955), which also take as primitive the conditional probability functor. Although Popper's systems are plainly a contribution more to abstract algebra than to philosophy, they are a splendid philosophical achievement too, revealing close and subtle connections between logic and probability theory. Popper explained his embarking on such a technical exercise on the grounds that only if the definitive properties of probability are established can we hope to evaluate the inductivist thesis that the conditional probability $p(h \mid e)$ of a hypothesis h on evidence e is identical with its degree of corroboration. It is questionable whether his underlying aim—to crush that thesis—has been achieved. In any case it is odd that when introducing the end-products of his long investigation Popper should have disparaged them as 'somewhat technical—too much so for my taste, at least' (14, proem to new appendices), since the zest with which he worked on them, and the quite proper pride he took in them, were well known. He always rightly objected to the cult of precision for its own sake, but these beautiful axiomatic systems are not an instance of precision for its own sake.

In the early 1950s the Poppers had bought Fallowfield, the house in Penn (Buckinghamshire) that they were to live in for 35 years. (Previously they had lived in East Barnet [293.7].) On the outskirts of London, only just within the 30-mile radius ordained by the University of London (Bartley 1982, p. 254), barely accessible by public transport, Penn was still a village, Fallowfield a relatively small house set in a large garden in a private road. So began a process of withdrawal from metropolitan life that, by the end of the decade, was almost eremitic. Never comfortable in smoke-filled rooms, Popper attended committee and board meetings at the L.S.E. as seldom as he could. New recruits to the Department, Joseph Agassi in 1957, J.W.N. Watkins (who transferred from the Department of Government) in 1958, W.W. Bartley III and Imre Lakatos in 1960, when Agassi departed, and Alan Musgrave in 1964, helped markedly to reduce his teaching load (in some years it was no more than one lecture and one seminar each week). He was relieved of much of the administrative burden too, largely by Agassi and Watkins. From the early 1950s until the early 1980s, the Nuffield Foundation and others supported Popper's employment of a research assistant, a singular luxury for a philosopher, and one of the main tasks of the assistant, in daily contact by telephone, was to procure books and periodicals; this was often a considerable task, since Popper's interests in natural science were only exiguously served by the British Library of Political & Economic Science (indeed in one year the assistant was the source of 50% of the Library's requests to the Inter-Library Loan Service). Popper was often able to restrict his attendance at the L.S.E. to one day each week.

With the publication of *The logic of scientific discovery* (14), *Logik der Forschung* was available in English translation. *The postscript* was planned to follow, as soon as the proofs could be amended to incorporate the extensive suggestions of a new critic who had arrived at the L.S.E. in 1958, W.W. Bartley. But Popper's health—always a worry to him—took a serious downward turn in 1960 when his eyesight started to fail. Anxiously returning to Vienna, he had operations on both eyes for detached retinas. The operations were successful, but work on the proofs of *The postscript* was pushed aside, and by 1962 abandoned. The revised typescript was retyped in 1969, but again put aside. Several parts were extracted and published independently; and by the time that the book was published in full in the early 1980s it had become almost a historical document. It is appropriate to turn to its contents here, and to resume in the following section the story of Popper's career at the L.S.E. No deep significance attaches to the year 1960 as the moment of division.

THE POSTSCRIPT

The unifying theme of the original unsupplemented *Postscript* is the propensity interpretation of probability, with which Popper proposed to overcome some outstanding problems faced by the frequency interpretation and, above all, to resolve the main paradoxes of quantum theory. The new interpretation is underwritten by metaphysical indeterminism, the doctrine that the cosmos is ineradicably infected with an element of chance. But it holds that chance is more than mere chance: chance events are not rank accidents, unintelligible depositions of causality; they are outcomes partly controlled by the physical situations from which they emanate. It is unfortunate that the key section introducing the propensity interpretation, 'The case for propensities' 31, Part II, §20, signally fails to do justice to its originality; elsewhere in *The postscript* too Popper tends to represent it as simply an adjustment to or correction of the

frequency interpretation (which, for want of any other, physicists, such as von Neumann 1936, Chapter IV.1, often settle for). But, right or wrong, the propensity interpretation is a radical addition to our understanding of physical probability, the crucial improvement being the authentication of single-case probabilities. According to the frequency interpretation, a single outcome may be ascribed a probability only if it is glossed as a member of a collective; for example, the probability that an individual particle is transmitted at a barrier makes sense only if the particle is assigned to some ensemble of particles. Since the frequency of transmission may be different in different ensembles, there is no unique collective and there is no unambiguous single-case probability. We might suggest that the true probability of an outcome is its frequency in the most tightly defined collective containing it, but this collapses all single-case probabilities to 0 or 1, a result agreeable only to determinists. There is to be sure a subjectivist angle, which understands the probability of the outcome of a single trial to be its frequency in the most tightly defined collective to which, prior to the trial, it was known to belong. On this view, it is appropriate to talk of probability only because we do not know enough. This position must in turn be distinguished carefully from the subjectivist position with regard to chance that Popper, anxious to avoid metaphysical speculation, fell back on in (1, §69). It was one of the theses of §62 of that book, much elaborated in §§3–6 of (30), that we resort to probability not as a relief from ignorance but because that is what the problem situation demands: statistical questions—such as those concerning the intensity of spectral lines—demand probabilistic or statistical answers. An inability to make adequate sense of single-case probabilities is an inevitable consequence of combining an objectivist approach to probability with a subjectivist construe of chance. *The postscript* renounced the latter, and set out to provide a theory of objective single-case probabilities.

For once we acknowledge indeterminism, which implies that the present state of the world can in principle evolve into different possible future states, we shall want to allow that these different possibilities are not necessarily equal; and if we are persevering, and willing to dig behind the appearances, we may want also to attribute these inequalities to some feature of the present state of the world. What Popper suggests is that the present state is endowed with propensities, *active forces whose resultant is indeterminate*; and it is the strength of the propensity of a particular outcome that we have in mind when we ascribe to it a single-case probability. As each particle in a stream arrives at a barrier, for example, there is a propensity of transmission (as well as propensities of reflection and absorption); and hence a unique probability of transmission that is independent of other particles (though if they are similar particles, the probability may be the same). We cannot measure directly the propensity of a single outcome, since it either happens or does not happen; single-case probabilities may therefore properly be called metaphysical. Constant propensities may indeed be measured more or less accurately by frequencies, but this does not mean a return to or a variant of the frequency interpretation. What the propensity interpretation does, in brief, is bring to life the uninterpreted term '*P*' ('probability') in the usual measure-theoretic formalism. The publication of extracts from *The postscript* many years before the book itself appeared has led to a widespread acceptance of Popper's innovation in one form or another. Yet there are two particularly common misunderstandings needing correction, both stemming from a naive assimilation of propensities to those dispositional properties (such as mass or conductivity) that are usually attributed to material bodies. First, propensities do not reside in individual things; they are 'objective physical properties of the physical situation under consideration and, ultimately, of the whole physical world' (30, Introduction, §4). Secondly, the

fundamental propensities are not propensities to produce frequencies. The propensity interpretation, unlike the frequency interpretation, is not restricted to repeatable effects or experiments (30, Introduction, §3, note 63). Even what counts as a repetition of an experiment is different: propensity theorists treat two situations as similar because, they conjecture, they have the same probability (propensity) distribution; for frequentists it is the other way round.

The propensity interpretation would be invalidated if the universe could be shown to be deterministic; invalidated, that is, as an interpretation of probabilities other than 0 and 1. One of the principal tasks of *The postscript* was to demonstrate that the debate between determinism and indeterminism, though inescapably metaphysical, is amenable to rational discussion. *Logik der Forschung* had tried hard to steer clear of metaphysics (see 14, §69, note *2), and declined to endorse the fashionable indeterminism of the Copenhagen interpretation of quantum mechanics. Unpredictability of some phenomena was admitted—indeed, as noted below, this was the substance of Popper's interpretation of the uncertainty formulas as scatter relations—but it was argued that this spoke neither for nor against indeterminism (§78). In other words, Popper held open the door to a deterministic interpretation of chance phenomena in terms of hidden variables. But since the matter was not empirically decidable either way, he regarded it as out of bounds; not meaningless, but best left undiscussed. By the time of the writing of *The postscript* this wariness of metaphysics had been allayed, and the empiricist leanings of falsificationism generalized to a critical rationalism (4, chapter 24) that was prepared to entertain any hypothesis that could be criticized (whether or not the criticism made appeal to empirical evidence). Given the affinity between determinism and historicism, it is not very surprising that Popper should have been won over to indeterminism. In (4), chapter 13, and in arguments with Einstein at Princeton in 1950, he had described indeterminism ('free will', if you like) as the position of common sense, and by no means 'a necessary prerequisite of a science which can make predictions'. In *The open universe* (29) he maintained that the only arguments against it are those (Laplace's demon) that extrapolate from the predictive success of science, especially classical physics; and that to expose determinism as an unsupported dogma in conflict with common sense, it is therefore enough to show that there are limits to what science can predict. He presented three main arguments, none of them alluding to the interpretation of the uncertainty formulas as scatter relations. One, a familiar one today, appeals to the existence of unstable dynamical systems (§14); relying heavily on a prescient discussion by Duhem (1906, Part II, chapter III, §3), Popper cited a result of Hadamard (1898) that shows that on certain surfaces of negative curvature qualitatively different trajectories may depart from starting points that are arbitrarily close together. The second argument, more original and still not well understood, attacks the possibility of prediction from within, the possibility, that is, that a predicting device predicts correctly its own future activity (§22; and also 7, 8); there are connections here with the phenomena of self-fulfilling (and self-frustrating) predictions, discussed briefly in (13, §§5f). A third argument appeals to special relativity; to the possibility that events outside the light cone, and therefore unknowable, at the time *at which* a prediction is made, may be within the light cone, and causally efficacious, at the time *for which* the prediction is made (§19). None of these arguments is sufficient to eliminate determinism, since the closedness of the future is, as Popper well knew, compatible with our not being able to predict the future. He therefore offered a variety of more intuitive arguments, as well as an argument, due to Landé (1955), specifically designed to undo the possibility of deterministic hidden variables (§29). This argument assumes a favourite thesis of Popper's, that a statistical conclusion always requires

statistical premises; a thesis that is rendered doubtful, if not contradicted, by the pseudo-random evolutions of some non-linear dynamical processes.

The subjectivist alternative, that a statistical conclusion always derives from a supposition of nescience, was the main target of *Quantum theory and the schism in physics* (30), the third volume of *The postscript*. Popper vehemently contested the infiltration of this subjectivism into statistical mechanics, where it had produced pronouncements such as that of Pauli: ‘the entropy of a system ... depends upon our *knowledge* of the system. ... If this knowledge of ours is maximal ... then the entropy is always zero’ (1954, pp. 114f., quoted in §5 of (30)). Against this view that, in an ideal state of knowledge, the statistics would wither away, Popper wrote that ‘it is clearly absurd to believe that pennies fall or molecules collide in a random fashion *because we do not know* the initial conditions’ (§3). The attack on subjectivism was continued into quantum theory, whose outstanding riddles, Popper argued, sprang from two mistakes: the positivist dismissal of what is hidden, and the attempt to graft Born’s statistical interpretation on to an underlying determinism. With the help of the propensity interpretation of probability he proposed to extend the realist discussion of (1), where it had been argued that the uncertainty relations, for instance

$$\Delta x \Delta p_x \geq h/4\pi,$$

set absolute limits not to the precise states of individual particles, but to the statistical homogeneity of ensembles that we can prepare. The uncertainty relations say nothing about what can be measured, let alone about what is truly the case, but only about what can be predicted. It is now recognized that Popper was here at least on the right track; the unpredictability, beyond a certain level of precision, of both the position and momentum of a particle does not preclude the possibility that more precise values of these quantities might be calculated from the results of later measurements. The uncertainty relations do not imply that precise positions and momenta can never be known—let alone, as the positivism of the Copenhagen interpretation pretends, that to talk about a particle’s path is to talk nonsense—only that there are details that we cannot know in advance. The propensity interpretation makes good sense here; indeed, the uncertainty relations can be understood as stating conditions on the propensity that a particle will take up a certain momentum (position), given that it is forced into a certain position (momentum). But the new outlook enables us, Popper suggested, also to deal with such conundrums as the two-slit experiment (§18). The secret is simply that propensities are a feature of ‘the physical situation under consideration and, ultimately, of the whole physical world’ (already quoted). A screen with one slit open evokes a different propensity distribution from a screen with two slits open. That the distribution may change instantaneously or ‘with superluminal velocity’ when a slit is opened or closed is not a mysterious quantum effect, but a trite fact of probability theory. (Why it changes in the way that it does is indeed something for quantum theory to explain.) A pinboard provides a simple classical example of exactly the same kind of effect (theses 8 and 9 of (18), included as Introduction to (30)). This is in general all that there is to the ‘reduction of the wave packet’ (1, §76; 30, §§8, 23). Of course, the instantaneous character of the change is problematic for those who interpret probability subjectivistically; for probability on this interpretation should change only with a change of subjective information, which would therefore have to be transmitted with superluminal velocity.

In (1), §77, Popper had hoped to seal his criticism of Heisenberg’s interpretation of the

uncertainty relations with a thought experiment demonstrating its inconsistency with the quantum-mechanical formalism. Although this experiment was vitiated by an error, it may have influenced the thinking that led to the Einstein–Podolsky–Rosen thought experiment of 1935 (Jammer 1974, pp. 174–178; (30), Preface, §V, note 20). The EPR experiment itself argues that instantaneous action at a distance is a consequence of any interpretation of the formalism that denies that particles possess sharp positions and momenta simultaneously. But later experiments, such as those suggested by Bohm, and those actually performed by Aspect, indicate that there may exist genuine infringements of the principle of locality. Popper’s response to these later developments was to concede that action at a distance may have to be accepted in modern physics (as it was accepted in classical physics). Aspect’s experiment, he declared, would then be ‘the first crucial experiment between Lorentz’s and Einstein’s interpretations of the Lorentz transformations’ (Author’s note to (30)).

For many years Popper was almost alone in his fight against what he saw as a stifling orthodoxy dominating quantum theory, though he remained heartened that Einstein and de Broglie were among the doubters. He was greatly delighted at the publication of Ballentine (1970) in *Reviews of Modern Physics*. He fought not only against the intrusion of ‘the observer’ into objective science, but also against the supposed revelation that quantum mechanics had brought at least one area of physics to completion. Not only was he convinced that this was based on a misinterpretation of the quantum formalism; he resisted it as wilfully obscurantist and, if taken seriously, in danger of becoming a self-fulfilling prophecy. The inspiring message with which (1) had ended was that science is a permanently unfinished enterprise, and that the opportunity of making revolutionary new discoveries must not be compromised. This ‘persistent and recklessly critical *quest* for truth’ (§85) he never abandoned.

Although *The postscript* is, as noted, largely a treatise on the philosophy of physical science, tied together by the propensity interpretation of probability, the first volume *Realism and the aim of science* (31) contains nearly 300 pages of renewed discussion of the problems of induction and demarcation, a defence of realism and a criticism of instrumentalism, and a further offensive against the idea that degree of corroboration can be identified with probability. Here it is possible to mention only two highlights: an important argument against the possibility of inductive logic, which takes notice of the fact that new theories in physics characteristically contradict the evidence on which they are supposed to be based; and a witty criticism of the positivist doctrine that the division between science and non-science is a division between what can and what cannot be said. The former point can be quickly summarized: according to Newton’s theory, for example, the elliptical orbits postulated by Kepler’s laws are perturbed by interplanetary forces. Yet Kepler’s laws offer no hint of what adjustments are needed to proceed to the more general theory. As Whewell (1860, p. 509) had said: ‘Newton’s proof that this [the elliptical orbit] was the result of a central force ... was the solution of a problem at which his contemporaries had laboured in vain.’ Duhem too noted that the process involved is in no sense logical (1906, Part II, chapter VI, §4); it is the invention of a deeper hypothesis. Popper went on to suggest that we may regard the correction of its predecessor (or of the evidence) by a new theory as a sufficient condition for an increase in depth (§15). In contrast to the craven shallowness of positivism and instrumentalism, realism encourages us to penetrate more and more deeply into the constitution of the universe.

If the signature tune of the Vienna Circle was based on Waismann’s theme that ‘the meaning of a statement is the method of its verification’ (and hence metaphysics is

meaningless), its most ambitious orchestration was due to Carnap who, following Wittgenstein, proposed the thesis that in an ideal language ('the language of science') only verifiable scientific statements would be capable of formulation, and what pretended to be metaphysical statements would be exposed as pseudo-statements, ungrammatical noises. In its primitive form this idea was destroyed by the recognition (for which Popper deserves much credit, even though it goes back to Hume) that scientific hypotheses, being spatiotemporally universal, are never verifiable by experience. But the meaninglessness of metaphysics was soon to be identified by Carnap with its untestability, and the dogma of the language of science dressed in a new costume. Popper's side of this story is painstakingly told in a paper written around this time (16, Chapter 11). The discussion in (31) of what Popper called 'the changing philosophy of sense and nonsense' contains at least one gem (§24). To establish that belonging to science has little to do with belonging to a scientific language, and that testability has little to do with logical form, Popper showed that two of the most famous unresolved conjectures of elementary number theory can be expressed in almost the same vocabulary, and cast in exactly the same logical form, yet one is falsifiable by inspection of finitely many positive integers and the other (according to present knowledge) is unfalsifiable. We may formulate the two conjectures in this way (the range of the variables is the set of positive integers).

- (1) $\forall x \exists z [\text{prime}(x + z) \wedge \text{prime}((x + 2) - z)]$,
- (2) $\forall x \exists z [\text{prime}(x + z) \wedge \text{prime}((x + 2) + z)]$.

Popper notes that if Goldbach's conjecture (eqn 1 above) is a grammatically correct expression in a meaningful language, then so is the twin prime conjecture (eqn 2 above). But whereas the former, if false, is finitely falsifiable, the latter is not. Testability is not to be identified with meaningfulness. Indeed, it had long been Popper's view that although some metaphysics is of dubious merit, in general it is not meaningless, and may even be an essential component of science. Atomism was the example usually given of a metaphysical theory that gradually became testable and scientific (1, §4). *The postscript* itself found metaphysical research programmes to be ubiquitous, and suggested that most scientific activity, from the selection of problems to the endorsement of solutions, takes place under the guidance of some programme or another. In its concluding chapter, 'A metaphysical epilogue', Popper listed 10 programmes, from the static world of Parmenides to the statistical world of Born, that had coloured the development of the theory of matter, and offered a particularly useful brief history of the change in the metaphysical problem situation from Descartes to Faraday (30, §20). A vision of the world in terms of fields of propensities was proposed as a new research programme that would solve many of the existing problems of the interpretation of physical theories. The idea is further developed in (36).

It is hard to assess what the impact of *The postscript* might have been had it been published 25 years earlier. Some of its central theses—that there are limits to predictability even in classical physics, that the Copenhagen interpretation is not the last word in microphysics, that scientific theories have to be falsifiable, that science is shot through with metaphysics—are among today's commonplaces. This was hardly true in the mid-1950s, when the book was written. But there is one stark exception: the thesis that induction is sheer sophistry has yet to become a commonplace, even among those whose work, in Popper's view, best illustrates what sophistry induction is.

AT THE L.S.E., 1960–69

For many years Fallowfield was the site of an intellectual cottage industry. In his study, overlooking the lawn, Popper wrote always in longhand, producing complex coiling pages crammed with corrections and recorections, cancellations and insertions. Hennie gave everything written her incisive critical attention, then typed it with four or five carbon copies, ready for the next round of revisions. When there was time, she worked in the garden, but much of her day was spent in what was called the bookroom, trying to control the ceaseless flow of paper from the external world. Few items were ever discarded, but there was not even a filing cabinet to organize the heaps of folders and envelopes precariously stacked on shelves. In later years Popper also had a secretary at the L.S.E., who played a crucial role in the production of the final typescripts. Through the research assistant's hands passed each version of each piece of philosophical writing, and comments and suggestions for improvements were encouraged and introduced at all levels and at all stages. In effect the assistant became an apprentice, except for Hennie the most influential and active partner in the business (Agassi 1993, chapter 4). In what spare time he had, Popper enjoyed reading the novels of Austen and Trollope—for which he had an almost photographic memory—but in truth there was little time allowed for relaxation at Penn. On normal days even meals were distractions; Hennie did not much like to cook, and visitors usually had to make do with sandwiches or biscuits, instant coffee or tea. Popper himself never touched alcohol—sometimes mentioning the degradation that he had witnessed in Vienna in his youth—and it was rarely offered. But visitors (not colleagues or assistants) could on occasion be served a glass of decidedly good sherry, in which Hennie would join. She was much happier entertaining at home than going out, and as Popper's aversion to smoke increased, not to mention his difficulties in hearing what was said if there was background conversation, they stayed more and more at Penn, always graciously receiving visitors but paying few return calls. Yet for all the mass of work being done, there was little pressure. Bondi has spoken of 'the purely intellectual atmosphere in which non-intellectual thoughts or actions had no place whatsoever ... exemplified there with clarity, beauty and kindness' (1995, p. 7). In summer many pleasant afternoons would be spent working at the table on the terrace, or sauntering round the fallow field, a large grassy plot adjoining the garden, listening—and intermittently intervening—as Popper tried out new ideas.

Popper was elected a Fellow of the British Academy in 1958. In the Academy's Annual Philosophical Lecture in January 1960 (16, Introduction) he explicitly compared the epistemological question 'How do you know?' with the political question 'Who should rule?', maintaining that each was responsible for untold authoritarian mischief. The political question, it had been argued in chapter 7 of *The open society*, is potentially paradoxical, and must be replaced by the more sensible question 'How can we organize political institutions so that bad or incompetent rulers can be prevented from doing too much damage?' Similarly, the sensible answer to the question 'How do you know?' is 'I do not know. How can I improve my guess?'. The way in which hypotheses and evidence in science adjudicate hesitantly and flexibly on each other's worth is, he suggested, parallel to the way in which the government and the electorate in a liberal democracy exercise limited but genuine control on each other's actions. What political theorists call the theory of checks and balances, though fraught with danger, works surprisingly successfully in practice once the appropriate institutions are in place. It seems fair to conclude that science too should also be able to flourish without appeal to ultimate authorities. Anarchy [= scepticism] and tyranny [= dogmatism] do not exhaust the alternatives.

Another important step, quite different in character, that was taken in 1960 was Popper's attempt to give a technical definition of the idea of 'verisimilitude' or 'closeness to the truth', an idea sharply to be distinguished from probability, which is a measure of closeness to certainty (see 16, chapter 10, §§xi–xiv). This attempt was—if the expression may be used—a splendid failure. For although Popper's own definition was shown in 1974 to be unsatisfactory, its refutation inspired much work by others, providing a timely illustration of the thesis that the way that our knowledge grows is through the invention of new hypotheses and their critical examination. The original problem here was that of explaining how progress is possible in science if so many scientific theories, even the most empirically successful ones—such as Kepler's laws or Newton's theory—are in time refuted. (The instrumentalist whitewash, which pleads that a refuted theory remains 'true within its field of application', is not countenanced by falsificationists.) Defining the *truth content*, B_T , and *falsity content*, B_F , of a theory B to be the sets of its true consequences and false consequences respectively, Popper proposed a seductively simple elucidation of the idea that C has greater verisimilitude than a rival A has (or is closer to the whole truth than A is). In brief, $A_T \subseteq C_T$ and $C_F \subseteq A_F$; that is, C must add to A 's successes (or, at worst, not lose any), and lose some of its errors (or, at worst, not add any). It can be shown that the relation between false theories is, unfortunately, never as simple as this; indeed, the definition does not allow one false theory to be objectively closer to the truth than is a rival theory (true or false). Popper responded in different ways to the impact on his philosophy of this result and related negative results: (24) and (27), appendix 2, were puzzled but appreciative, (31), Introduction, §V, more defiant. It is perhaps fair to say that, enchanted by a definition that he found 'very beautiful indeed' [294. 6], he had neglected to subject it to stringent criticism, and that, had he done so, he would quickly have discovered its unsatisfactoriness. A voluminous subsequent discussion has hardly resolved the matter, and the situation at present seems to be that although realists (those who believe in truth at all) are for the most part confident that it makes good sense to claim that one theory (say, general relativity) is objectively closer to the truth than is another (say, classical mechanics), and hence an instance of scientific progress, no one can give a logically watertight account of what such progress amounts to.

A third advance at the beginning of the 1960s was into the philosophy of biology. Biological ideas lay behind many of Popper's early methodological proposals, for learning by trial and error is, as he had realized, a Darwinian process of selection by the environment, while induction—if it existed—would resemble a Lamarckian process of instruction. (Just as natural selection is often misread as the inheritance of acquired characteristics, so learning by trial and error is often misread as induction.) It is perhaps surprising that Darwin's theory, and biology generally, had until this point been mentioned so little in Popper's writings, the most weighty appearance having been the claim in (13), §27, that evolutionary theory is a historical hypothesis, and that there can be no law of evolution; there is also an interesting reference in (28), §S4, written in the early 1930s, to Jennings (1906). Darwin had been the subject of one of the portraits in his father's study (the other was of Schopenhauer), and he retained a fond memory of it; and among scientists biologists (Eccles, Medawar, Wächtershäuser) have been notably vocal votaries of Popper's ideas. Doubtless it was his preoccupation with the logical analysis of the process of scientific discovery, rather than its description, that kept him so long from articulating what was later to be known as evolutionary epistemology. The similarity between falsificationism and Darwinism was brought to the fore in the 1961 Herbert Spencer Lecture at Oxford (22, chapter 7), where Popper talked freely of the 'natural selection of

hypotheses', insisting that the term was meant quite literally. Of course, there are differences between the growth of science and the evolution of life; he remarked, in particular, on the sharp contrast between the increasing differentiation typical of biological evolution and the increasing integration and unification typical of pure scientific knowledge. At this stage he had not developed the more subtle application of evolutionary categories to scientific knowledge that was evident, for example, in his second Spencer Lecture of 1973 (41, chapter 1); there science was no longer likened to a population of hypotheses struggling for survival but to a human organ, exosomatically developed and subject to unnatural (critical) selection. The crucial point, however, is the process, not the medium. Harking back to the psychology of the Würzburg school, Popper began to stress more and more the similarity of the problem-solving phases of learning and of biological development, eventually (22, chapter 6, §xviii) condensed into the scheme

$$P_1 \rightarrow TS \rightarrow EE \rightarrow P_2$$

(problem $\rightarrow$ tentative solution $\rightarrow$ error elimination $\rightarrow$ new problem(s)).

In the 1961 lecture Popper also sought to make a contribution to evolutionary biology itself, by offering a hypothetical Darwinian explanation of orthogenesis; first by emphasizing (as Baldwin and Hardy had emphasized) that the preferences of an organism may, by remoulding its ecological niche, have a direct impact on evolutionary development (without, of course, having any impact on the genes)—this he called 'active Darwinism'—and secondly, more particularly, by proposing a model ('the spearhead theory') that makes it possible for a genotype to evolve in a seemingly goal-directed manner. The idea here is that each genotype incorporates more or less independently both preferences and ways of indulging those preferences; that the organism need not be imperilled if the former (the aim-structure) evolves without a corresponding evolution in the latter (the skill-structure); but that later mutations in the genetic basis of the skill-structure may enable it to catch up. Mutations that would almost certainly have been lethal may therefore turn out to be favourable if the ground (of aims and preferences) has already been prepared for them. A familiar example is the woodpecker's beak, whose successful establishment, the spearhead theory suggests, would have been the easier the more the bird had already cultivated a taste for the bugs living deep in the bark of trees. The spearhead theory has been examined in some detail, and applied to the body/mind problem, by Watkins (1995).

Part of the motive for a substantial international congress in the philosophy of science that Lakatos, with much help from others, organized at Bedford College (London) in July 1965, was to bring Popper, recently knighted, face to face with two of his most prominent opponents in the USA. One was Carnap, a friend and disputant since the time of *Logik der Forschung*, with whom he had been battling in the prints about inductive logic since the early 1950s; the other T.S. Kuhn, a historian of science who had attended Popper's seminars at Harvard in 1950 and had recently published a philosophical best-seller, *The structure of scientific revolutions* (1962). Although the tone of the public controversy between Popper and Carnap, and between their esquires, had often been bellicose, they liked and admired each other, and many considerate letters passed between them over the years. Popper wrote later in an obituary notice that Carnap's 'influence was based on his great constructive powers and on the unique charm of his personality' (20). The confrontation between them at Bedford College was strangely artificial, and on induction they were as far apart as ever. Hardly more

conciliatory was the session devoted to Kuhn's work. Kuhn repeated the celebrated contention (due originally to M. Polanyi 1958) that progress in science depends not, as Popper had always claimed, on imaginative new ideas subjected to harsh criticism, but on the suspension of criticism and its replacement by routine puzzle-solving under the auspices of what he called a paradigm; and suggested that the major developments, especially the revolutions, in the history of science, are understood better in socio-psychological than in rationalistic terms (1970). Popper's comments emphasized the oddity, when the understanding of the working of empirical science is at issue, of deferring to sociology and psychology, hardly the best developed of the empirical sciences. More significant for Popper at the congress were his Opening Address and a distressing dispute with Bartley, who had been one of his favourite pupils. This was not the first time, and not the last, that Popper and one of his followers were driven into painful and almost incomprehensible estrangement. In this instance the suspicion persists that the sharp tone of Bartley's paper, which Popper felt as an aggressive personal affront, was encouraged by a third party determined to make mischief. But in this instance, unlike others, the damage was eventually repaired. By 1980 Popper and Bartley were friends again, and Bartley undertook the considerable task of seeing *The postscript* through the press. The Opening Address, originally entitled 'Rationality and the search for invariants', returned to the question of metaphysical research programmes in physics, tracing the history and the dominance of the programme (which originated with the speculations of the pre-Socratic philosopher Parmenides) of explaining change by concentrating on what does not change, of explaining change by explaining it away. Yet another work whose publication has been too long delayed, the infinitely revised lecture will be published in a forthcoming volume (46) on pre-Socratic thought.

Popper's last major original philosophical venture, his 'three worlds philosophy', which he sometimes called 'the philosophy of my old age', started to take recognizable shape in 1965 in his Arthur Holly Compton Lecture at Washington University, St. Louis. Entitled 'Of clouds and clocks' (22, chapter 6), the lecture set out to contribute to our understanding of human freedom in a world that many regard as under the iron control of the laws of physics. Drawing again on material in *The postscript* (29)—Popper was an implacable cannibalizer—the lecture contemplated the nightmare of physical determinism, and firmly turned its back on it. Yet for genuine human freedom and creativity to exist, Popper stressed, it is not enough that our behaviour is not completely determined. To be free—though not, of course, completely—we must be able to control to some extent—though not, of course, completely—some aspects of the physical world. The physical world must be causally open to influences from outside, in particular psychological influences. These considerations simply reformulate the ancient problem of the interaction of body and mind ('Descartes's problem'). But Popper was alert to another problem, less often tackled, which he called 'Compton's problem', the problem of how abstract entities such as rules and decisions, theories and melodies, are able to bring about changes in the physical world. Materialists, who regard the world as essentially a world of matter (or, more generally, physics), formulate these problems as problems of reduction; the task, as they see it, is one of explaining how supposedly mental and abstract entities are to be construed as manifestations of matter alone. Popper had never cared for such views, or related ones such as behaviourism, if only because—though metaphysical in the highest degree—they appear to stem from a positivistic distaste for metaphysics rather than from any serious interest in the world. (For all that, of course, they may be interesting, and even true.) The Compton Lecture proposed that both problems may yield to the evolutionary hypothesis that the mind,

language and consciousness are components of a hierarchy of controls that has evolved (in such a way as had been proposed in (22), chapter 7) precisely because it allows the organism more freedom to pose and to evaluate competently solutions to the problems that it encounters. Popper reminded his audience that 50 years earlier his teacher Bühler had singled out disparate functions, hierarchically organized, of animal and human languages (to Bühler's expressive, signalling and descriptive functions Popper added others, especially the argumentative function). The controls are plastic, neither clockwork nor cloudage, resembling a soap bubble that both controls and is controlled by the air inside (in §ii the similarity to the mechanism of checks and balances in a democratic state was fleetingly noted). Popper insisted that the higher control units, the mind, language, consciousness, though evolved from and embedded in a world of pure physics, are not themselves purely physical, any more than the problems that organisms struggle with are purely physical.

From this it was a relatively short step to the picture of three distinct domains, the physical, the mental and the products of (human) mental activity, world 3 (22, chapter 3). World 3, Popper suggested, is created by minds (world 2), and acts on the physical world (world 1) only through the intervention of minds. But it is autonomous, in the sense that we may discover in it items that have not been consciously placed there; for example, new objective problems, and unseen consequences of previously invented theories. The principle of least action in classical mechanics, to take a simple example, was invented when Newton invented it, but discovered only when Maupertuis (or perhaps Euler) discovered it. The theory of three worlds has had many detractors, who have derided it as ontologically extravagant, mulish [= obstinate and barren], and even unphilosophical. Popper himself drew attention to some superficial similarities between world 3 and Plato's world of forms; but, as he noted, Plato's forms are eternal, unlike the human constructions that inhabit world 3. The similarity between world 3 and Hegel's Absolute Spirit, which he also mentioned, seems closer, though the evolution of world 3 is more of a hit-and-miss affair than the almost necessary advance that is supposed to characterize Absolute Spirit. As it happens, Popper's theory provides room for a much-needed synthesis—dialectical, if you must, but not Hegelian—of platonism and constructivism in the philosophy of mathematics. The autonomy of world 3 is most clearly revealed in the way that research continues to turn up new objective problems in its interior. The idea of an objective problem may sound contradictory, but Wächtershäuser (1995) has drawn attention to the important position that it has in German patent law. Illustrating his thesis with the example of Galileo's theory of the tides (22, chapter 4, §9), Popper himself suggested that a better understanding of important events in the history of science, in some respects resembling what is known as hermeneutics, can be reached if an effort is made to reconstruct the objective problem situation in which the scientist worked. Scientific theories cannot be understood unless we understand the objective problems to which they respond. It is important to appreciate that this thesis, which relativizes understanding to problems and to historical context, is quite compatible with the anti-justificationist thesis that the origin of a theory has no relevance to its truth value.

Popper made several more visits to the USA in the 1960s and 1970s, and attended and spoke at many international conferences. He continued to refine and to extend his ideas in many areas, especially the philosophy of physics. In 1965 he began work on what was planned as a monograph with the title *Exorcizing 'the observer' from modern physics*. He wrote the bulk of two long chapters on Einstein's two theories of relativity, and started a chapter on statistical mechanics. Quantum mechanics jumped the queue when he received an urgent call

for a paper (18) in that area, and work on the rest of the monograph was never resumed (some fragments were included in §§34–36 of (23)). Other papers of interest in this period are (19), mentioned earlier, and ‘The moral responsibility of the scientist’ (41, chapter 6), which contains, among other proposals, a ‘tentative restatement’ of the Hippocratic oath. The end of the decade was an unhappy time in university life, and even Popper could not remain entirely detached from such issues as the Vietnam war and student unrest. Nonetheless he went on working at Penn. He had had little influence within his own department for some years, partly by choice, partly because Lakatos was sweeping all power to himself. When his retirement and departure from the L.S.E. in 1969 passed almost unnoticed, an era quietly ended.

IN RETIREMENT, 1969–94

As with many others who work unremittingly hard, there had always been tasks that Popper could not bring himself to complete, or even to get under way. One of these now began to lour: the behemoth of the Popper volume in *The library of living philosophers*, a series that in the previous 30 years had produced some useful books, including the excellent Einstein volume (Schilpp 1949). The format of the books was standard: an intellectual autobiography from the victim, a substantial helping of critical and descriptive essays on his work, his replies to criticism, and an exhaustive bibliography. Popper had successfully completed the autobiography (later published separately as (23)) in 1968–70. Thirty three solicited papers had arrived, many of them by the May 1965 deadline. Hansen had compiled a richly detailed bibliography (1974). But Popper had for years been unable to write more than two or three replies and a cursory introduction to them. The threat from Schilpp in June 1971 that, unless he had a complete hand by 1 September, he would shelve the Popper volume indefinitely, roused Popper to concerted action. A small team of assistants and typists was assembled. It was agreed that Popper would draft the outstanding replies, and check the final versions, and that most of the intermediate work would be executed elsewhere. In about six weeks the replies were drafted, knocked into shape, supplied with references, checked and rechecked; at the same time the autobiography was brought editorially into line. (In Schilpp 1974, which amounts to over 1300 pages of printed text, the replies account for about 250 pages, the autobiography and bibliography for about 250 more.) The relief felt at Fallowfield at the end of August when Popper despatched a telegram to Schilpp announcing ‘666 pages mailed today’ was stupendous.

The reply that caused Popper the most difficulty—it was more than once rewritten—was the reply to Lakatos (1974). This paper by an ex-colleague, while speaking the language of admiration and benevolence, criticized Popper’s solutions of the problems of demarcation and induction with startlingly unoriginal and unsophisticated arguments. According to Lakatos, scientific theories are rarely falsifiable, for in the event of a failed prediction it is always permissible to blame not the theory under test but some auxiliary hypothesis used in the testing process. That *ad hoc* adjustment may rescue a falsified theory is a well-recognized point (going back at least to Duhem 1906, Part II, chapter VI, §3, and fully acknowledged in (1), §6), but it does not show that theories cannot be falsified; only that falsification often requires hard work, because ‘nature does not give an answer unless pressed for it’ (1, §85). Popper’s solution of the problem of induction was dismissed by Lakatos on the specious grounds that, in the absence of some metaphysical inductive principle connecting the past to the future, it reduces the application of science to the level of irrational dabbling. This is an

objection that only those who identify rationality with justification can make. (Few who do identify rationality with justification have managed to avoid making it.) Popper's point always was that we connect the past to the future by an empirical theory, and it is simply gratuitous to postulate a metaphysical connection as well. It is astonishing that someone who had been a colleague for a decade should have been unable to do better than this; much less astonishing that Popper was enraged to find the classic panorama of (1) being disfigured by such a philosophical folly. There was thereafter a permanent rift between Popper and Lakatos, who died shortly before Schilpp (1974) appeared.

One day in May 1971, Peter Krohn casually asked Medawar why Popper was not a Fellow of the Royal Society, and was told that the only defensible excuse was that no one had ever thought to nominate him. Medawar at once undertook this task, and began to enlist supporters. The election was eventually secured in June 1976, and reported in the press in the unfortunate words '[Popper] was considered exceptional in that his work had fundamentally altered the methods of deduction used by scientists' [405.9]. Noting that his sponsors had included 'some tough-minded experimental scientists', Medawar wrote to him 'I think you can take it that everybody who supported your name was a little bit prouder of being a Fellow of the Royal Society because you are now one of our number' [325.27]. Popper himself was hugely pleased with his Royal, feeling that his contribution to science had at last been recognized. Apparently he did not fully take in that he had not been elected as a fellow scientist, for some years later he telephoned Medawar, upset to see his name included in the list of Fellows elected under Statute 12 along with politicians and others not actively involved in science. Medawar's reply was a masterpiece of tact. Noting that Churchill (whom Popper very much admired, not only for his courage and for his political wisdom, but also for his philosophical good sense) had been elected under Statute 12, he reported the view of the Librarian that 'you could not be elected as a straightforward Sectional candidate and that the Society was so determined to have your name on their books that they decided there was nothing for it but Statute 12'. 'The matter is really too trivial for you to allow it to be a cause of concern' he soothed [325. 29]. But Popper did not see it in exactly that way.

In the early 1970s, in the aftermath of a Dutch television discussion in which they had participated in 1971, Popper and Eccles began a joint book on the body/mind problem, 'an attempt in interdisciplinary co-operation' (p. IX). The outcome, *The self and its brain* (25), is not genuinely a joint work; it contains five philosophical chapters and a summary by Popper, eight chapters by Eccles reporting on the current state of brain research and relating it to the problem of brain/mind interaction, and the edited transcripts of 12 dialogues recorded in September 1974 at the Villa Serbelloni on Lake Como, where the authors were guests of the Rockefeller Foundation. Some of what he nominates as 'the high points of the Dialogues' are mentioned in Eccles (1982). Dualists both, even pluralists, the authors acknowledged in their preface some serious divergences of opinion. In particular, they disagreed about religion. Although ethical issues had always constituted an important part of the background to Popper's philosophy (Kiesewetter 1995; Wächtershäuser 1995), in later life he discussed ethics rarely, and religious questions hardly at all. Indeed one of the aspects of Bartley (1962) that he most admired was the way that this book, which was a significant addition to the literature on critical rationalism, made the dilemmas of contemporary protestantism sound interesting. But he sympathized with the religious stance of others, and was not prepared to endorse various humanist and secular offensives: 'Although I am not for religion, I do think that we should show respect for anybody who believes honestly ...' [297.11].

Popper's contribution to (25) is in the main a development of the preliminary ideas of 'Of clouds and clocks', strengthened by a more explicit doctrine of how the human self, the self-conscious mind, is anchored in world 3; that is, of how personal identity is sustained and in part controlled by the descriptive and argumentative uses of language. As Adam Smith had recognized two centuries before, social life—of which the use of language is a critical part—is an essential condition for the development of full self-consciousness. The book contains a thorough criticism of materialism and physicalism and an enthralling potted history of the subject that, by tracing body/mind dualism back to *The Odyssey*, attempts to silence the modern doctrine that consciousness of the problem of self-consciousness began with Descartes. But Popper privately regarded his contribution to the book as unsatisfactory, and indeed not properly finished. When it was discovered in March 1977 that Hennie, who had been unwell for some time, had a malignant tumour, he stopped writing and contented himself with tidying and preparing for publication what he had already written. The next eight years were terrible years for them both. Every kind of treatment was considered, in several countries; sometimes the treatment was temporarily successful, sometimes of no apparent benefit at all. During the better spells Popper worked as unflinchingly as ever. Late in 1977 in Cambridge he gave the first Darwin lecture (26), in which he retracted the view—expressed in (23), §37—that the theory of natural selection is not a scientific theory, but only a metaphysical research programme for biology. The second edition of (27), published in 1979, contains some important new material, particularly on verisimilitude. His writings were often being anthologized, and his books were being translated into many languages. Popper almost never employed the services of a literary agent, preferring to undertake all the negotiation himself. This was certainly a way of keeping busy, but he gradually refined the basic contractual terms that he insisted on and, as one overseas publisher said grudgingly a few years ago, 'he knows what he wants'. The other great demand on his time was the increasing quantity of correspondence that he received. Educators and engineers, epidemiologists and ethnographers, all wanted to share their thoughts with him, and to be rewarded with his response. In the matter of replying to letters—at least to those that he opened and did not at once mislay—he was a model of generosity, though not always of gentleness, often writing in longhand many pages in reply to some quite elementary request.

The second and third volumes of the repackaged *Postscript* were published in the summer of 1982. In a new preface to the third volume (30) he proposed what he hoped was a genuinely realizable experiment that would test the correctness of the Copenhagen interpretation of quantum mechanics. This proposal received some critical attention in the scientific literature later (Collett & Loudon 1987; (35); see also Redhead 1995, pp. 167–170). At about the same time, shortly before his 80th birthday, he was appointed to the Order of Companions of Honour. He had received several other prestigious awards and prizes by this time, and the financial rewards that had accompanied some of them had enabled him to begin at last to satisfy his passion for collecting rare books. In 1937 he had managed to save some books from his father's magnificent library, but now he plunged into purchases in an almost reckless spirit. It was his only vice. At the time of his death his library contained first editions of Gilbert's *De magnete*, Kepler's *Harmonices mundi*, Galileo's *Dialogue concerning the two chief world systems*, Hobbes's *Leviathan*, Hume's *Treatise*, Rousseau's *Discours*, each of Kant's three *Critiques*, Lavoisier's *Mémoires de chimie* and Lyell's *Elements*; second editions of Newton's *Principia* and Boyle's *Sceptical chymist*; and some 200 other antiquarian works on scientific, philosophical and classical subjects, including a copy of the Aldine Aristophanes of 1498 (Sotheby's 1995).

In January 1982 George Soros, a Hungarian emigré who had attended Popper's lectures at the L.S.E. in the 1950s, established, with Popper's warm approval, the Open Society Fund, offering Karl Popper Bursaries to black students in South Africa. This was the first in a series of acknowledgements by Soros of the profound importance, especially for those not permitted to implement it, of the idea of an open society, culminating in the founding of the Central European University in 1991 in Budapest, Prague, and Warsaw, and the Open Society Prize in 1994. 'His work has been an inspiration to the Central European University' said Soros in his presentation speech to Popper, the first recipient of the prize (Soros *et al.* 1995, p. 34). *The open society* was widely studied in samizdat translations in eastern Europe in the 1970s and 1980s, and continues to be more deeply appreciated there (Havel 1995) than it is in better-established liberal democracies. 'The little revolution in my country in 1980–81 had its theoretician. It was you, Sir,' wrote a Polish student feelingly in 1982 [284.6]. Following the rapid and almost peaceful disintegration of the socialist tyrannies in 1989 and 1991, which he insisted that he had not foreseen, Popper took a close interest in the preparation of the Russian translation of (38) *The open society*, published by the Soros Foundation in Moscow in 1992. In his message to Russian readers, and repeatedly in conversations with friends, he spoke of the dangers of converting the Russian economy to a market system in advance of the setting up of a working and respected tradition of civil law (Simkin 1993, pp. 198f.; (45), chapter 4).

By the late summer of 1985 Hennie's condition had so much worsened that the Poppers decided to return for a while to Vienna, where several members of her family still lived. She spent some weeks in hospital, after which they rented a large house in the western suburbs of the city, close to what had once been the Emperor's hunting chase. Here in November she died. Popper, loving and romantic, but not sentimental, quickly came to terms with his loss, and renewed his manifold intellectual interests, above all in quantum physics, the theory of probability and evolutionary epistemology. For some months his plans for the future were undecided. There was much well-meaning local enthusiasm to keep 'the great Austrian philosopher' in his home country. In 1976 both he and Hennie had resumed Austrian nationality, though the motive for this had been Popper's firm expectation that he would die before his wife, who would then have been able to return home. But an attempt to capture him by establishing a new branch of the Boltzmann Institut [*für Wissenschaftstheorie* = for the philosophy of science] in the house in Kramer Glöckner Straße, with Popper as Director, did not prosper. In the summer of 1986 he returned to England for good.

For the next eight years he lived in Kenley (Surrey), in another house, like those in Penn and Vienna, that was perched on the edge of nowhere. (No doubt this was the most that he could concede to utopianism.) Fallowfield had been sold, and the new house bought on his behalf, by his secretary Melitta Mew and her husband Raymond, who were to be his principal support, and his unshakable friends, for the remainder of his life. Much more than a secretary, Mrs Mew (who had started working for him in 1982 in a part-time capacity at the L.S.E.) energetically but gently reorganized nearly every aspect of Popper's life and his household. Filing cabinets were installed, and order brought to his business affairs (which had been neglected for some years). A daily housekeeper (for whom he was always touchingly just 'Professor'—never 'Professor Popper' or 'Sir Karl') kept house. He lived in comfort, though hardly in luxury. His birthdays were celebrated with small lunch parties for his closest friends. He was very well looked after. Not always physically well, suffering many small tachycardias and minor strokes, as well as some more serious problems, he nonetheless maintained a noble independence, and an undiminished vitality of mind. 'I am having a very happy old age' he said

on many occasions. His friends shared in his well-being, and were time and time again affected by his warmth and by his exceptional generosity. Many visitors came to the house prepared for battle, and departed hours later disarmed, not by argument but by an utterly unsuspected charm. At the end of *Unended quest* Popper had described himself as 'one who feels that in struggling with ideas he has found more happiness than he could ever deserve'. In his last few years, perhaps for the first time, his whole life was a source of genuine happiness.

In 1986 Popper was appointed the first Medawar Lecturer at the Royal Society. He had a large audience, but the lecture, 'A new interpretation of Darwinism' (33), was sadly not one of his best. This is perhaps understandable given the turmoil of the previous year. In the lecture, which was not properly written, and was unfortunately never prepared for publication, he set himself a large agenda, which he was unable to cope with fully. Returning to the theme of active Darwinism, and identifying biological adaptation with the acquisition of knowledge, he suggested that the difficulties in the establishment of life (both naturally and artificially) may lie more in the failure of life-forms to be adapted to their environment than in their original creation. On the grounds that what characterizes life is problem-solving (as depicted in the scheme above, something that cannot be reduced, he claimed, to chemistry, he defended the use in biology of anthropomorphic and even teleological language, aversion to the latter being likened to the Victorian aversion to talking about sex. The constant theme, that problem-solving is the key to a general theory of life (a theory to be contrasted with that of Schrödinger 1967, that feeding on negative entropy is the characteristic feature; see (23), §30) was incorporated into the title of a book of essays (43) published shortly after his death. Some (for example, Gellner 1996, pp. 80, 84) have dismissed these ideas as a return to a romanticism scarcely distinguishable from nineteenth-century *Naturphilosophie*. A criticism of Popper's Medawar lecture is to be found in Perutz (1986).

A feature of Popper's intellectual life was that the discovery of a wealth of new problems hardly ever had much effect on his fascination with old ones; time and again he returned with relish to a topic of years past. A striking example is the return in (32) and its extended sequel (34) to the doctrine that the problem of induction can be solved by means of considerations of probability. The renewed attack on this doctrine began with the discovery in 1981, while (31) was being prepared for publication, of an unnoticed minor error in Part II, §15; and grew into the claim that, even if the probability of a scientific theory h can be increased by empirical evidence e , that increase is a purely deductive effect, depending crucially on the extent to which the contents of h and e overlap. Indeed it was proved that the ampliative part $h \leftarrow e$ of a hypothesis h (that part of h that goes beyond what the evidence e says) is never positively supported by e ; in brief, $p(h \leftarrow e | e) \leq p(h \leftarrow e)$ for every h and every e . The conclusion drawn was that, even if there is such a thing as probabilistic support, there is nothing inductive about it: *there is no inductive logic*. This was a deeply satisfying result for someone who had spent half a century fighting this most inexplicably adhesive of philosophical prejudices.

But if there was any one intellectual problem that dominated Popper's thought and captured his imagination in his last decade it was not the problem of induction, nor any other philosophical problem, but the problem of the origin of life, mentioned briefly in the Medawar Lecture and, Popper tells us in (23), §5, a problem that began to bother him at the age of 12 or 13. From the outset he was deeply involved in the development of Wächtershäuser's theory of the 'iron-sulphur world' (1988), according to which life began with acellular, effectively two-dimensional, organisms bonded to positively charged surfaces such as pyrite. Wächtershäuser, a chemist who had become a patent attorney, had met Popper in 1982 in Alpbach, a meeting

that, he says, 'drew him back to science'. For years they spent many hours weekly on the telephone discussing every aspect, substantial and methodological, of Wächtershäuser's theory; and it was with tremendous enthusiasm and pride that Popper would keep his friends up to date about the progress of the theory, his intellectual godson, especially when there were new experimental results to report (37). His contribution to the success of the theory is not to be underestimated (Wächtershäuser 1995, 1997). It was an exemplary instance of his unquenchable curiosity, and of his utter devotion to the search for truth.

As Popper became more celebrated, more sought after, more honoured, he turned once more to face the outside world. Deeply anguished as a youth, and disappointed in middle age, but for years not actively political, in his old age he began to speak out on issues of public concern, from the enormity of the Russian nuclear arsenal to the evil effects of violence on television (45, chapters 3 and 7). On many occasions he held forth on the vanity and the dangers of nationalism (44). He spoke with increasing frequency of the moral duty of intellectuals to be optimistic but not complacent, to adopt the modest critical rationalist motto 'I may be wrong and you may be right, and by an effort, we may get nearer to the truth' (4, chapter 24, §i), and to open their eyes to the beauties of the world. He disliked intensely the anti-scientific ideology that he saw in the ecological movement, and seemed confident that, although science had admittedly been responsible for much damage, it alone had the power to correct these mistakes (45, chapter 9). The population explosion he viewed with the greatest concern. Earnest and moralistic he may often have been, but he retained his impish humour. On the way to Barcelona in 1989 to receive the Catalunya Prize, for example, he disconcerted his companions by insisting that, since one of the conditions of the prize was that it could not be awarded posthumously, he must buy instant travel insurance for the equivalent of the prize money. Lecturing on Feynman's suggestion that a positron is an electron travelling backwards in time, he expressed the doubt that his bank manager would willingly regard his mounting overdraft as a credit balance moving backwards in time. He had a good eye for the incongruous, for example the way in which so many papers published in the *Proceedings of the National Academy of Sciences of the USA* are labelled as 'advertisements'. When he gave a lecture to the L.S.E. Alumni Association in 1989, he solemnly announced that, since this was his first public lecture at the School, he regarded it as a belated inaugural lecture; 'It is an occasion to which I have been looking forward eagerly for the last 40 years' he declared (36, p. 29).

Working to the very end of his life, he was preparing, among other things, a short note (to appear in (46)) on the proof given by Aristotle (*Metaphysics* 1051a 28f.) that the angle in a semicircle is a right angle; and sharing the writing of a short response to an article critical of the main thesis of (34). The end came quickly. On Saturday 10 September 1994 he was taken into hospital for what was plainly to be a serious, even dangerous, operation. This went well, but it revealed unexpectedly grave problems. He was thereafter in great pain for much of the time, though characteristically able to make his presence felt throughout the ward. For a few days he was as conscious and alert as ever, and when not in pain peacefully cheerful. He died early in the morning of 17 September. His body was cremated, and some weeks later his ashes were laid, as he had requested, in his wife's grave in the Lainzer Friedhof, a small cemetery in Vienna. In 1992 Karl Popper had been made an honorary citizen of the City of Vienna, and the grave is to be maintained in perpetuity by this city of innumerable illustrious names.

PROFESSIONAL CAREER

- 1937–40 Lecturer in Philosophy, Canterbury College, University of New Zealand.
1941–46 Senior Lecturer in Philosophy, Canterbury College, University of New Zealand.
1945–48 Reader in Logic and Scientific Method, London School of Economics.
1949–69 Professor of Logic and Scientific Method, University of London.
1951–53 Chairman, Philosophy of Science Group of the British Society for the History of Science (now the British Society for the Philosophy of Science).
1958–59 President, Aristotelian Society.
1959–61 President, British Society for the Philosophy of Science.
1969–94 Emeritus Professor of Logic and Scientific Method, University of London.
1986–94 Senior Research Fellow, Hoover Institution on War, Revolution and Peace (Stanford).
1986–94 Visiting Professor, University of Vienna.

HONOURS, MEDALS AND PRIZES

- 1965 Knight Bachelor.
 Prize of the City of Vienna, Geisteswissenschaften.
1973 Sonning Prize of the University of Copenhagen.
1976 Lippincott Award of the American Political Science Association.
 Grand Decoration of Honour in Gold (Austria).
1978 Dr Karl Renner Stiftung Prize of the City of Vienna.
1979 Gold Medal for Distinguished Service to Science of the American Museum of Natural History (New York).
1980 Ehrenzeichen für Wissenschaft und Kunst (Austria).
 Orden Pour le Mérite für Wissenschaften und Künste (Germany).
1981 Dr Leopold Lucas Prize of the University of Tübingen.
1982 Companion of Honour.
1983 Grosses Verdienstkreuz mit Stern und Schulterband (Germany).
 Ehrenring of the City of Vienna.
1984 Prix Alexis de Tocqueville (Valognes).
1986 Wissenschaftsmedaille der Stadt Linz (Austria).
1988 Xth International Prize 'Friedrich Nietzsche' (Palermo).
1989 Premi Internacional Catalunya (Barcelona).
1992 Goethe Medaille (Goethe Institut, Munich).
 Kyoto Prize (Japan).
 Ehrenbürger (Honorary citizen) der Stadt Wien.
1993 Otto-Hahn-Friedensmedaille (Berlin).
1994 Open Society Prize (CEU, Prague).

MEMBERSHIPS AND HONORARY MEMBERSHIPS

- 1949 Membre de l'Académie Internationale de Philosophie des Sciences.
 1958 Fellow of The British Academy.
 1964 Honorary Member, Harvard Chapter of Phi Beta Kappa.
 1965 Honorary Member, The Royal Society of New Zealand.
 1966 Foreign Honorary Member, American Academy of Arts and Sciences.
 1972 Honorary Fellow, London School of Economics.
 1974 Correspondant de l'Institut de France.
 1976 Fellow of The Royal Society of London.
 Associate Member, l'Académie Royale de Belgique.
 1977 Honorary Member, l'Académie Internationale d'Histoire des Sciences (Paris).
 1979 Honorary Member, Deutsche Akademie für Sprache und Dichtung.
 Honorary Member, Allgemeine Gesellschaft für Philosophie in Deutschland.
 1980 Membre de l'Académie Européenne des Sciences, des Arts et des Lettres (Paris)
 (Delegation of Great Britain).
 Membre de l'Institut de France.
 Honorary Fellow, Darwin College, Cambridge.
 1981 Socio Straniero dell'Accademia Nazionale dei Lincei.
 1982 Ehrenmitglied, Österreichische Akademie der Wissenschaften.
 Honorary Research Fellow, Department of History and Philosophy of Science,
 King's College, London.
 1986 Foreign Associate, National Academy of Sciences (Washington).
 Ehrenmitglied, Gesellschaft der Ärzte Wien.
 1991 Emeritus Member Academia Europaea (London).
 Ehrenmitglied, Academia Scientiarum et Artium Europaea (Salzburg).
 1992 Ehrenmitglied, Hochschule für Musik und Darstellende Kunst (Vienna).
 1994 Honorary Member, Scientific Council, 3rd Medical Faculty, Charles University
 (Prague).

HONORARY DEGREES

Hon. LL.D., Chicago 1962; Hon. LL.D., Denver 1966; Hon. Lit.D., Warwick 1971; Hon. Lit.D., Canterbury (NZ) 1973; Hon. D.Litt., Salford 1976; Hon. D.Litt., The City University 1976; Dr.rer.nat.h.c., Vienna 1978; Renewal of the degree of Doctor of Philosophy by the Faculty of Fundamental Sciences, Vienna 1928/1978; Dr.phil.h.c., Mannheim 1978; Hon. D.Litt., Guelph 1978; Dr.rer.pol.h.c., Frankfurt am Main 1979; Dr.phil.h.c., Salzburg 1979; Hon. Litt.D., Cambridge 1980; Hon. D.Sc., Gustavus Adolphus College 1981; Hon. D.Litt., Oxford 1982; Hon. D.Sc., London 1986; Dr.phil.h.c., Catholic University of Eichstätt 1991; Dr.phil.h.c., Madrid 1991; Dr.phil.h.c., Athens 1993; Dr.med.sc.h.c., Charles University 1994.

ENDOWED LECTURESHIPS

- 1950 William James Lectures, Harvard University.
Woodward Lecture, Yale University.
- 1956 Eleanor Rathbone Memorial Lecture, University of Bristol.
- 1960 Annual Philosophical Lecture, Henrietta Hertz Trust, British Academy.
- 1961 Herbert Spencer Lecture, University of Oxford.
Shearman Memorial Lectures, University College, London.
- 1963 Farnum Lecture, Princeton University.
- 1965 Arthur Holly Compton Lecture, Washington University, St. Louis
- 1969 Kenan Lectures, Emory University.
- 1971 James Scott Lecture, Royal Society of Edinburgh.
- 1972 Romanes Lecture, University of Oxford.
- 1973 Henry Dan Broadhead Memorial Lecture, University of Canterbury, Christchurch.
Herbert Spencer Lecture, University of Oxford.
- 1977 Darwin Lecture, Darwin College, Cambridge.
- 1978 Obert C. Tanner Lecture, University of Michigan.
- 1979 Frank Nelson Doubleday Lecture, Smithsonian Institution.
- 1981 J.B. Morrell Memorial Lecture, University of York.
- 1986 Medawar Lecture, Royal Society of London.
- 1989 Vera and Walter Morris Foundation Philosophy of Science Lecture, University of Arkansas for Medical Sciences, Little Rock.

CONFERENCES DEDICATED TO POPPER'S PHILOSOPHY

- 1968 Simposio de filosofía de la ciencia: homenaje a Karl R. Popper, Burgos.
- 1969 University centennial symposium in honor of Sir Karl Popper, Boston.
- 1973 Methodologies: Bayesian and Popperian, Columbia SC.
- 1980 Symposium on the philosophy of Sir Karl Popper, London.
Symposium on the philosophy of Sir Karl Popper, Beijing.
- 1981 Karl Popper et la science d'aujourd'hui, Cerisy-la-Salle.
Karl Popper: science et philosophie, Strasbourg.
Popper e la società aperta, Assisi.
- 1982 The open society and its friends, New York
- 1983 'Individuale-collettivo', Club Turati, Turin.
Wiener Popper-Symposium, Vienna.
- 1984 V Jornades de filosofia. La filosofía de K. Popper, Barcelona.
One-day conference on the Philosophy of Karl Popper, University of Manchester
(repeated annually in Manchester, Coventry or London)
Symposium on the philosophy of Sir Karl Popper, Madrid.
- 1985 Kritischer Rationalismus Heute, Arnoldsheim.
- 1986 Popper and political thought, Edinburgh.
- 1987 Conference on the philosophy of Karl Popper, Wuhan.
- 1994 L'epistemologia de Karl Popper, Cesena.
Teoría y método de las ciencias sociales, Mexico City.

- 1995 Karl Popper 1902–1994, Warsaw.
Conference to celebrate the fiftieth anniversary of the publication of *The open society and its enemies*, Prague.

ACKNOWLEDGEMENTS

I should like to record my indebtedness to those who have assisted in the preparation of this Memoir. I thank especially Mr & Mrs A.R. Mew, the executors of the estate of Karl Popper, for much information and much encouragement; and also for permission to quote from unpublished material in the Hoover Institution Archives, and to reproduce a simplified version of Popper's list of publications. Thanks are due also to Sir Hermann Bondi, F.R.S., Dr A. Chmielewski, Sir Ernst Gombrich, Lady Medawar, Dr J. Musafia, McMaster University and the estate of the 3rd Earl Russell, F.R.S., Harvard University and Dr David C. Williams, and the University of Canterbury, for permission to quote from unpublished and informal writings; Professor M.H. Hacothen for generously sharing with me the results of his research into Popper's years in Vienna; Sir Brian Follett, F.R.S.; the Archive Department of the British Library for Political & Economic Science and the University Library at Klagenfurt, for allowing free access to their copies of the Popper Archive; the Library of the Royal Society; Mr J. Field of the public relations office of the University of Canterbury; and Professor J. Agassi, Mr O.A.S. Bueno, Professor S.G. Houlgate, Professor I.C. Jarvie, Mme M.-I. Brudny-de Launay, Mr B. Magee, Dozent K.M. Milford, Dr A.F. Petersen, Dr J.F.G. Shearmur, Professor C.G.F. Simkin, Professor Dr G. Wächtershäuser, and Professor J.W.N. Watkins for information and valuable criticism.

The frontispiece photograph by Herr Michael von Haugwitz was taken in Weimar in March 1992, and is reproduced with permission.

REFERENCES TO OTHER AUTHORS

- Agassi, J. 1993 *A philosopher's apprentice: in Karl Popper's workshop*. Amsterdam & Atlanta: Rodopi.
- Ballentine, L.E. 1970 The statistical interpretation of quantum mechanics. *Rev. Mod. Phys.* **42**, 358–381.
- Bar-Hillel, Y. & Carnap, R. 1954 Semantic information. *Br. J. Phil. Sci.* **4**, 145–157.
- Bartley, W.W., III 1962 *The retreat to commitment*. London: Chatto & Windus.
- Bartley, W.W., III. 1982 A Popperian harvest. In *In pursuit of truth. Essays in honour of Karl Popper's 80th birthday* (ed. P. Levinson), pp. 249–289. Atlantic Highlands: Humanities.
- Beaglehole, J.C. *et al.* 1952 Three obituary notices for I.L.G. Sutherland. *J. Polynesian Soc.* **61**, 120–129.
- Berkeley, G. 1721 *De motu*. London: Tonson. English translation contained in *De motu and The analyst* (ed. D.M. Jesseph), pp. 73–107. Dordrecht: Kluwer.
- Bernard, C. 1865 *Introduction à l'étude de la médecine expérimentale*. Paris: Baillière.
- Bondi, H. 1992 The philosopher for science. *Nature* **358**, 363.
- Bondi, H. 1995 Sir Karl Popper's impact on the scientific community. In *Tributes paid at the Memorial Celebration for Sir Karl Popper, 12 December 1994*, pp. 5–7. London School of Economics.
- Collett, M.J. & Loudon, R. 1987 Analysis of a proposed crucial test of quantum mechanics. *Nature* **326**, 671f.; **328**, 675f.
- Crawshay-Williams, R. 1970 *Russell remembered*. London, New York & Toronto: Oxford University Press.
- Dahrendorf, R. 1995 *LSE: a history of the London School of Economics and Political Science, 1895–1995*. Oxford University Press.
- Delahaye, J.-P. 1994. *Information, complexité et hasard*. Paris: Hermès.
- Duhem, P.M.M. 1906 *La Théorie Physique, son objet, sa structure*. Paris: Chevalier et Rivière. English translation: *The aim and structure of physical theory*. Princeton University Press, 1954.
- Eccles, J.C. 1982 My living dialogue with Popper. In *In pursuit of truth. Essays in honour of Karl Popper's 80th birthday* (ed. P. Levinson), pp. 221–236. Atlantic Highlands: Humanities.

- Gardner, W.J., Beardsley, E.T. & Carter, T.E. 1973 *A history of the University of Canterbury, 1873–1973*. Christchurch: University of Canterbury.
- Geier, M. 1994 *Karl Popper*. Hamburg: Rowohlt.
- Gellner, E.A. 1959 *Words and things*. London: Gollancz.
- Gellner, E.A. 1996 Karl Popper—the thinker and the man. In *The significance of Popper's thought*. Proceedings of the conference Karl Popper 1902–1994, 10–12 March 1995, Graduate School of Social Research, Warsaw (ed. S. Amsterdamski), pp. 75–85. Amsterdam & Atlanta: Rodopi.
- Godsiff, P. 1972 Sir Karl Popper in New Zealand. *The NZ Listener*, 24 January 1972, 12.
- Gombrich, E.H.J. 1995 *The open society and its enemies*: remembering its publication fifty years ago. London: LSE Centre for the Philosophy of the Natural and Social Sciences.
- Grattan-Guinness, I. 1992 Russell and Karl Popper: their personal contacts. *Russell* n.s. **12**, 3–18.
- Hacohen, M.H. 1996 Karl Popper in exile: the Viennese progressive imagination and the making of *The open society*. *Phil. Soc. Sci.* **26**, 452–492.
- Hadamard, J. 1898 Les surfaces à courbures opposées et leur lignes géodésiques. *J. Math. Pures Appl.* (5), **iv**, 27–73.
- Hansen, T.E. 1974 Bibliography of the writings of Karl Popper [to 1973]. In *The philosophy of Karl Popper* (ed. P.A. Schilpp), pp. 1201–1287. La Salle: Open Court.
- Havel, V. 1995 Karl Popper's *The open society and its enemies in the contemporary global world*. The Chancellor's Lecture for 1995 at the Victoria University of Wellington. *Stout Centre Rev.* **5**, 4–6.
- Jammer, M. 1974 *The philosophy of quantum mechanics*. New York: Wiley.
- Jennings, H.S. 1906 *The behaviour of the lower organisms*. New York: Columbia University Press.
- Kiesewetter, H. 1995 Ethical foundations of Popper's philosophy. In *Karl Popper: philosophy and problems* (ed. A. O'Hear), pp. 275–288. Cambridge University Press.
- Kraft, J. 1957 *Von Husserl zu Heidegger. Kritik der phänomenologischen Philosophie*. 2nd edn. Frankfurt a. Main: Verl. Öffentlich. Leben.
- Kuhn, T.S. 1962 *The structure of scientific revolutions*. University of Chicago Press.
- Kuhn, T.S. 1970 Logic of discovery or psychology of research? In *Criticism and the growth of knowledge* (eds I. Lakatos & A.E. Musgrave), pp. 1–23. Cambridge University Press.
- Lakatos, I. 1974 Popper on demarcation and induction. In *The philosophy of Karl Popper* (ed. P.A. Schilpp), pp. 241–273. La Salle: Open Court.
- Landé, A. 1955 *Foundations of quantum theory*. New Haven: Yale University Press.
- Leblanc, H. 1989 Popper's formal contributions to probability theory. In *Perspectives on psychologism* (ed. M.A. Notturmo), pp. 341–367. Leiden, New York & Köln: Brill.
- Liebig, J. von 1865 *Induktion und Deduktion*. Munich: Königl. Akademie.
- Magee, B. 1997 *Confessions of a philosopher*. London: Weidenfeld & Nicolson.
- Medawar, P.B. 1967 Hypothesis and imagination. In *The art of the soluble*. London: Methuen.
- Medawar, P.B. 1986 *Memoirs of a thinking radish: an autobiography*. Oxford University Press.
- Menger, K. 1974 Postscript to the English edition of *Morality, decision and social organization* (1934). Dordrecht: Reidel.
- Neumann, J. von 1931 *Mathematische Grundlagen der Quantenmechanik*. Berlin: Springer. English translation: *Mathematical foundations of quantum mechanics*. Princeton University Press, 1955.
- Pauli, W. 1954 Wahrscheinlichkeit und Physik. *Dialectica* **8**, 112–124.
- Perutz, M.F. 1986 A new view of Darwinism. *New Scientist*, 2 October 1986, 36–38.
- Polanyi, M. 1958 *Personal knowledge. Towards a post-critical philosophy*. London & Henley: Routledge & Kegan Paul.
- Redhead M.L.G. 1995 Popper and the quantum theory. In *Karl Popper: philosophy and problems* (ed. A. O'Hear), pp. 163–176. Cambridge University Press.
- Rényi, A. 1955 On a new axiomatic theory of probability. *Acta Math. Hungar.* **6**, 286–335.
- Russell, B.A.W. 1948 *Human knowledge. Its scope and limits*. London: Allen & Unwin.
- Schilpp, P.A. (ed.) 1949 *Albert Einstein: philosopher-scientist*. New York: Tudor.
- Schilpp, P.A. (ed.) 1974 *The philosophy of Karl Popper*. La Salle: Open Court.
- Schrödinger, E. 1967 *What is life? and Mind and matter*. Cambridge University Press.

- Schroeder-Heister, P. 1984 Popper's theory of deductive inference and the concept of a logical constant. *Hist. Phil. Logic* **5**, 79–110.
- Shearmur, J.F.G. 1996 *The political thought of Karl Popper*. London: Routledge.
- Simkin, C.G.F. 1993 *Popper's views on natural and social science*. Leiden, New York, & Köln: Brill.
- Soros, G. et al. 1995 Sir Karl Popper receiving the Open Society Prize. In *Karl Popper in Prague: In Memoriam*, pp. 32–37. Prague: 3rd Medical Faculty, Charles University, and Central European University Foundation.
- Sotheby's 1995 *The Library of Sir Karl Popper* (catalogue for a planned auction on 19 May 1995). London: Sotheby's.
- Tarski, A. 1936 O pojęciu wynikania logicznego. *Przegląd Filozoficzny* **39**, 58–68. English translation: On the concept of logical consequence. In A. Tarski, *Logic, semantics, metamathematics*, pp. 409–420. Oxford: Clarendon Press, 1956.
- Ville, J. 1939 *Étude critique de la notion de collectif*. Paris: Gauthier-Villars.
- Wächtershäuser, G. 1988 Before enzymes and templates: theory of surface metabolism. *Microbiol. Rev.* **52**, 452–484.
- Wächtershäuser, G. 1995 Karl Popper, mentor of science—a personal view. In *Tributes paid at the Memorial Celebration for Sir Karl Popper, 12 December 1994*, pp. 21–25. London School of Economics.
- Wächtershäuser, G. 1997 The origin of life and its methodological challenge. *J. theor. Biol.* **187**, 483–494.
- Watkins, J.W.N. 1995 Popper and Darwinism. In *Karl Popper: philosophy and problems* (ed. A. O'Hear), pp. 191–206. Cambridge University Press.
- Whewell, W. 1840 *Philosophy of the inductive sciences*. London: Parker.
- Whewell, W. 1860 *On the philosophy of discovery*. London: Parker.

BIBLIOGRAPHY

The following publications are those referred to directly in the text. A full bibliography appears on the accompanying microfiche, numbered as in the second column. A photocopy is available from the Royal Society Library at cost. Where possible, reference has been made to reprints in easily available collections, rather than to original publications, and to chapter and section numbers, rather than to page numbers.

- | | | | |
|------|-------|------|---|
| (1) | (7) | 1934 | <i>Logik der Forschung</i> . Vienna: Julius Springer. |
| (2) | (9) | 1938 | A set of independent axioms for probability. <i>Mind</i> 47 , 275–277. (See also pp. 415 and 552 for corrections.) |
| (3) | (18) | 1945 | <i>The open society and its enemies</i> , vol. I (<i>The spell of Plato</i>). London: George Routledge. |
| (4) | (19) | | <i>The open society and its enemies</i> , vol. II (<i>The high tide of prophecy: Hegel, Marx, and the aftermath</i>). London: George Routledge. |
| (5) | (21) | | (With others) <i>Research and the University</i> . Christchurch, NZ: Caxton Press. |
| (6) | (30) | 1947 | Logic without assumptions. <i>Proc. Aristotelian Soc.</i> 47 , 251–292. |
| (7) | (47) | 1950 | Indeterminism in quantum physics and in classical physics I. <i>Br. J. Phil. Sci.</i> 1 , 117–133. |
| (8) | (48) | | Indeterminism in quantum physics and in classical physics II. <i>Br. J. Phil. Sci.</i> 1 , 173–195. |
| (9) | (50) | 1952 | <i>The open society and its enemies</i> . 2nd edn. London: Routledge & Kegan Paul. |
| (10) | (62) | 1954 | Degree of confirmation. <i>Br. J. Phil. Sci.</i> 5 , 143–149. |
| (11) | (67) | 1955 | Two autonomous axiom systems for the calculus of probabilities. <i>Br. J. Phil. Sci.</i> 6 , 51–57. |
| (12) | (75) | 1956 | The arrow of time. <i>Nature</i> 177 , 538. |
| (13) | (89) | 1957 | <i>The poverty of historicism</i> . London: Routledge & Kegan Paul. |
| (14) | (104) | 1959 | <i>The logic of scientific discovery</i> . London: Hutchinson. |
| (15) | (138) | 1962 | Julius Kraft 1898–1960. <i>Ratio</i> (Oxford) 4 , 2–12. |

- (16) (153) 1963 *Conjectures and refutations: the growth of scientific knowledge*. London: Routledge & Kegan Paul.
- (17) (213) 1967 Zum Thema Freiheit. In *Die Philosophie und die Wissenschaften: Simon Moser zum 65. Geburtstag* (ed. E. Oldemeyer), pp. 1–12. Meisenheim/Glan: Anton Hain.
- (18) (219) Quantum mechanics without 'the observer'. In *Quantum theory and reality* (ed. M. Bunge), pp. 7–44. Berlin, Heidelberg & New York: Springer.
- (19) (248) 1968 Birkhoff and von Neumann's interpretation of quantum mechanics. *Nature* **219**, 682–685.
- (20) (325) 1970 Unsigned obituary notice for Professor Rudolf Carnap. *The Times* 57973, 17 September 1970, 12.
- (21) (338) 1971 The philosophy of Russell: II—discussion among Karl Popper, Peter Strawson and Geoffrey Warnock. In *Modern British philosophy* (ed. B. Magee), pp. 131–149. London: Secker & Warburg.
- (22) (355) 1972 *Objective knowledge: an evolutionary approach*. Oxford: Clarendon Press.
- (23) (485) 1976 *Unended quest: an intellectual autobiography*. London: Fontana.
- (24) (486) A note on verisimilitude. *Br. J. Phil. Sci.* **27**, 147–159.
- (25) (536) 1977 (With J.C. Eccles) *The self and its brain. An argument for interactionism*. Berlin, Heidelberg, London & New York: Springer International.
- (26) (571) 1978 Natural selection and the emergence of mind. *Dialectica* **32**, 339–355.
- (27) (582) 1979 *Objective knowledge: an evolutionary approach*, 2nd edn. Oxford: Clarendon Press.
- (28) (609) *Die beiden Grundprobleme der Erkenntnistheorie* (ed. T.E. Hansen). Tübingen: Mohr (Paul Siebeck).
- (29) (715) 1982 *The open universe: an argument for indeterminism* (ed. W.W. Bartley III). London: Hutchinson.
- (30) (716) *Quantum theory and the schism in physics* (ed. W.W. Bartley III). London: Hutchinson.
- (31) (745) 1983 *Realism and the aim of science* (ed. W.W. Bartley III). London: Hutchinson.
- (32) (749) (With D.W. Miller) A proof of the impossibility of inductive probability. *Nature* **302**, 687f.
- (33) (872) 1986 A new interpretation of Darwinism (the first Medawar Lecture, given 12 June 1986). (Unpublished.)
- (34) (875) 1987 (With D.W. Miller) Why probabilistic support is not inductive. *Phil. Trans. R. Soc. Lond. A* **321**, 569–596.
- (35) (892) Popper versus Copenhagen. *Nature* **328**, 675.
- (36) (988) 1990 *A world of propensities*. Bristol: Thoemmes.
- (37) (989) Pyrite and the origin of life. *Nature* **344**, 387.
- (38) (1096) 1992 *Otkrytoie obschestwo i iego wragi*. Moscow: Soros Foundation.
- (39) (1099) Contribution to programme note, concert #44 (5 August 1992). Paisley International Organ Festival.
- (40) (1141) 1994 *Logik der Forschung*, 10th edn. Tübingen: Mohr (Paul Siebeck).
- (41) (1154) *The myth of the framework. In defence of science and rationality* (ed. M.A. Notturmo). London: Routledge.
- (42) (1160) (With D.W. Miller) Contributions to the formal theory of probability. In *Patrick Suppes: scientific philosopher*, vol. I (*Probability and probabilistic causality*) (ed. P. Humphreys), pp. 3–23. Dordrecht: Kluwer.
- (43) (1161) *Alles Leben ist Problemlösen: über Erkenntnis, Geschichte und Politik*. Munich: Piper.
- (44) (1174) The Prague Lecture. In *Karl Popper in Prague: In Memoriam*, pp. 12–15. Prague: 3rd Medical Faculty, Charles University, and Central European University Foundation.
- (45) (1237) 1996 *The lesson of this century. Karl Popper interviewed by Giancarlo Bosetti*. London: Routledge.
- (46) (1264) 1998 *The world of Parmenides. Essays on the pre-Socratic enlightenment* (ed. A.F. Petersen). London: Routledge. (In preparation.)



Biographical Memoirs of Fellows of the Royal Society

Please Note: The following pages for this article were scanned from microfiche. Although the original microfiche images were not of highest quality, JSTOR has made a best effort to present the material in legible condition. Please contact jstor-info@umich.edu if you have any questions.

**Please click on "Next Page" (at the top of the screen)
to begin viewing this article.**

Karl Raimund Popper

Bibliography

SIR KARL RAIMUND POPPER C.H., F.R.S.

LIST OF PUBLICATIONS

- 1925(a)** Über die Stellung des Lehrers zu Schule und Schüler. Gesellschaftliche oder individualistische Erziehung?. *Schulreform* (Vienna) **4**, No. 4, April 1925, pp. 204-208.
- 1927(a)** Zur Philosophie des Heimatgedankens. *Die Quelle* (Vienna) **77**, No. 10, October 1927, pp. 899-908.
- 1931(a)** Die Gedächtnispflege unter dem Gesichtspunkt der Selbsttätigkeit. *Die Quelle* (Vienna) **81**, No. 6, June 1931, pp. 607-619.
- 1932(a)** Pädagogische Zeitschriftenschau. *Die Quelle* (Vienna) **82**, 3, March 1932, pp. 301-303; *ibid.*, 7, July 1932, pp. 580-582; *ibid.*, 8, August 1932, pp. 646f.; *ibid.*, 9, September 1932, pp. 712f.; *ibid.*, 10, October 1932, pp. 778-781; *ibid.*, 11, November 1932, pp. 846-849; and *ibid.*, 12, December 1932, pp. 930f.
- 1933(a)** Ein Kriterium des empirischen Charakters theoretischer Systeme. *Erkenntnis* **3**, Nos 4/6, 1932-1933, pp. 426f. English translation contained in **1959(a)**; German reprint in **1966(e)**.
- 1934(a)** Zur Kritik der Ungenauigkeitsrelationen. *Die Naturwissenschaften* **22**, No. 48, 30th November 1934, pp. 807f.
- 1934(b)** *Logik der Forschung*. Julius Springer Verlag, Vienna (with the imprint '1935'). *Schriften zur wissenschaftlichen Weltauffassung*, edited by Philipp Frank & Moritz Schlick, Volume 9.
- Further impressions, editions, and translations of **1934(b)** into other languages are indicated in the year in which they are published.
- 1935(a)** 'Induktionslogik' und 'Hypothesenwahrscheinlichkeit'. *Erkenntnis* **5**, Nos 2/3, 1935, pp. 170-172. English translation contained in **1959(a)**; German reprint in **1966(e)**.
- 1938(a)** A Set of Independent Axioms for Probability. *Mind* **47**, N.S., No. 186, April 1938, pp. 275-277. Corrections: *ibid.*, No. 187, July 1938, p. 415 and No. 188, October 1938, p. 552. Contained in part in **1959(a)**.
- 1938(b)** Isaac Newton. A Notable Biography. *The Press* (Christchurch, New Zealand), 15th October 1938, p. 20. Review of *Isaac Newton 1642-1727* by J.W.N. Sullivan.
- 1940(a)** What is Dialectic?. *Mind* **49**, N.S., No. 196, October 1940, pp. 403-426. *Errata*: To the Editor of *Mind*, in *Mind* **50**, July 1941, No. 199, pp. 311f.

Contained in **1963(a)**. German translations: **1965(b)**, **1965(n)**, **1966(n)**, and **1966(o)**; Italian: **1981(z₃)**; Chinese (unauthorized): **1981(z₃₂)**; Greek: contained in **1990(z₄)**; Russian: **1995(t)**.

- 1940(b)** Interpretation of Nebular Red-Shifts. *Nature* **145**, No. 3663, 13th January 1940, pp. 69f. *Errata: ibid.*, No. 3679, 4th May 1940, p. 701.
- 1943(a)** Are Contradictions Embracing?. *Mind* **52**, N.S., No. 205, January 1943, pp. 47-50.
- 1944(a)** The Poverty of Historicism I (Parts I and II). *Economica* **11**, N.S., No. 42, May 1944, pp. 86-103.
- 1944(b)** The Poverty of Historicism II (Part III). *Economica* **11**, N.S., No. 43, August 1944, pp. 119-137.
- 1944(c)** Tankyū Ronrigaku Zyosetu. In *Kagaku Ronrigaku, Vin Gakudan [The Logic of Science, by the Vienna Circle]*, edited by Katsumi Nakamura, pp. 311-349. Nisshin Shoin, Tokyo. Japanese translation by Syūitiros Yosioka of **Part I of 1934(b)**.
- 1945(a)** The Poverty of Historicism III (Part IV). *Economica* **12**, N.S., No. 46, May 1945, pp. 69-89.
- 1945(b)** *The Open Society and Its Enemies*. Volume I: *The Spell of Plato*. George Routledge & Sons Ltd, London.
- 1945(c)** *The Open Society and Its Enemies*. Volume II: *The High Tide of Prophecy: Hegel, Marx, and the Aftermath*. George Routledge & Sons Ltd, London.
- Further impressions, editions, and translations of **1945(b)** and **1945(c)** into other languages are indicated in the year in which they are published.
- 1945(d)** The Growth of German Historicism. *Economica* **12**, N.S., No. 48, November 1945, pp. 259-261. Review of *The Growth of German Historicism* by F. Engel-Janosi.
- 1945(e)** (with R.S. Allan, J.C. Eccles, H.G. Forder, J. Packer, & H.N. Parton) *Research and the University: A Statement by a Group of Teachers in the University of New Zealand*. *Carlton Press*, Christchurch, New Zealand, 10th July 1945.
- 1945(f)** Place of Research in the University. *Christchurch Press* (Christchurch, New Zealand), 28th July 1945. A reprint of **1945(e)**.
- 1945(g)** Research and the University. *The Critic* (Dunedin, New Zealand) **21**, No. 11, 13th September 1945, p. 1. An abridged version of **1945(e)**.
- 1945(h)** Principles of Scientific Method. In *Principles of Scientific Method: Notes on Lectures by Dr. K.R. Popper given at the University of Otago from 22-26th May, 1945*, pp. 1-20. Unpublished typescript, edited by J.C. Eccles.

- 1945(i)** Principles of Indeterminacy. *Ibid.*, pp. 21f. Notes on an informal talk to a group of physicists. 19th May 1945. University of Otago, New Zealand.
- 1945(j)** Atomic Theory and Biology. *Ibid.*, pp. 23-26. Notes on an informal talk to a group in the Medical School. University of Otago, New Zealand.
- 1946(a)** A Scientist Looks at History. *Time and Tide* **27**, No. 5, 2nd February 1946, p. 107. A letter to the Editor.
- 1946(b)** Why are the Calculuses of Logic and Arithmetic Applicable to Reality?. *Aristotelian Society. Supplementary Volume XX: Logic and Reality*, pp. 40-60. Harrison & Sons, Ltd. London. Contained in **1963(a)**.
- 1947(a)** New Foundations for Logic. *Mind* **56**, N.S., No. 223, July 1947, pp. 193-235. Corrections & additions: *ibid.* **57**, pp. 69f.
- 1947(b)** Logic Without Assumptions. *Proceedings of the Aristotelian Society* **47**, 1947, pp. 251-292.
- 1947(c)** Functional Logic without Axioms or Primitive Rules of Inference. *Proceedings of the Koninklijke Nederlandsche Akademie van Wetenschappen* **50**, No. 9, 1947, pp. 1214-1224, and *Indagationes Mathematicae* **9**, 1947, Fasciculus 5, pp. 561-571.
- 1947(d)** L'Utopie et la violence. *Les rencontres philosophiques des Bruzelles*. French translation of **1948(a)**.
- 1947(e)** *The Open Society and Its Enemies*. Volumes I and II. George Routledge & Sons, London. A reprint of **1945(b)** and **1945(c)**.
- 1947(f)** To the Editor of *Philosophy*. *Philosophy* (London) **22**, No. 82, July 1947, p. 191.
- 1948(a)** Utopia and Violence. *The Hibbert Journal* **46**, No. 2, January 1948, pp. 109-116. Contained in **1963(a)**. French translation: **1947(d)**; Japanese **1963(j)**; German **1968(x)**; Russian **1977(z₄)**; Italian **1981(z₄)**.
- 1948(b)** On the Theory of Deduction. Part I: Derivation and its Generalizations, in *Proceedings of the Koninklijke Nederlandsche Akademie van Wetenschappen* **51**, No. 2, 1948, pp. 173-183, and *Indagationes Mathematicae* **10**, Fasciculus 1, 1948, pp. 44-54.
- 1948(c)** On the Theory of Deduction. Part II: The Definitions of Classical and Intuitionist Negation, in *Proceedings of the Koninklijke Nederlandsche Akademie van Wetenschappen* **51**, No. 3, 1948, pp. 322-331, and *Indagationes Mathematicae* **10**, Fasciculus 2, 1948, pp. 111-120.
- 1948(d)** Prediction and Prophecy and their Significance for Social Theory. In *Proceedings of the Xth International Congress of Philosophy* (Amsterdam

- 1948), volume 1, pp. 82-91. Contained in **1963(a)**. German translation: **1965(m)**; Portuguese: **1969(x)**; Chinese (unauthorized): **1981(z₃₃)**.
- 1948(e)** The Trivialization of Mathematical Logic. *Proceedings of the Xth International Congress of Philosophy* (Amsterdam 1948), volume 2, pp. 722-727. *Errata: ibid.*, p. 1259.
- 1948(f)** What can Logic do for Philosophy?. *Aristotelian Society, Supplementary Volume XXI: Logical Positivism and Ethics*, pp. 141-154. Harrison & Sons Ltd, London.
- 1948(g)** An important correction. *The Hibbert Journal* **46**, No. 3, April 1948, p. 275.
- 1949(a)** A Note on Natural Laws and so-called 'Contrary-to-Fact Conditionals'. *Mind* **58**, N.S., No. 229, January 1949, pp. 62-66.
- 1949(b)** Towards a Rational Theory of Tradition. *The Rationalist Annual 1949*, London, pp. 36-55. Contained in **1963(a)**. Chinese translation: **1977(z₁)**.
- 1949(c)** *The Open Society and Its Enemies*. Volumes I and II. George Routledge & Sons, London. A reprint of **1947(e)**.
- 1949(d)** Naturgesetze und theoretische Systeme. In *Gesetz und Wirklichkeit*, edited by S. Moser, pp. 43-60. Tyrolia Verlag, Innsbruck. English translation contained in **1972(a)**.
- 1950(a)** *The Open Society and Its Enemies*. Princeton University Press, Princeton, N.J. Revised and enlarged edition in one volume of **1945(b)** and **1945(c)**.
- 1950(b)** Indeterminism in Quantum Physics and in Classical Physics I. *The British Journal for the Philosophy of Science* **1**, No. 2, August 1950, pp. 117-133.
- 1950(c)** Indeterminism in Quantum Physics and in Classical Physics II, in *The British Journal for the Philosophy of Science* **1**, No. 3, November 1950, pp. 173-195.
- 1950(d)** *De Vrije Samenleving en Haar Vijanden*. Volumes I and II. F.G. Kroonder, Bussum, Netherlands. Dutch translation by Justus Meijer of **1945(b)** and **1945(c)**.
- 1952(a)** *The Open Society and Its Enemies*. Routledge & Kegan Paul, London. Further revised and enlarged English edition in two volumes of **1945(b)** and **1945(c)**, and of **1950(a)**.
- 1952(b)** Humanism and Reason. *The Philosophical Quarterly* **2**, No. 7, April 1952, pp. 166-171. Abridged review of *Von Ursprung und Grenzen der Geisteswissenschaften und Naturwissenschaften* by Ernesto Grassi & Thure von Uexküll. Contained in **1963(a)**.

- 1952(c)** The Nature of Philosophical Problems and Their Roots in Science. *The British Journal for the Philosophy of Science* **3**, No. 10, August 1952, pp. 124-56. Contained in **1963(a)**.
- 1952(d)** La povertà dello storicismo. *L'industria* (Milan), No. 3, Summer 1952, pp. 285-307. Italian translation by Camilla Roatta of a revised version of part of **1944(a)**.
- 1953(a)** Language and the Body-Mind Problem. In *Proceedings of the XIth International Congress of Philosophy* (Brussels 1953), volume 7, pp. 101-107. Contained in **1963(a)**.
- 1953(b)** The Principle of Individuation. *Aristotelian Society, Supplementary Volume XXVII*, pp. 97-120. Harrison & Sons, Ltd, London.
- 1953(c)** The Sociology of Knowledge: A Critique. In *Readings in Philosophy of Science*, edited by P.P. Wiener, pp. 357-366. Scribner, New York. An excerpt from chapter 23 of **1950(a)**.
- 1953(d)** A Note on Berkeley as Precursor of Mach. *The British Journal for the Philosophy of Science* **4**, No. 13, May 1953, pp. 26-36. Contained in **1963(a)**.
- 1953(e)** La povertà dello storicismo. *L'industria* (Milan), No. 1, January 1953, pp. 69-93; No. 2, March/May 1953, pp. 283-307; No. 3, Summer 1953, pp. 451-491. Italian translation of a revised version of the final part of **1944(a)**, and of **1944(b)**, and **1945(a)**.
- 1954(a)** Immanuel Kant: Philosopher of the Enlightenment. *The Listener* **51**, 18th February 1954, pp. 291f. and 303. Contained in **1963(a)** and in **1992(t)**. German translation: **1954(b)**.
- 1954(b)** Immanuel Kant. Zu seinem 150. Todestag. In *Englische Rundschau* (Cologne), No. 4, 19th March 1954, pp. 166f. and 171. German translation of **1954(a)**. Contained in **1957(k)**.
- 1954(c)** Self-Reference and Meaning in Ordinary Language. *Mind* **63**, N.S., No. 250, April 1954, pp. 162-169. Contained in **1963(a)**.
- 1954(d)** Degree of Confirmation. *The British Journal for the Philosophy of Science* **5**, No. 18, August 1954, pp. 143-149. Errata: *ibid.* **5**, No. 20, February 1955, pp. 334 and 359. Continued in **1957(b)** and **1958(a)**. Contained in **1959(a)**.
- 1954(e)** *Miseria dello storicismo*. Editrice L'industria, Milan. Book edition of **1952(d)** and **1953(e)**.
- 1954(f)** The Advocate of Democratic Criticism. In *The State versus Socrates: A Case Study in Civic Freedom*, edited by J.D. Montgomery, pp. 160-166. The Beacon Press, Boston. An excerpt from chapter 10 of **1950(a)**.

- 1954(g)** Has History Any Meaning?. In *Contemporary Philosophy*, edited by J.L. Jarrett & S.M. McMurrin, pp. 433-440. Henry Holt & Co., New York. An excerpt from chapter 25 of **1945(c)**.
- 1955(a)** On a Proposed Solution of the Paradox of the Liar. *The Journal of Symbolic Logic* **20**, No. 1, March 1955, pp. 93f.
- 1955(b)** Two Autonomous Axiom Systems for the Calculus of Probabilities. *The British Journal for the Philosophy of Science* **6**, No. 21, May 1955, pp. 51-57. *Errata: ibid.* **6**, No. 22, August 1955, p. 176 and *ibid.* **6**, No. 24, February 1956, p. 351.
- 1955(c)** A Note on the Body-Mind Problem. Reply to Professor Wilfrid Sellars. *Analysis* **15**, N.S., No. 6, June 1955, pp. 131-135. Contained in **1963(a)**.
- 1955(d)** A Note on Tarski's Definition of Truth. *Mind* **64**, N.S., No. 255, July 1955, pp. 388-391. Contained in **1972(a)**.
- 1955(e)** 'Content' and 'Degree of Confirmation': A Reply to Dr. Bar-Hillel. *The British Journal for the Philosophy of Science* **6**, No. 22, August 1955, pp. 157-163. Continued in **1956(i)**.
- 1955(f)** Verso una teoria liberale dell'opinione pubblica. *Il Politico* (Milan) **20**, No. 2, 1955, pp. 181-189. Italian translation of **1956(e)**. English version contained in **1963(a)** and in **1992(t)**.
- 1955(g)** The Demarcation between Science and Metaphysics. A contribution (**1964(b)**) to *The Philosophy of Rudolf Carnap, The Library of Living Philosophers*, edited by P.A. Schilpp. Distributed in mimeographed form. Contained in **1963(a)**.
- 1955(h)** The Illusion of the Epoch. *The Times Literary Supplement* **54**, No. 2786, 22nd July 1955, p. 413. A letter to the Editor.
- 1956(a)** *Misère de l'historicisme*. Librairie Plon, Paris. French translation of revised and enlarged version of **1944(a)**, **1944(b)**, and **1945(a)**.
- 1956(b)** The Arrow of Time. *Nature* **177**, No. 4507, 17th March 1956, p. 538. Continued in **1956(g)**, **1957(d)**, **1957(f)**, **1958(b)**, **1965(f)**, **1967(b)**, and **1967(h)**.
- 1956(c)** *The Open Society and Its Enemies*. Princeton University Press. A reprint of **1950(a)**.
- 1956(d)** *The Open Society and Its Enemies*. The Library of Science, Basic Books, New York. A special book club edition of **1950(a)**.
- 1956(e)** Die öffentliche Meinung im Lichte der Grundsätze des Liberalismus. *Ordo* **8**, Helmut Küpper Verlag, Düsseldorf & Munich, pp. 7-17. English version contained in **1963(a)**. Italian translation: **1955(f)**; Spanish: **1963(f)**.

- 1956(f)** Three Views Concerning Human Knowledge. In *Contemporary British Philosophy*, edited by H.D. Lewis, pp. 355-388. Allen & Unwin, London. Contained in **1963(a)**. Italian translations: **1958(d)**, **1969(c)**, and **1969(d)**; Japanese **1968(m)**; Portuguese: **1980(w)**.
- 1956(g)** Irreversibility and Mechanics. *Nature* **178**, No. 4529, 18th August 1956, p. 382. A continuation of **1956(b)**.
- 1956(h)** Reply to Professor Carnap. *The British Journal for the Philosophy of Science* **7**, No. 27, November 1956, pp. 244f.
- 1956(i)** Adequacy and Consistency: A Second Reply to Dr. Bar-Hillel. *The British Journal for the Philosophy of Science* **7**, No. 27, November 1956, pp. 249-256. A continuation of **1955(a)**.
- 1957(a)** Philosophy of Science: A Personal Report. In *British Philosophy in the Mid-Century*, edited by C.A. Mace, pp. 155-191. Allen & Unwin, London. Contained in **1963(a)**. Italian translation: **1969(d)**; Swedish: **1980(zg)**.
- 1957(b)** A Second Note on Degree of Confirmation. *The British Journal for the Philosophy of Science* **7**, No. 28, February 1957, pp. 350-353. Addition and correction, *ibid.* **8**, No. 32, February 1958, pp. 294f. A continuation of **1954(d)**. Contained in **1959(a)**.
- 1957(c)** *La Sociedad Abierta y Sus Enemigos*. Biblioteca de Psicología Social y Sociológica, Editorial Paidós, Buenos Aires. Spanish translation by Eduardo Loedel of **1950(a)**.
- 1957(d)** Irreversible Processes in Physical Theory. *Nature* **179**, No. 4573, 22nd June 1957, p. 1297. A continuation of **1956(b)** and **1956(g)**.
- 1957(e)** The Propensity Interpretation of the Calculus of Probability, and the Quantum Theory. In *Observation and Interpretation, Proceedings of the Ninth Symposium of the Colston Research Society, University of Bristol*, edited by S. Körner in collaboration with M.H.L. Pryce, pp. 65-70 and 88f. Butterworths, London, 1957. Volume 9 of the *Colston Papers*. Errata: *British Journal for the Philosophy of Science* **8**, No. 32, February 1958, p. 301, note 1.
- 1957(f)** Irreversibility; or Entropy since 1905. *The British Journal for the Philosophy of Science* **8**, No. 30, August 1957, pp. 151-155. Erratum: *The British Journal for the Philosophy of Science* **8**, No. 31, November 1957, p. 258. A continuation of **1956(b)**, **1956(g)**, and **1957(d)**.
- 1957(g)** *The Poverty of Historicism*, Routledge & Kegan Paul, London, and The Beacon Press, Boston. Revised English book-edition of **1944(a)**, **1944(b)**, and **1945(a)**.

Further impressions, editions, and translations of **1957(g)** into other languages are indicated in the year in which they are published.

- 1957(h)** *The Open Society and Its Enemies*. Volumes I and II. Routledge & Kegan Paul, London. Third edition, newly revised, of **1952(a)**, with a new *Addendum* ('Plato and Geometry') to volume I.
- 1957(i)** The Aim of Science. *Ratio* (Oxford) **1**, No. 1, December 1957, pp. 24-35. Revised English edition, **1969(k)**, contained in **1972(a)**. German translations: **1957(j)** and **1964(d)**; Italian: contained in **1962(j)** and **1969(c)**.
- 1957(j)** Über die Zielsetzung der Erfahrungswissenschaft. *Ratio* (Frankfurt am Main) **1**, No. 1, December 1957, pp. 21-31. German translation of **1957(i)**.
- 1957(k)** *Die offene Gesellschaft und ihre Feinde*. Volume I: *Der Zauber Platons*. Francke Verlag, Berne. German translation by P.K. Feyerabend of volume I of **1952(a)**. Continued in **1958(i)**.
- 1957(l)** Probability Magic, or Knowledge out of Ignorance. *Dialectica* **11**, Nos 3/4, September/December 1957, pp. 354-372.
- 1958(a)** A Third Note on Degree of Corroboration or Confirmation. *The British Journal for the Philosophy of Science* **8**, No. 32, February 1958, pp. 294-302. A continuation of **1954(d)** and **1957(b)**. Contained in **1959(a)**.
- 1958(b)** Irreversible Processes in Physical Theory. *Nature* **181**, No. 4606, 8th February 1958, pp. 402f. A continuation of **1956(b)**, **1956(g)**, **1957(d)**, and **1957(f)**.
- 1958(c)** Das Problem der Nichtwiderlegbarkeit von Philosophien. *Deutsche Universitätszeitung* **13**, January 1958, pp. 7-13. German translation of part II of **1958(f)**.
- 1958(d)** Tre concezioni sulla conoscenza umana. *L'industria* (Milan), pp. 60-73 and 183-198. Italian translation of **1956(f)**. See also pp. 138f., 198-201, and 299 for English summaries and a list of publications. Contained in **1969(c)**. Published also as booklet by Editrice L'industria, Milan.
- 1958(e)** Back to the Pre-Socratics. *Proceedings of the Aristotelian Society* **59**, N.S., 1958-1959, pp. 1-24. Contained in **1963(a)**.
- 1958(f)** On the Status of Science and of Metaphysics, *Ratio* (Oxford) **1**, No. 2, December 1958, pp. 97-115. Contained in **1963(a)**. German translation: **1958(g)**; Greek: **1980(l)**.
- 1958(g)** Über die Möglichkeit der Erfahrungswissenschaft und der Metaphysik. *Ratio* (Frankfurt am Main) **1**, 1958, No. 2, December 1958, pp. 1-16. German translation of **1958(f)**. Part II first published as **1958(c)**.
- 1958(h)** On Mr. Roy Harrod's New Argument for Induction. *The British Journal for the Philosophy of Science* **9**, No. 35, November 1958, pp. 221-224.

- 1958(i)** *Die offene Gesellschaft und ihre Feinde. Volume II: Falsche Propheten: Hegel, Marx und die Folgen.* Francke Verlag, Bern. 1958. German translation by P.K. Feyerabend of volume II of **1952(a)**. Continuation of **1957(k)**.
- 1959(a)** *The Logic of Scientific Discovery.* Hutchinson, London (January), and Basic Books, New York (March). Contains English translations of **1934(b)**, **1933(a)** **1935(a)**; amended reprints of **1938(a)**, **1954(d)**, **1957(b)**, and **1958(a)**; new *Appendices*; and other new material.
- Further impressions, editions, and translations of **1959(a)** into other languages are indicated in the year in which they are published.
- 1959(b)** *'Uqm al-madhab al-tārikhī.* Munsha'at al-ma'ārif. Alexandria. Arabic translation by Abdulhamid Sabra of **1957(g)**.
- 1959(c)** *A sociedade democrática e seus inimigos.* Editora Itatiaia, Belo Horizonte. Portuguese translation of **1957(h)**.
- 1959(d)** Über die Unwiderlegbarkeit philosophischer Theorien. *Forum* (Vienna) **6**, No. 61, January 1959, pp. 15-18. A revised version of **1958(g)**. Contained in **1963(k)** and **1969(t)**.
- 1959(e)** The Propensity Interpretation of Probability. *The British Journal for the Philosophy of Science* **10**, No. 37, May 1959, pp. 25-42.
- 1959(f)** Testability and 'Ad-Hocness' of the Contraction Hypothesis. *The British Journal for the Philosophy of Science* **10**, No. 37, May 1959, p. 50.
- 1959(g)** Woran glaubt der Westen?. In *Erziehung zur Freiheit*, edited by A. Hunold, pp. 237-262. Eugen Rentsch Verlag, Zürich & Stuttgart. Contained in **1992(t)**.
- 1959(h)** On the Sources of Our Knowledge. *The Indian Journal of Philosophy* **1**, No. 1, August 1959, pp. 3-7. An excerpt from **1960(d)**.
- 1959(i)** On Subjunctive Conditionals with Impossible Antecedents. *Mind* **68**, N.S., No. 272, October 1959, pp. 518-520.
- 1959(j)** Has History Any Meaning?. In *The Philosophy of History in Our Time*, edited by Hans Meyerhoff, pp. 301-311. Doubleday Anchor Books, New York. An excerpt from of chapter 25 of **1950(a)**.
- 1959(k)** Critical Rationalism. In *Philosophy for a Time of Crisis: An Interpretation with Key Writings by Fifteen Great Modern Thinkers*, edited by Adrienne Koch, pp. 262-275. Dutton & Co., New York. An excerpt of chapters 24 and 25 of **1950(a)**.
- 1959(l)** Prediction and Prophecy in the Social Sciences. In *Theories of History*, edited by Patrick Gardiner, pp. 276-285. The Free Press, Glencoe, Illinois. A revised version of **1948(d)**. Contained in **1963(a)**. German translation: **1965(m)**.

- 1959(m)** *The Logic of Scientific Discovery. The New Scientist* **5**, No. 124, 2nd April 1959, p. 763. A letter to the Editor.
- 1959(n)** *The Logic of Scientific Discovery*. University of Toronto Press, Toronto. A reprint of **1959(a)**. Unauthorized.
- 1960(a)** *The Logic of Scientific Discovery*. Hutchinson, London. Second impression, revised, of **1959(a)**.
- 1960(b)** Probabilistic Independence and Corroboration by Empirical Tests. *The British Journal for the Philosophy of Science* **10**, No. 40, February 1960, pp. 315-318.
- 1960(c)** *The Poverty of Historicism*. Routledge & Kegan Paul, London. Second edition, revised, of **1957(g)**.
- 1960(d)** On the Sources of Knowledge and of Ignorance. *Proceedings of The British Academy* **46**, 1960, pp. 39-71. Published separately as **1961(f)**. Contained in **1963(a)** and **1966(k)**. Spanish translation: **1962(q)**; Japanese: **1963(b)**; Italian: **1964(h)** and **1969(c)**.
- 1960(e)** Philosophy and Physics: The Influence on Physics of some Metaphysical Speculations on the Structure of Matter. *Atti del XII Congresso Internazionale di Filosofia* (Venice 1958), volume 2, Florence, pp. 367-374. Contained in **1994(r)**. French translation: **1987(z₁)**.
- 1961(a)** The Moral Theory of Historicism. In *Society, Law, and Morality*, edited by Frederick A. Olafson, pp. 286-297. Prentice-Hall Inc., Englewood Cliffs, N.J. An excerpt from of chapter 22 of **1950(a)**.
- 1961(b)** *The Poverty of Historicism*. Routledge & Kegan Paul, London. A paperback edition of **1960(c)**.
- 1961(c)** *Rekishii-shugi no Hinkon* [*The Poverty of Historicism*]. Chuo-Koron Sha, Inc., Tokyo, 1961. First, second, and third impressions. Japanese translation by Saburo Ichii & Osamu Kuno of **1960(c)**.
- 1961(d)** Selbstbefreiung durch das Wissen. In *Der Sinn der Geschichte*, edited by Leonhard Reinisch, pp. 100-116. C.H. Beck, Munich. English translation: **1968(t)**.
- 1961(e)** *The Logic of Scientific Discovery*. Science Editions Inc., New York. A paperback edition of **1959(a)**.
- 1961(f)** *On the Sources of Knowledge and of Ignorance*. Oxford University Press, London. Annual Philosophical Lecture, Henrietta Hertz Trust, British Academy 1960. Separate edition of **1960(d)**.
- 1961(g)** *La miseria del historicismo*. Taurus Ediciones, Madrid. Spanish translation by Pedro Schwartz of **1960(c)**.

- 1961(h)** Has History Any Meaning?. In *The Philosophy of History in Our Time: An Anthology*. Doubleday Anchor Books, New York, pp. 300-311. Second edition, revised, of **1959(j)**, containing expanded excerpts from chapter 25 of **1950(a)**.
- 1961(i)** *Evolution and the Tree of Knowledge*. Herbert Spencer Lecture, Oxford. Copy of lecture deposited at the Bodleian Library on October 30th, 1961. Contained in **1972(a)**.
- 1961(j)** *The Logic of Scientific Discovery*. Basic Books, New York. A reprint of **1960(a)**.
- 1962(a)** On Carnap's Version of Laplace's Rule of Succession. *Mind* **71**, N.S., No. 281, January 1962, pp. 69-73. Continued in **1967(c)**.
- 1962(b)** Historical Explanation: An Interview. *Cambridge Opinion* **28**, 1962, pp. 21-25. Replies to questions by Michael Tanner, John Dunne, & Alistair Young.
- 1962(c)** *The Open Society and Its Enemies*. Routledge & Kegan Paul, London. Fourth edition, revised, of **1945(b)** and **1945(c)**, in two volumes, with two new *Addenda* ('The Dating of the Theaetetus' and 'Reply to a Critic') to volume I, and a new *Addendum* ('Facts, Standards, and Truth: A Further Criticism of Relativism') to volume II.
- 1962(d)** *The Open Society and Its Enemies*. Volumes I and II. Routledge Paperbacks, London. A paperback edition of **1962(c)**.
- 1962(e)** Selbstbefreiung durch das Wissen. In *Der Sinn der Geschichte*. C.H. Beck, Munich, 1962, pp. 100-116. Second edition, revised, of **1961(d)**.
- 1962(f)** Julius Kraft 1898-1960. *Ratio* (Oxford) **4**, No. 1, June 1962, pp. 2-12.
- 1962(g)** Julius Kraft. *Ratio* (Frankfurt am Main) **4**, No. 1, June 1962, pp. 2-10. German translation of **1962(f)**.
- 1962(h)** Some Comments on Truth and the Growth of Knowledge. In *Logic, Methodology and Philosophy of Science: Proceedings of the 1960 International Congress*, edited by E. Nagel, P. Suppes, & A. Tarski, pp. 285-292. Stanford University Press, Stanford. An excerpt from chapter 10 of **1963(a)**.
- 1962(i)** Über Geschichtsschreibung und über den Sinn der Geschichte. In *Geist und Gesicht der Gegenwart*, edited by Otto Molden, pp. 111-142. Europa Verlag, Zürich. A revised version of part of chapter 25 of **1958(i)**. Contained in **1994(y)**.
- 1962(j)** Lo Scopo della Scienza. *La Scuola in Azione* (Milan) **13**, 1962, pp. 5-22. Italian translation of **1957(i)**; another translation is contained in **1969(c)**.
- 1962(k)** Die Logik der Sozialwissenschaften. *Kölner Zeitschrift für Soziologie und Sozialpsychologie* **14**, No. 2, 1962, pp. 233-248. Contained in **1969(m)**.

English translation **1976(b)**, contained in **1992(t)**. Italian: **1972(x)**;
Hungarian: **1976(k)**; French: **1979(z₁₀)**.

- 1962(l)** Critical Rationalism. In *Philosophy for a Time of Crisis* by Albert Einstein, E.M. Forster, Karl R. Popper, and Bertrand Russell, edited by Kôzô Tada, pp. 36-53. Kinseido Ltd, Tokyo. A reprint, with Japanese notes, of four of the essays contained in **1959(k)**.
- 1962(m)** A Comment on the New Prediction Paradox. *The British Journal for the Philosophy of Science* **13**, No. 49, May 1962, p. 51.
- 1962(n)** On the Sources of Knowledge and of Ignorance. *Encounter* **108**, No. 3, September 1962, pp. 42-57. Abbreviated version of **1960(d)** and **1961(f)**.
- 1962(o)** The Propensity Interpretation of the Calculus of Probability, and the Quantum Theory. In *Observation and Interpretation*. Dover Publications, New York, pp. 65-70 and 88f. A paperback edition of **1957(e)**.
- 1962(p)** *The Logic of Scientific Discovery*. Hutchinson, London. Third impression of **1959(a)**.
- 1962(q)** Las Fuentes del Conocimiento y de la Ignorancia. *Cuadernos de Epistemología* (Buenos Aires), No. 50, 1962. Spanish translation by Emilio O. Colombo of **1960(d)**.
- 1962(r)** *La lógica de la investigación científica*. Editorial Tecnos, Madrid. Spanish translation by Víctor Sánchez de Zavala, of **1960(a)**, incorporating some changes.
- 1962(s)** *The Open Society and Its Enemies*. Princeton University Press, Princeton. Two-volume American edition of **1962(c)**.
- 1962(t)** *Conjectures and Refutations*. Basic Books, New York.
- 1963(a)** *Conjectures and Refutations: The Growth of Scientific Knowledge*, Routledge & Kegan Paul, London.
- Further impressions, editions, and translations of **1963(a)** into other languages are indicated in the year in which they are published.
- 1963(b)** Kindai Ninshikiron no Shinwa: Chishiki to Muchi no Gensen ni tsuite [The Myth of Modern Epistemology: On the Sources of Knowledge and of Ignorance]. *Jiyu* **5**, No. 1, January 1963. Japanese translation of **1960(d)** by Tooru Yoshimura.
- 1963(c)** Plato as Enemy of the Open Society. In *Plato: Totalitarian or Democrat?*, edited by Thomas L. Thorson, pp. 41-102. Prentice-Hall Inc., Englewood Cliffs, N.J. A Spectrum book. An excerpt from **1950(a)**.

- 1963(d)** Kirk on Heraclitus, and on Fire as the Cause of Balance. *Mind* **72**, N.S., No. 287, July 1963, pp. 386-392. An abridged version of the *appendix* to chapter 5 of **1963(a)**.
- 1963(e)** The Erewhonians and the Open Society. *ETC: A Review of General Semantics* **20**, May 1963, No. 1, pp. 5-22.
- 1963(f)** La opinión pública a la luz de los principios del liberalismo. In *La Economía de Mercado*. Sociedad de Estudios y Publicaciones, Madrid, pp. 137-152. Spanish translation of **1956(e)**.
- 1963(g)** Creative and Non-Creative Definitions in the Calculus of Probability. *Synthese* **15**, No. 2, June 1963, pp. 167-186. Correction: **1970(v)**.
- 1963(h)** Science: Problems, Aims, Responsibilities. *Federation Proceedings*, Federation of American Societies for Experimental Biology, **22**, No. 4, July/August 1963, Part I, pp. 961-972. Contained in **1994(r)**. Italian translations: **1966(h)** and **1969(c)**.
- 1963(i)** *Jiyu Shakai no Tetsugaku to sono Ronteki* [The Philosophy of the Free Society and its Enemies]. The Izumiya Co., Inc., Osaka City. Japanese translation by Hiromichi Takeda of chapters 11-17 of **1962(c)**.
- 1963(j)** Yūtopia to Bōryoku [Utopia and Violence]. *Shisō* [Thought], Iwanami Shoten, Tokyo, No. 469, July 1963, pp. 968-979. Japanese translation by Saburo Ichii of **1948(a)**.
- 1963(k)** Über die Unwiderlegbarkeit philosophischer Theorien. In *Club Voltaire*, I, edited by Gerhard Szczesny, pp. 271-279. Szczesny Verlag, Munich. A reprint of **1959(d)**.
- 1963(l)** *The Open Society and Its Enemies*. Princeton University Press, Princeton, N.J. A reprint of **1962(s)**.
- 1963(m)** *The Open Society and Its Enemies*. The Academy Library, Harper & Row, New York & Evanston. A paperback edition (Harper Torchbook) of **1962(s)**.
- 1963(n)** *The Open Society and Its Enemies*. Routledge Paperbacks, London. A reprint of **1962(d)**.
- 1963(o)** Problems of Scientific Knowledge. *Bulletin of the International House of Japan*, No. 12, Tokyo, October 1963, pp. 23-28. Japanese translation contained in *The Japanese Language Bulletin* [Kokusai Bunka Kaikan Kaiho], No. 11.
- 1963(p)** *The Poverty of Historicism*, Routledge & Kegan Paul, London. Second impression of **1961(b)**.
- 1964(a)** *The Poverty of Historicism*. The Academy Library, Harper & Row, New York & Evanston. A paperback edition (Harper Torchbook) of **1961(b)**.

- 1964(b)** The Demarcation between Science and Metaphysics. In *The Philosophy of Rudolf Carnap*, edited by Paul A. Schilpp, *The Library of Living Philosophers*, volume 11, pp. 183-226. The Open Court Publishing Co., La Salle, Illinois, and Cambridge University Press, London. Previously circulated as **1955(g)**. Contained in **1963(a)**. French translation: **1980(d)**.
- 1964(c)** Al-'Aql al-Naqid [The Critical Mind]. In *Al Falsafiyah fi Azmat al-'Asr*, edited by Adrienne Koch. Anglo, Cairo. Arabic translation by Mahmoud Mahmoud of **1959(k)**.
- 1964(d)** Die Zielsetzung der Erfahrungswissenschaft. In *Theorie und Realität*, edited by Hans Albert, pp. 73-86. J.C.B. Mohr (Paul Siebeck), Tübingen. Revised reprint of **1957(j)**.
- 1964(e)** Naturgesetze und theoretische Systeme. In *Theorie und Realität*, edited by Hans Albert, pp. 87-102. J.C.B. Mohr (Paul Siebeck), Tübingen, 1964. A reprint of **1949(d)**.
- 1964(f)** Creative and Non-Creative Definitions in the Calculus of Probability. In *Form and Strategy in Science: Studies dedicated to J.H. Woodger*, edited by John R. Gregg & F.T.C. Harris, pp. 171-190. D. Reidel Publishing Company, Dordrecht. Book edition of **1963(g)**. Correction: **1970(v)**.
- 1964(g)** On Socrates. In *The Socratic Enigma*, edited by H. Spiegelberg in collaboration with B.Q. Morgan. The Bobbs-Merrill Company, New York, pp. 119-122. Excerpts from chapters 7 and 10 of **1945(b)**.
- 1964(h)** Sulle fonti della conoscenza e dell'ignoranza. *L'industria* (Milan), No. 2, April/June 1964, pp. 219-236, and No. 3, July/September 1964, pp. 337-354. English Summary: *ibid.*, No. 2, pp. 282f., and No. 3, p. 459. Published separately in *Collana di moderne opere economiche*, edited by Ferdinando di Fenizio. Editrice L'industria, Milan, 1964. Italian translation of **1960(d)**; another translation is contained in **1969(c)**.
- 1964(i)** Anaximander's Theory of the Earth. *Christian Science Monitor*, New England Edition, 17th April 1964, p. 8. An excerpt from chapter 5 of **1963(a)**.
- 1965(a)** Hume's Explanation of Inductive Inference. In *Human Understanding, Studies in the Philosophy of David Hume*, edited by A. Sesonske & N. Fleming, pp. 69-74. Wadsworth Publishing Company, Belmont, California. An excerpt from **1957(a)**.
- 1965(b)** Was ist Dialektik?. *Student im Bild*, 10. Jahrgang, No. 2, May/June 1965, pp. 42-45. Continued in **1966(n)** and **1966(o)**. German translation of part of **1940(a)**.
- 1965(c)** *Conjectures and Refutations: The Growth of Scientific Knowledge*. Routledge & Kegan Paul, London, and Basic Books, New York. Second edition, revised and enlarged, of **1963(a)**, with a new *Addendum* ('A Historical Note on Verisimilitude') and new indices.

- 1965(d)** *The Logic of Scientific Discovery*. Hutchinson, London. Fourth impression, revised, of **1959(a)**.
- 1965(e)** *The Logic of Scientific Discovery*. The Science Library, Harper & Row, New York. A paperback edition (Harper Torchbook) of **1965(d)**.
- 1965(f)** Time's Arrow and Entropy. *Nature* **207**, No. 4994, 17th July 1965, pp. 233f. A continuation of **1956(b)**, **1956(g)**, **1957(d)**, **1957(f)**, and **1958(b)**.
- 1965(g)** *Das Elend des Historizismus*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. German translation by Leonhard Walentik of **1964(a)**, with a new Preface.
- 1965(h)** Hat die Weltgeschichte einen Sinn?. In *Versäumte Lektionen*, edited by Peter Glotz & Wolfgang R. Langenbucher, pp. 433-439. Sigbert Mohn Verlag, Gütersloh. Published also by Europäischer Buchklub, Europäische Bildungsgemeinschaft, Stuttgart, Zürich, & Salzburg, 1965. An excerpt from chapter 25 of **1958(i)**.
- 1965(i)** Was ist Aufklärung?. In *Versäumte Lektionen*, edited by Peter Glotz & Wolfgang R. Langenbucher, pp. 407-411. Sigbert Mohn Verlag, Gütersloh. An excerpt from **1954(b)**.
- 1965(j)** Unity of Method in the Natural and Social Sciences. In *Philosophical Problems of the Social Sciences*, edited by David Braybrooke, pp. 32-41. The Macmillan Company, New York, and Collier-Macmillan Limited, London. An excerpt from **1960(c)**.
- 1965(k)** Social Science and Social Policy. In *Philosophical Problems of the Social Sciences*, edited by David Braybrooke, pp. 99-107. The Macmillan Company, New York, and Collier-Macmillan Limited, London. An excerpt from **1960(c)**.
- 1965(l)** Historical and Generalizing Sciences. In *Philosophy of History*, edited by Alan Donagan & Barbara Donagan, pp. 86-89. The Macmillan Company, New York, and Collier-Macmillan Limited, London. An excerpt from **1957(h)**.
- 1965(m)** Prognose und Prophetie in den Sozialwissenschaften. In *Logik der Sozialwissenschaften*, edited by Ernst Topitz, pp. 113-125. Kiepenheuer & Witsch, Cologne & Berlin, first and second editions. German translation of **1948(d)**.
- 1965(n)** Was ist Dialektik?. *Ibid.*, pp. 262-290. German translation of **1948(a)**.
- 1965(o)** *Rekish-shugi no Hinkon*. Chuo-Koron Sha Inc., Tokyo. Fourth impression of **1961(c)**.
- 1966(a)** *The Open Society and Its Enemies*. Volumes I and II. Routledge Paperbacks. Fifth edition, further revised, of **1945(b)** and **1945(c)**, with two new *Addenda*, one (untitled) to volume I and one ('Note on Schwartzschild's Book on Marx') to volume II.

- 1966(b)** Foreword to *Two Kinds of Values* by L.M. Loring. Routledge & Kegan Paul, London, pp. vii-xii.
- 1966(c)** A Note on the Difference between the Lorentz-Fitzgerald Contraction and the Einstein Contraction. *The British Journal for the Philosophy of Science* **16**, No. 64, February 1966, pp. 332f.
- 1966(d)** A Comment on Miller's New Paradox of Information. *The British Journal for the Philosophy of Science* **17**, No. 1, May 1966, pp. 61-69. Continued in **1966(j)** Erratum: *ibid.*, No. 3, November 1966, p. 264.
- 1966(e)** *Logik der Forschung*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Second German edition, revised, with a new Preface, *Addenda* and indices, of **1934(b)** and **1959(a)**.
- 1966(f)** *Of Clouds and Clocks: An Approach to the Problem of Rationality and the Freedom of Man*. Washington University Press, St. Louis, Missouri. The Second Arthur Holly Compton Memorial Lecture, delivered April 1965. Contained in **1972(a)**. Italian translation: **1972(h)**.
- 1966(g)** A Theorem on Truth-Content. In *Mind, Matter, and Method: Essays in Philosophy and Science in Honor of Herbert Feigl*, edited by Paul K. Feyerabend & Grover Maxwell, pp. 343-363. University of Minnesota Press, Minneapolis, Minnesota.
- 1966(h)** Problemi, finalità e responsabilità della scienza. *L'industria* (Milan), No. 6, pp. 157-184. Italian translation by Giulia Sanvisenti of **1963(h)**. See also pp. 185-198 for an *Appendix*, 'Karl Popper's Life and Scientific Work (with a bibliographical note up to January 1966)', and pp. 321f. for English summaries. Another translation is contained in **1969(c)**. The appendix was published also as a booklet by Editrice L'industria, Milan.
- 1966(i)** Historical Explanation: An Interview. *University of Denver Magazine* **3**, No. 4, June 1966, pp. 4-7. A revised reprint of **1962(b)**.
- 1966(j)** A Paradox of Zero Information. *The British Journal for the Philosophy of Science* **17**, No. 2, August 1966, pp. 141-143. A continuation of **1966(d)**.
- 1966(k)** On the Sources of Knowledge and of Ignorance. In *Studies in Philosophy: British Academy Lectures*, edited by J.N. Findlay, pp. 169-212. Oxford University Press, London. A revised and enlarged reprint of **1960(d)**. Italian translation contained in **1969(c)**.
- 1966(l)** *Rekish-shugi no Hinkon*, Chuo-Koron Sha, Inc., Tokyo. Fifth impression of **1961(c)**.
- 1966(m)** *The Poverty of Historicism*, Routledge & Kegan Paul, London. Third impression of **1961(b)**.

- 1966(n)** Was ist Dialektik?. *Student im Bild*, 11. Jahrgang, No.1. A continuation of **1965(b)**.
- 1966(o)** Was ist Dialektik?. *Student im Bild*, 11. Jahrgang, No. 2, pp. 42-44. A continuation of **1965(b)** and **1966(n)**.
- 1966(p)** *The Open Society and Its Enemies*. Princeton Paperback, Princeton University Press. American edition of **1966(a)**.
- 1967(a)** *The Poverty of Historicism*. Basic Books, Inc., New York. Fourth impression of **1961(b)**.
- 1967(b)** Time's Arrow and Feeding on Negentropy. *Nature* **213**, No. 5073, 21st January 1967, p. 320. A continuation of **1956(b)**, **1956(g)**, **1957(d)**, **1957(f)**, **1958(b)**, and **1965(f)**.
- 1967(c)** The Mysteries of Udolpho: A Reply to Professors Jeffrey and Bar-Hillel. *Mind* **76**, N.S., No. 301, January 1967, pp. 103-110. *Erratum: ibid.*, No. 303, July 1967, p. 462.
- 1967(d)** La Rationalité et le Statut du Principe de Rationalité. In *Les Fondements Philosophiques des Systèmes Économiques: Textes de Jacques Rueff et essais rédigés en son honneur 23 août 1966*, edited by E.M. Claassen, pp. 142-150. Bibliothèque Economique et Politique, Payot, Paris. French translation of an excerpt ('Rationality and the Status of the Rationality Principle') from chapter 8 of **1994(r)**. English version contained in **1983(i)**. Spanish translation: **1968(v)**.
- 1967(e)** Zum Thema Freiheit. In *Die Philosophie und die Wissenschaften: Simon Moser zum 65. Geburtstag*, edited by Ernst Oldemeyer, pp. 1-12. Anton Hain, Meisenheim/Glan. Contained in **1994(y)**.
- 1967(f)** *De Armoede van het Historicisme*. Arbeiderspers, Amsterdam. Dutch translation of **1960(c)**.
- 1967(g)** Die Wissenschaft als Institution des Fortschritts. In *Sozialer Wandel: Zivilisation und Fortschritt als Kategorien der soziologischen Theorie, Soziologische Texte 41*, edited by Hans Peter Dreitzel, pp. 305-312. Luchterhand Verlag, Neuwied & Berlin. German translation of an excerpt from section 32 of **1963(p)**.
- 1967(h)** Structural Information and the Arrow of Time. *Nature* **214**, No. 5085, 15th April 1967, p. 322. A continuation of **1956(b)**, **1956(g)**, **1957(d)**, **1957(f)**, **1958(b)**, and **1965(f)**.
- 1967(i)** The Cosmological Origins of Euclidean Geometry. In *Problems in the Philosophy of Mathematics*, Proceedings of the International Colloquium in the Philosophy of Science, London, 1965, volume 1, edited by Imre Lakatos, pp. 18-20. North-Holland Publishing Company, Amsterdam.

- 1967(j)** Selbstbefreiung durch das Wissen. In *Der Sinn der Geschichte*, edited by Leonhard Reinisch, pp. 100-116. C.H. Beck, Munich. Third impression of **1961(d)**.
- 1967(k)** Quantum Mechanics without 'The Observer'. In *Quantum Theory and Reality*, edited by Mario Bunge, *Studies in the Foundations, Methodology, and Philosophy of Science*, pp. 7-44. Springer Verlag, Berlin, Heidelberg, & New York.
- 1967(l)** The Pre-Socratics and the Rationalist Tradition. *ETC: A Review of General Semantics* **24**, No. 2, June 1967, pp. 149-172. A reprint of **1958(e)**.
- 1967(m)** Plato as a Totalitarian Ideologist. In *Totalitarianism: Temporary Madness or Permanent Danger?*, edited by Paul T. Mason, pp. 17-21. D.H. Heath & Company. Boston. A selection from chapters 6, 7, 8 and 9 of **1963(l)**.
- 1967(n)** *Conjeturas y refutaciones: El desarrollo del conocimiento científico*. Editorial Paidós, Buenos Aires. Spanish translation by Néstor Míguez of **1965(c)**.
- 1967(o)** *La lógica de la investigación científica*. Editorial Tecnos, Madrid. Second impression of **1962(r)**.
- 1967(p)** Wieczny Ideal Społeczeństwa Otwartego [*The Open Society and Its Enemies*]. In *Demokracja a ideal społeczny*, edited by Jan Ostaszewski, pp. 373-400. London. 1967. Polish translation of chapter 10 of **1957(h)**.
- 1967(q)** Epistemology and Scientific Knowledge. In *Abstracts of Papers, Third International Congress for Logic, Methodology and Philosophy of Science*, Amsterdam, 1967, p. 72. Abstract of **1968(s)**.
- 1967(r)** Reply to a Critic (1961). In *Plato, Popper and Politics*, edited by Renford Bambrough, pp. 199-219. Heffer & Sons, Cambridge, and Barnes & Noble, New York. A reprint of *Addenda* III and IV of volume I of **1966(a)**.
- 1967(s)** Homage to Norwood Russell Hanson. In *Boston Studies in the Philosophy of Science*, volume 3, 1964-1966, edited by R.S. Cohen & M.W. Wartofsky, pp. xxxiii-xxxiv. D. Reidel Publishing Company, Dordrecht.
- 1967(t)** [Einstein's Influence on My Idea of Science: an interview]. In *Einstein: the Man and his Achievement*, BBC Third Programme Talks, edited by G.J. Whitrow, pp. 23-28. British Broadcasting Corporation, London. Spanish translation: **1969(w)**.
- 1967(u)** *Açık Toplum ve Düşmanlari*. Volume I: *Platon' un Büyüsü*. Turkish Political Science Association of Ankara, USIS, Ankara. Turkish translation of volume I of **1963(l)**. Continued in **1968(l)**.
- 1967(v)** *La Sociedad Abierta y sus Enemigos*. Editorial Paidós, Buenos Aires. A reprint in two volumes of **1957(c)** without footnotes, etc. Unauthorized.

- 1967(w)** *The Open Society and Its Enemies*. Princeton University Press. A reprint of **1966(p)**.
- 1968(a)** *The Logic of Scientific Discovery*. Hutchinson, London. Fifth impression, revised, of **1959(a)**. A Radius Book.
- 1968(b)** *Conjectures and Refutations*. Harper & Row, New York & Evanston. A paperback edition (Harper Torchbook) of **1965(c)**.
- 1968(c)** Remarks on the Problems of Demarcation and of Rationality. In *Problems in the Philosophy of Science*. Proceedings of the International Colloquium in the Philosophy of Science, London, 1965, volume 3, edited by I. Lakatos & A.E. Musgrave, pp. 88-102. North-Holland Publishing Company, Amsterdam.
- 1968(d)** Non-Apparent Depth, Depth, and Pseudo-Depth. *Ibid.*, pp. 139f.
- 1968(e)** Is there an Epistemological Problem of Perception?. *Ibid.*, pp. 163f.
- 1968(f)** On So-called Paradoxes in Physics. *Ibid.*, pp. 202-204.
- 1968(g)** Mathematics, Observation, and Physical Thought. *Ibid.*, pp. 242-244.
- 1968(h)** On Rules of Detachment and So-called Inductive Logic. In *The Problem of Inductive Logic*, Proceedings of the International Colloquium in the Philosophy of Science, London, 1965, volume 2, edited by I. Lakatos, pp. 130-139. North-Holland Publishing Company, Amsterdam.
- 1968(i)** Theories, Experience, and Probabilistic Intuitions. *Ibid.*, pp. 285-303.
- 1968(j)** A Revised Definition of Natural Necessity. *The British Journal for the Philosophy of Science* **18**, No. 4, February 1968, pp. 316-321.
- 1968(k)** Plato. In *International Encyclopedia of the Social Sciences*, edited by Alvin Johnson, David L. Sills, & W. Allen Wallis, volume 12, pp. 159-164. The Macmillan Company and The Free Press, New York, and Collier-Macmillan Ltd, London.
- 1968(l)** *Açık Toplum ve Düşmanları*. Volume II: *Hegel ve Marx*. Turkish Political Science Association of Ankara, USIS, Ankara. Turkish translation of volume II of **1963(l)**. A continuation of **1967(u)**.
- 1968(m)** Chishiki ni taisuru Mittsu no Mikata. In *Kagaku no Tetsugaku* [*Philosophy of Science*], edited, with an Introduction, by Saburo Ichii, pp. 271-308. Heibon-sha, Tokyo. Japanese translation of chapter 3 of **1963(a)**.
- 1968(n)** A Note on Berkeley as Precursor of Mach and Einstein. In *Berkeley's Principles of Human Knowledge*, edited by G.W. Engle & G. Taylor, pp. 90-100. Wadsworth Publishing Company, Belmont, California. A reprint of chapter 6 of **1965(c)**.

- 1968(o)** A Note on Berkeley as Precursor of Mach and Einstein. In *Locke and Berkeley: A Collection of Critical Essays*, edited by C.B. Martin & D.M. Armstrong, pp. 436-449. Doubleday-Anchor Original: Modern Studies in Philosophy, Anchor Books, Doubleday & Co. Inc., New York. A reprint of chapter 6 of **1965(c)**.
- 1968(p)** *The Open Society and Its Enemies*. Volumes I and II. Routledge & Kegan Paul, London. A reprint of **1966(a)**.
- 1968(q)** Birkhoff and von Neumann's Interpretation of Quantum Mechanics. *Nature* **219**, No. 5155, 17th August 1968, pp. 682-685.
- 1968(r)** On the Theory of the Objective Mind. In *Proceedings of the XIVth International Congress of Philosophy* (Vienna 1968), volume 1, pp. 25-53. Herder, Vienna. Contained in **1972(a)**. Spanish translation: **1970(j)**.
- 1968(s)** Epistemology Without a Knowing Subject. In *Proceedings of the Third International Congress for Logic, Methodology and Philosophy of Science* (Amsterdam 1967), edited by B. van Rootselaar & J.F. Staal, pp. 333-373. North-Holland Publishing Company, Amsterdam. Contained in **1972(a)**. Italian translation: **1972(h)**.
- 1968(t)** Emancipation through Knowledge. In *The Humanist Outlook*, edited by A.J. Ayer, pp. 281-296. Pemberton Publishing Company, London. English translation of **1967(j)**. Contained in **1992(t)**.
- 1968(u)** Has History Any Meaning?. In *Science, Faith, and Man: European Thought Since 1914*, edited by W. Warren Wagar, pp. 247-262. Harper Torchbooks, New York. A reprint of chapter 25 of **1963(i)**.
- 1968(v)** La explicación en las ciencias sociales (La racionalidad y el status del principio de racionalidad). *Revista de Occidente* (Madrid) **6**, No. 65, August 1968, pp. 133-146. Spanish translation of **1967(d)**.
- 1968(w)** The Autonomy of Sociology. In *Mill: A Collection of Critical Essays*, edited by J.B. Schneewind, pp. 426-442. Doubleday & Co., New York. A reprint of chapter 14 of **1966(a)**.
- 1968(x)** Utopie und Gewalt. In *Utopie*, edited by Arnhelm Neusüss. *Soziologische Texte* **44**, pp. 313-326. Hermann Luchterhand Verlag, Neuwied & Berlin. German translation of chapter 18 of **1963(a)**.
- 1968(y)** Quantum Theory, Quantum Logic and the Calculus of Probability. In *Proceedings of the XIVth International Congress of Philosophy* (Vienna 1968), volume 3, pp. 307-313. Herder, Vienna, 1969.
- 1968(z)** Summary [of a paper by Paul Weingartner, 'Modal Logics with Two Kinds of Necessity and Possibility']. *Notre Dame Journal of Formal Logic* **9**, No. 2, April 1968, pp. 152f.

- 1968(z₁)** *The Logic of Scientific Discovery*. Harper & Row, New York. A reprint of **1965(e)**.
- 1968(z₂)** The Seeds of Totalitarianism. In *Great Thinkers on Plato*, edited by Barry Gross, pp. 306-320. G.P. Putnam's & Sons, New York. An excerpt from volume I of **1966(p)**.
- 1969(a)** The Moral Responsibility of the Scientist. *Encounter* **32**, 3, March 1969, pp. 52-54, and 56f. Contained in **1994(r)**. German translation: **1970(z)**.
- 1969(b)** Popper's 'Moral Responsibility'. *Encounter* **33**, 1, July 1969, pp. 95f. A reply to Professor Aiken.
- 1969(c)** *Scienza e filosofia: Problemi e scopi della scienza*. Giulio Einaudi Editore, Turin. Italian translation by Mario Trincherò of chapter 3 of **1965(c)**, chapter 3; **1957(i)**; **1966(k)**; **1963(h)**; and chapter 10 of **1965(c)**.
- 1969(d)** Tre orientamenti epistemologici: Il criterio della rilevanza scientifica; La tematica induttiva: congetture e confutazioni; Enunciati probabilistici, teorii, spiegazione. In *Il Neoempirismo (Classici della Filosofia, No. 4)*, edited by Alberto Pasquinelli, pp. 659-751. Unione Tipografico-Editrice Torinese, Turin. Italian translation of chapters 1 and 3 of **1965(c)**.
- 1969(e)** *Logik der Forschung*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Third German edition, revised, with a new Preface and eight new *Addenda*, of **1934(b)** and **1966(e)**.
- 1969(f)** *Das Elend des Historizismus*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Second edition of **1965(g)**.
- 1969(g)** *The Poverty of Historicism*. Routledge & Kegan Paul, London. Fifth impression of **1961(b)**.
- 1969(h)** *Conjectures and Refutations: The Growth of Scientific Knowledge*. Routledge & Kegan Paul, London. Third edition, with two new *Addenda*, of **1963(a)** and **1965(c)**.
- 1969(i)** *Conjectures and Refutations: The Growth of Scientific Knowledge*. Routledge & Kegan Paul, London. A paperback edition of **1969(h)**.
- 1969(j)** A Pluralist Approach to the Philosophy of History. In *Roads to Freedom: Essays in Honour of F.A. von Hayek*, edited by E. Streissler, pp. 181-200. Routledge & Kegan Paul, London. Contained in **1994(r)**.
- 1969(k)** The Aim of Science. In *Contemporary Philosophy: A Survey*, edited by R. Klibansky, III: *Metaphysics, Phenomenology, Language and Structure*, pp. 129-142. La Nuova Italia Editrice, Firenze. A revised version of **1957(i)**. Contained in **1972(a)**.

- 1969(l)** The Hypothetical-Deductive Method and the Unity of Social and Natural Science. In *The Nature and Scope of Social Science: A Critical Anthology*, edited by L.I. Krimerman, pp. 47-53. Appleton-Century-Crofts, New York. A reprint of section 29 of **1961(b)**.
- 1969(m)** Die Logik der Sozialwissenschaften. In *Der Positivismusstreit in der deutschen Soziologie (Soziologische Texte 58)*, edited by H. Maus & F. Fürstenberg, pp. 103-123. Hermann Luchterhand Verlag, Neuwied & Berlin. A reprint of **1962(k)**.
- 1969(n)** Epistemology Without a Knowing Subject. *Philosophy Today*, No. 2, edited by J.H. Gill, pp. 225-277. The Macmillan Company, New York, and Collier-Macmillan Ltd, London. A revised reprint of **1968(s)**.
- 1969(o)** *Misère de l'historicisme*. Librairie Plon, Paris. A reprint of **1956(a)**.
- 1969(p)** The Autonomy of Sociology. In *Mill: A Collection of Critical Essays*, edited by J.B. Schneewind, pp. 426-442. University of Notre Dame Press, Notre Dame, Indiana (hardbound), and Macmillan & Co. Ltd, London (paperback). A reprint of **1968(w)**.
- 1969(q)** *The Open Society and Its Enemies*. Volumes I and II. Routledge & Kegan Paul, London. A reprint of **1966(a)**.
- 1969(r)** *The Open Society and Its Enemies*. Volumes I and II. Routledge Paperbacks, London. A reprint of **1966(a)**.
- 1969(s)** *The Poverty of Historicism*. Routledge Paperbacks, London. Sixth impression of **1961(b)**.
- 1969(t)** Über die Unwiderlegbarkeit philosophischer Theorien einschliesslich jener, welche falsch sind. In *Club Voltaire: Jahrbuch für kritische Aufklärung*, I, edited by Gerhard Szczesny, pp. 271-279. Rowohlt Taschenbuch Verlag, Reinbek bei Hamburg. A reprint of **1963(k)**.
- 1969(u)** Of Clouds and Clocks. *Architectural Design* **39**, 9, September 1969, pp. 491f. An excerpt from **1966(f)** made by the Editor.
- 1969(v)** Einstein's Influence on my view of Science: An Interview. In *Einstein: The Man and his Achievement*, edited by G.J. Whitrow; edited with notes by H. Nakayama & T. Fuse, pp. 20-22. Asahi Press, Tokyo. An abridged version of **1967(t)**.
- 1969(w)** Einstein's Influence on my view of Science: An Interview. In *Einstein, el hombre y su obra*, edited by G.J. Whitrow, pp. 46-53. Siglo XXI Editores, S.A., Mexico. Spanish translation by Julieta Campos of **1967(t)**.
- 1969(x)** Previsão e Profecia nas Ciências Sociais. In *Theories of History*, edited by P. Gardiner. Fundação Calouste Gulbenkian, Lisbon. Portuguese translation of

1948(d).

- 1969(y)** Two Letters to Hans Lenk [in German]. *Conceptus* **3**, 1969, pp. 43-46.
- 1970(a)** Dialectical Methodology. *The Times Literary Supplement* **69**, No. 3552, 26th March 1970, pp. 338f.
- 1970(b)** Has History Any Meaning?. In *The Nature of Historical Enquiry*, edited by L.M. Marsak, pp. 155-157. Holt, Rinehart & Winston, New York. An excerpt from chapter 25 of **1963(i)**.
- 1970(c)** The Open Society and Its Enemies. In *Philosophical Perspectives for Education*, edited by Carlton H. Bowyer, pp. 261-270. Scott, Foresman & Co., Glenview, Illinois. A reprint of chapter 22 of **1966(a)**.
- 1970(d)** Plato, *Timaeus* 54E-55A. *The Classical Review* **XX**, N.S., No. 1, March 1970, pp. 4f.
- 1970(e)** *Logica della scoperta scientifica: Il carattere autocorrettivo della scienza*. Giulio Einaudi Editore, Turin. Italian translation by Mario Trincherio of **1968(a)**, with a new Preface.
- 1970(f)** Adevăr, raționalitate și progresul cunoașterii științifice. In *Logica Științei*, edited by Gh. Enescu & Cornel Popa, pp. 99-155. Editura Politică, Bucharest. Romanian translation of chapter 10 of **1963(a)**.
- 1970(g)** A Note on Berkeley as Precursor of Mach and Einstein. In *Berkeley: Principles of Human Knowledge*, edited by Colin Murrey Turbayne, pp. 129-144. The Bobbs-Merrill Company, Indianapolis & New York. A reprint of chapter 6 of **1965(c)**.
- 1970(h)** Eine objektive Theorie des historischen Verstehens. *Schweizer Monatshefte* **50**, No. 3, June 1970, pp. 207-215.
- 1970(i)** Palabras de apertura del simposio. In *Simposio de Burgos: Ensayos de Filosofía de la Ciencia en torno a la obra de Sir Karl R. Popper*, edited by Miguel Boyer, Víctor Sánchez de Zavala & Pedro Schwartz, pp. 13f. Editorial Tecnos, Madrid.
- 1970(j)** Sobre la teoría de la inteligencia objetiva. *Ibid.*, pp. 202-237. Spanish translation by Víctor Sánchez de Zavala of **1968(r)**.
- 1970(j₁)** [Remarks on Norman Barracrough's paper 'El principio de simetría como origen de una realidad matematicamente estructurada']. *Ibid.*, pp. 37f.
- 1970(j₂)** [Remarks on Víctor Sánchez de Zavala's paper 'Sobre las ciencias de "complexos"']. *Ibid.*, pp. 64-69.
- 1970(j₃)** [Remarks on Luis Angel Rojo's paper 'El método empírico y el conocimiento económico']. *Ibid.*, pp. 110-116.

- 1970(j₄)** [Remarks on Pedro Schwartz's paper 'El individualismo metodológico y los historiadores']. *Ibid.*, pp. 150-152.
- 1970(k)** The Sociology of Knowledge. In *The Sociology of Knowledge: A Reader*, edited by James E. Curtis & John W. Petras, pp. 649-660. Praeger Publishers, New York. A reprint of chapter 23 of **1966(a)**.
- 1970(l)** A Realist View of Logic, Physics, and History. In *Physics, Logic and History*, edited by Wolfgang Yourgrau & Allen D. Breck, pp. 1-30 and 35-37. Plenum Press. New York & London. Contained in **1972(a)**.
- 1970(m)** [Remarks on André Mercier's paper 'Knowledge and Physical Reality']. *Ibid.*, pp. 56f.
- 1970(n)** [Remarks on Willard V.O. Quine's paper 'Existence']. *Ibid.*, pp. 98 and 103.
- 1970(o)** [Remarks on Jaakko Hintikka's paper 'On Semantic Information']. *Ibid.*, pp. 169f.
- 1970(p)** [Remarks on Czesław Lejewski's paper 'Quantification and Ontological Commitment']. *Ibid.*, p. 189.
- 1970(q)** [Remarks on Jean-Pierre Vigiér's paper 'Possible Internal Subquantum Motions of Elementary Particles']. *Ibid.*, pp. 199f.
- 1970(r)** [Remarks on Alfred Landé's paper 'Non-Quantal Foundations of Quantum Mechanics']. *Ibid.*, p. 308.
- 1970(s)** [Remarks on György Ránki's paper 'Some Problems of the Connection Between Technical Development and Economic History']. *Ibid.*, pp. 316f.
- 1970(t)** Back to the Presocratics. In *Studies in Presocratic Philosophy*, volume I, edited by R.E. Allen & David J. Furley, pp. 130-153. Routledge & Kegan Paul, London. A reprint of chapter 5 of **1963(a)**.
- 1970(u)** Selbstbefreiung durch das Wissen. In *Der Sinn der Geschichte*, edited by Leonhard Reinisch, pp. 100-116. C.H. Beck, Munich. Fourth impression of **1961(d)**.
- 1970(v)** Correction. *Synthese* 21, No. 1, March 1970, p. 107. A correction to **1963(g)** and **1964(f)**.
- 1970(w)** Normal Science and Its Dangers. In *Criticism and the Growth of Knowledge*, Proceedings of the International Colloquium in the Philosophy of Science, London. 1965, volume 4, edited by I. Lakatos & A.E. Musgrave, pp. 51-58. Cambridge University Press, Cambridge. German translation: **1974(z₁₀)**; Spanish: **1975(z₅)**.
- 1970(x)** Normal Science and Its Dangers. A paperback edition of **1970(w)**.

- 1970(y)** Three Views Concerning Human Knowledge. In *The Foundations of Knowledge*, Central Issues in Philosophy Series, edited by C. Landesman, pp. 93-123. Prentice-Hall, Englewood Cliffs, N.J. A reprint of chapter 3 of **1965(c)**.
- 1970(z)** Die moralische Verantwortlichkeit des Wissenschaftlers. *Schweizer Monatshefte* **50**, No. 7, October 1970, pp. 561-570. A revised translation of **1969(a)**.
- 1970(z₁)** Einstein: Early Years. In *Physics and Man*, edited by Robert Karplus, pp. 47-52. W.A. Benjamin Inc., New York. A reprint of **1967(t)**.
- 1970(z₂)** [Contributions to a discussion of a paper by A. Grünbaum]. In *Induction, Physics and Ethics: Proceedings and Discussions of the 1968 Salzburg Colloquium in the Philosophy of Science*, edited by Paul Weingartner & Gerhard Zecha, pp. 167f. D. Reidel Publishing Company, Dordrecht.
- 1970(z₃)** The Moral Responsibility of the Scientist. *Ibid.*, pp. 329-336. A reprint of **1969(a)**.
- 1970(z₄)** Reason or Revolution?. *Archives européennes de sociologie* **XI**, 2, November 1970, pp. 252-262. An expanded version of **1970(a)**. Contained in **1994(r)**. Hungarian translation: **1976(j)**; French: **1979(z₁₁)**.
- 1970(z₅)** On the Philosophy of Bertrand Russell: Karl Popper, Peter Strawson and Geoffrey Warnock Take Part in a Discussion Chaired by Bryan Magee. *The Listener* **83**, No. 2146, 14th May 1970, pp. 633-636 and 638; and *ibid.*, No. 2147, 21st May 1970, pp. 685f. A radio broadcast. Continued in **1970(z₆)**.
- 1970(z₆)** Russell, Popper, and Oxford. *The Listener* **83**, No. 2149, 4th June 1970, p. 753. A continuation of **1970(z₅)**.
- 1970(z₇)** Die Logik der Sozialwissenschaften. In *Der Positivismusstreit in der deutschen Soziologie (Soziologische Texte 58)*, edited by H. Maus & F. Fürstenberg, pp. 103-123. Hermann Luchterhand Verlag, Neuwied & Berlin. Second impression of **1969(m)**.
- 1970(z₈)** A Pluralist Approach to the Philosophy of History. In *Roads to Freedom: Essays in Honour of F.A. von Hayek*, edited by E. Streissler, pp. 181-200. Routledge & Kegan Paul, London. A reprint of **1969(j)**.
- 1970(z₉)** *Der Zauber Platons. Die offene Gesellschaft und ihre Feinde*, volume I. Francke Verlag, Berne & Munich. Second edition of **1957(k)**.
- 1970(z₁₀)** *Falsche Propheten: Hegel, Marx und die Folgen. Die offene Gesellschaft und ihre Feinde*, volume II. Francke Verlag, Berne & Munich. Second edition of **1958(i)**.
- 1970(z₁₁)** Unsigned obituary notice for Professor Rudolf Carnap. *The Times*, No. 57973, 17th September 1970, p. 12.

- 1971(a)** Conversations with Philosophers. Sir Karl Popper talks about some of his basic ideas with Bryan Magee. *The Listener* **85**, No. 2180, 7th January 1971, pp. 8-12. Slightly expanded in **1971(l)**. Danish translation: **1972(n)**; Italian: **1981(z₂)**.
- 1971(b)** *The Open Society and Its Enemies*. Princeton Paperbacks Nos. 219 and 220, Princeton, N.J. A paperback reprint of **1966(p)**.
- 1971(c)** Nachrevolutionäres Risiko – oder freie, offene Gesellschaft? Gegensätzliche politische Entwürfe von Herbert Marcuse und Karl Popper zum Thema 'Revolution oder Reform'. *Süddeutsche Zeitung am Wochenende*, No. 14, 16th/17th January 1971, pp. 1f. An abridged version of **1971(g)**.
- 1971(d)** *Samfunnsvitenskap og profeti [The Poverty of Historicism]*. Gyldendal Norsk Forlag, Oslo. Norwegian translation of **1961(b)**.
- 1971(e)** Die Logik der Sozialwissenschaften. In *Der Positivismusstreit in der deutschen Soziologie (Soziologische Texte 58)*, edited by H. Maus & F. Fürstenberg, pp. 103-123. Hermann Luchterhand Verlag, Neuwied & Berlin. Third impression of **1969(m)**.
- 1971(f)** Utopia and Violence. In *The Good Society*, edited by Anthony Arblaster & Steven Lukes, pp. 380-386. Methuen & Co., London. An abridged reprint of **1948(a)**.
- 1971(g)** Revolution oder Reform?. In *Revolution oder Reform? Herbert Marcuse und Karl Popper – Eine Konfrontation*, edited by Franz Stark, pp. 3, 9f., 22-29, 34-39, and 41. Kösel-Verlag, Munich. For an abridged version see **1971(c)**. English version contained in **1972(f)** and **1972(m)**. Dutch translation: **1971(w)**; Norwegian: **1972(s)**; Italian: **1972(t)**, **1977(c)**, **1981(z_{2s})**; Japanese: **1972(u)**; Icelandic: **1973(t)**; Spanish: **1973(p)** (partial): **1976(l)**.
- 1971(h)** Science. Problems, Aims, Responsibilities. In *Science and Public Policy*, edited by Louise B. Young & William J. Trainor, pp. 20-35. Oceana Publications, New York. An excerpt from **1963(h)**.
- 1971(i)** Conjectural Knowledge: My Solution of the Problem of Induction. *Revue Internationale de Philosophie* **25**, fasc. 1-2, *Hypothèse et Confirmation*, pp. 167-197. Contained in **1972(a)**.
- 1971(j)** (with P.B. Medawar) Einstein's Greatness. *The Sunday Telegraph*, 23rd May 1971, p. 11. A letter to the Editor.
- 1971(k)** Einstein's Error. *The Listener* **86**, 1971, No. 2208, 22nd July 1971, p. 115. A letter.
- 1971(l)** Conversation with Karl Popper. In *Modern British Philosophy*, edited by Bryan Magee, pp. 66-82. Secker & Warburg, London, and St. Martin's Press, New York. A slightly expanded reprint of **1971(a)**.

- 1971(m)** The Philosophy of Russell. II: Discussion among Karl Popper, Peter Strawson and Geoffrey Warnock. In *Modern British Philosophy*, edited by Bryan Magee, pp. 131-149. Secker & Warburg, London, and St. Martin's Press, New York. A reprint of **1970(z₅)**.
- 1971(n)** Particle Annihilation and the Argument of Einstein, Podolsky, and Rosen. In *Perspectives in Quantum Theory: Essays in Honor of Alfred Landé*, edited by Wolfgang Yourgrau & Alwyn van der Merwe, pp. 182-198. M.I.T. Press, Cambridge, Massachusetts, & London.
- 1971(o)** *Logik der Forschung*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Fourth German edition, revised and slightly enlarged, of **1934(b)** and **1966(e)**.
- 1971(p)** *Das Elend des Historizismus*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Third German edition, revised, of **1965(g)**.
- 1971(q)** The Moral Responsibility of the Scientist. *Bulletin of Peace Proposals* (Oslo) **2**, 3, 1971, pp. 279-283. A revised version of **1969(a)**.
- 1971(r)** Wider die groen Worte: Ein Plädoyer für intellektuelle Redlichkeit. *Die Zeit* (Hamburg) **26**, 39, 24th September 1971, p. 8. An excerpt from **1971(u)**. Unauthorized. Original version contained in **1984(b)**.
- 1971(s)** The Moral Responsibility of the Scientist. In *Proceedings of the XIVth International Congress of Philosophy* (Vienna 1968), volume 6, pp. 489-496. Herder, Vienna. First published as **1969(a)**.
- 1971(t)** *Kagakuteki Hakken no Ronri* [*The Logic of Scientific Discovery*], volume I. Koseisha-Koseikaku, Tokyo. First part of Japanese translation by I.T. Mori & G. Ouchi of **1959(a)**. Contains chapters I-VIII of **1959(a)**, the Preface to **1966(e)**, and the first three new *Addenda* to **1969(e)**. Continued in **1972(v)**.
- 1971(u)** Philosophische Selbstinterpretation und Polemik gegen die Dialektiker. In Claus Grossner, *Verfall der Philosophie: Politik deutscher Philosophen*, pp. 278-289. Christian Wegner Verlag, Reinbek bei Hamburg. Original version of **1971(r)**.
- 1971(v)** The Nature of Philosophical Problems and their Roots in Science. In *Plato's Meno*, edited by Malcolm Brown, pp. 128-179. The Bobbs-Merrill Co., New York. A reprint of chapter 2 of **1965(c)**.
- 1971(w)** Sociale revolutie of Sociale hervorming?. In *Herbert Marcuse/Karl Popper, Sociale revolutie of sociale hervorming – een confrontatie*, edited by Franz Stark, pp. 5, 9-11, 22-29, 34-38, and 40. Het Werelvenster, Baarn, The Netherlands. Dutch translation of **1971(g)**.
- 1971(x)** Kritik des Sozialutopismus. In *Wort und Sinn: Lesebuch für den Deutsch-unterricht*, edited by Karl Ernst Jeismann & Gustav Muthmann, pp. 221-231. Ferdinand Schöningh, Paderborn. A reprint of chapter 9 of **1957(k)**.

- 1971(y)** Utopia and Violence. In *The Good Society*, edited by Anthony Arblaster & Steven Lukes, pp. 380-386. Harper & Row, New York. American edition of **1971(f)**.
- 1971(z)** Revolution oder Reform?. In *Revolution oder Reform? Herbert Marcuse und Karl Popper – Eine Konfrontation*, edited by Franz Stark, pp. 3, 9f., 22-29, 34-39, and 41. Kösel-Verlag, Munich. Second impression of **1971(g)**.
- 1971(z₁)** Die Logik der Sozialwissenschaften. In *Methodendiskussion: Arbeitsbuch zur Literaturwissenschaft*, edited by Jürgen Hauff et al., pp. 71-79. Athenäum Fischer Taschenbuch Verlag, Frankfurt am Main. An excerpt from **1979(m)**.
- 1971(z₂)** *La lógica de la investigación científica*. Editorial Tecnos, Madrid. Third impression of **1962(r)**.
- 1971(z₃)** *Faghre Tarichigari*. Kharazmi, Tehran. Persian translation by A. Aram of **1961(b)**.
- 1972(a)** *Objective Knowledge: An Evolutionary Approach*. Clarendon Press, Oxford.
- Further impressions, editions, and translations of **1972(a)** into other languages are indicated in the year in which they are published.
- 1972(b)** *The Poverty of Historicism*. Routledge & Kegan Paul, London. Seventh impression of **1961(b)**.
- 1972(c)** *The Logic of Scientific Discovery*. Radius Books, Hutchinson, London. Sixth impression, revised and enlarged, of **1959(a)**. A paperback edition.
- 1972(d)** *Conjectures and Refutations*. Routledge & Kegan Paul, London. Fourth edition, revised, of **1963(a)**.
- 1972(e)** *Congetture e confutazioni*. Società editrice il Mulino, Bologna. Italian translation by Giuliano Pancaldi of **1969(h)**.
- 1972(f)** On Reason and The Open Society. *Encounter* **38**, 5, May 1972, pp. 13-18. Revised and augmented English translation of **1971(g)**.
- 1972(g)** Revolution oder Reform?. In *Revolution oder Reform? Herbert Marcuse und Karl Popper – Eine Konfrontation*, edited by Franz Stark, pp. 3, 9f., 22-29, 34-39, and 41. Kösel-Verlag, Munich. Third impression of **1971(g)**.
- 1972(h)** *Epistemologia, Razionalità e Libertà*, Armando Armando Editore, Rome. Italian translation of **1966(f)** and **1968(s)**.
- 1972(i)** Die Zielsetzung der Erfahrungswissenschaft. In *Theorie und Realität*, edited by Hans Albert, pp. 29-41. J.C.B. Mohr (Paul Siebeck), Tübingen. A reprint of **1964(d)**.

- 1972(j)** Naturgesetze und theoretische Systeme. In *Theorie und Realität*, edited by Hans Albert, pp. 43-58. J.C.B. Mohr (Paul Siebeck), Tübingen, 1964. A reprint of **1964(e)**.
- 1972(k)** Utopia and Violence. In *The Good Society*, edited by Anthony Arblaster & Steven Lukes, pp. 380-386. Harper & Row, New York. A paperback edition of **1971(y)**.
- 1972(l)** Die Logik der Sozialwissenschaften. In *Der Positivismusstreit in der deutschen Soziologie (Soziologische Texte 58)*, edited by H. Maus & F. Fürstenberg, pp. 103-123. Hermann Luchterhand Verlag, Neuwied & Berlin. Fourth impression of **1969(m)**.
- 1972(m)** On Reason and The Open Society – A Conversation. *Congressional Record* (Washington) **118**, No. 83, 23rd May 1972, pp. E 5625-E 5627. A reprint of **1972(f)**.
- 1972(n)** Den kritiske metode til eliminering af fejltagelser: En dialog om sandheden. *Berlingske Aftenavis: Weekendavisen* **224**, 21, 2nd June 1972, pp. 14f. Danish translation of **1971(l)**.
- 1972(o)** Basisprobleme. In *Ideologiekritik in der Erziehungswissenschaft*, edited by Heinrich Kanz, pp. 37-49, 158-160. Akademische Verlagsgesellschaft, Frankfurt am Main. A reprint of chapter 5 of **1969(e)**.
- 1972(p)** The Open Society and Its Enemies: [Utopian Engineering]. In *Experimentation with Human Beings*, edited by Jay Katz, pp. 266-268. Russell Sage Foundation, New York. An excerpt from chapter 9 of **1966(a)**.
- 1972(q)** Science: Conjectures and Refutations. In *Challenges to Empiricism*, edited by Harold Morick, pp. 127-160. Wadsworth Publishing Company Inc., Belmont, California. A reprint of chapter 1 (excluding appendix) of **1969(h)**.
- 1972(r)** *Scienza e filosofia: Problemi e scopi della scienza*. Giulio Einaudi Editore, Turin. Second impression of **1969(c)**.
- 1972(s)** Revolusjon eller reform?. In *Herbert Marcuse og Karl Popper: en konfrontasjon*, edited by Franz Stark. Dreyers Forlag, Oslo. Norwegian translation of **1971(g)**.
- 1972(t)** Una conversazione con Karl Popper. *Biblioteca della Libertà* **IX**, No. 39, July-August 1972, pp. 51-62. Italian translation of **1972(f)**.
- 1972(u)** Kakumei Kairō Ka [Revolution or Reason]. *Asahi Jaanaru* **14**, No. 20, 28th July 1972, pp. 43-50. Japanese translation of **1971(g)**.
- 1972(v)** *Kagakuteki Hakken no Ronri [The Logic of Scientific Discovery]*, volume II. Koseisha-Koseikaku, Tokyo. Second part of Japanese translation by I.T. Mori & G. Ouchi of **1959(a)**. Contains chapters IX-X and *Appendices*, presumably

of **1968(a)**, and also a bibliography of Popper's writings to 1971, and some secondary literature, also to 1971, including Japanese secondary literature. Continuation of **1971(t)**.

- 1972(w)** Die Logik der Sozialwissenschaften. In *Methodendiskussion: Arbeitsbuch zur Literaturwissenschaft*, edited by Jürgen Hauff et al., pp. 71-78. Athenäum Fischer Taschenbuch Verlag, Frankfurt am Main. Second edition of **1971(z₁)**.
- 1972(x)** La logica delle scienze sociali. In *Dialettica e positivismo in sociologia*, edited by H. Maus & F. Fürstenberg, pp. 105-123. Giulio Einaudi, Turin. Italian translation by Anna Marietti Solmi of **1962(k)**.
- 1972(y)** Normal Science and Its Dangers. In *Criticism and the Growth of Knowledge*, Proceedings of the International Colloquium in the Philosophy of Science, London, 1965, volume 4, edited by I. Lakatos & A.E. Musgrave, pp. 51-58. Cambridge University Press, Cambridge. A reprint of **1970(w)**.
- 1973(a)** Indeterminism is Not Enough. A Philosophical Essay. *Encounter* (London) **40**, No. 4, April 1973, pp. 20-26.
- 1973(b)** The Poverty of Historicism. In *Modes of Individualism and Collectivism*, edited by John O'Neill, pp. 68-87. Heinemann Educational Books, London. A reprint of sections 29-32 of **1961(b)**.
- 1973(c)** *Logica naučnog otkrića*. Nolit, Belgrade. Serbo-Croat translation by Staniša Novaković of **1968(a)** and addenda *iv and *v of **1971(o)**.
- 1973(d)** *La logique de la découverte scientifique*. Payot, Paris. French translation by Nicole Thyssen-Rutten & Philippe Devaux of **1968(a)**. Contains a Preface by Jacques Monod, and material translated from **1970(e)** and **1971(o)**.
- 1973(e)** *Objective Knowledge: An Evolutionary Approach*. Clarendon Press, Oxford. Second impression, with some corrections, of **1972(a)**.
- 1973(f)** *The Open Society and Its Enemies*. Volumes I and II. Routledge & Kegan Paul, London. A reprint of **1966(a)**.
- 1973(g)** *Logik der Forschung*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Fifth German edition of **1934(b)** and **1966(e)**.
- 1973(h)** *La lógica de la investigación científica*. Editorial Tecnos, Madrid. Fourth impression of **1962(r)**.
- 1973(i)** *Objektive Erkenntnis: Ein evolutionärer Entwurf*, Hoffman & Campe, Hamburg. German translation by Hermann Vetter of **1972(a)**.
- 1973(j)** *La miseria del historicismo*. Taurus Ediciones, Alianza Editorial, Madrid. Spanish translation by Pedro Schwartz of **1957(g)** (and possibly **1960(c)**).

- 1973(k)** *Kritisk rationalisme: Udvalgte essays om videnskab og samfund*. Nyt Nordisk Forlag/Arnold Busck, Copenhagen. Danish translation by Knud Haakonssen & Niels Chr. Stefansen of eight essays published between 1945 and 1972.
- 1973(l)** *Jivuu Shaki No Tetsugaku To Sono Ronteki [The Open Society and Its Enemies]*. Sekai Shisosha, Tokyo. Japanese translation in two volumes, probably of **1969(q)**. The text of the work is contained in volume I of the translation, the notes in volume II.
- 1973(m)** *La società aperta e i suoi nemici*. Volumes I and II. Armando Armando Editore, Rome. Italian translation by Renato Pavetto of **1969(q)**. Edited by Dario Antiseri.
- 1973(n)** *Die offene Gesellschaft und ihre Feinde*. Francke Verlag, Berne & Munich. Third edition of **1957(k)** and **1958(i)**.
- 1973(o)** A Biographical Sketch. In *Empiricism and Sociology: Otto Neurath*, edited by Marie Neurath & Robert S. Cohen, pp. 51-56. D. Reidel Publishing Company, Dordrecht & Boston.
- 1973(p)** Problemas de la Sociedad Contemporánea. *Orientación Económica* (Caracas) **41**, September 1973, pp. 25-27. Spanish translation of excerpts from **1971(g)**.
- 1973(q)** Zur Theorie der Politik: Bemerkungen zu einer Arbeit bei Heiner Flohr. *Rechtstheorie, Zeitschrift für Logik, Methodenlehre, Kybernetik und Theorie des Rechts* **4**, No. 1, 1973, pp. 88f.
- 1973(r)** Conversation with Karl Popper. In *Modern British Philosophy*, by Bryan Magee, pp. 88-107. Granada Publishing. A reprint of **1971(l)**.
- 1973(s)** The Philosophy of Russell: II; Discussion Among Karl Popper, Peter Strawson and Geoffrey Warnock. In *Modern British Philosophy*, by Bryan Magee, pp. 165-186. Granada Publishing. A reprint of **1971(m)**.
- 1973(t)** Um skynsemi og opid þjóðfélag. Úr brezka tímaritinu. *Lesbók Morgunbladsins*, 24th February 1973, pp. 1-3. Icelandic translation of **1972(f)**.
- 1973(u)** The Logic of Scientific Discovery. In *Truth* (Problems of Philosophy, units 9-10, Arts: A Third Level Course), edited by Susan Wilson, pp. 15f. The Open University Press. An excerpt from **1972(c)**.
- 1973(v)** L'emancipazione attraverso la conoscenza. *Controcorrente* **5**, No. 2, 1973, pp. 16-18. Italian translation of **1968(t)**, **1967(j)**.
- 1973(w)** Einstein's Influence on My View of Science: An Interview. In *Einstein: the Man and his Achievement*, edited by G.J. Whitrow. Dover, New York. A reprint of **1967(t)**.

- 1973(x)** *Previsão e Profecia nas Ciências Sociais*. In *Theories of History*, edited by P. Gardiner. Fundação Calouste Gulbenkian, Lisbon. A reprint of **1969(x)**.
- 1973(y)** *The Open Society and Its Enemies*. Pirated edition, produced in Taiwan, of **1956(c)**.
- 1974(a)** *Objective Knowledge: An Evolutionary Approach*. Clarendon Press, Oxford. Third impression of **1972(a)**.
- 1974(b)** Autobiography of Karl Popper. In *The Philosophy of Karl Popper, The Library of Living Philosophers*, volume I, edited by Paul Arthur Schilpp, pp. 1-181. The Open Court Publishing Co., La Salle, Illinois.
- Further impressions, editions, and translations of **1974(b)** into other languages are indicated in the year in which they are published.
- 1974(c)** Replies to my Critics. In *The Philosophy of Karl Popper, The Library of Living Philosophers*, volume II, edited by Paul Arthur Schilpp, pp. 959-1197. The Open Court Publishing Co., La Salle, Illinois.
- 1974(d)** *De armoede van het historicisme*. Uitgeverij Het Spectrum, Utrecht/Antwerp. A reprint of **1967(f)** as an Aula-boeken paperback.
- 1974(e)** *Objektive Erkenntnis: Ein evolutionärer Entwurf*. Hoffman & Campe, Hamburg. Second impression of **1973(i)**.
- 1974(f)** *Conocimiento objetivo: Un enfoque evolucionista*. Editorial Tecnos, Madrid. Spanish translation by Carlos Solis Santos of **1972(a)**.
- 1974(g)** *Logica della scoperta scientifica*. Giulio Einaudi Editore, Turin. Second impression of **1970(e)**.
- 1974(h)** *Scienza e filosofia: Problemi e scopi della scienza*. Giulio Einaudi Editore, Turin. Third impression of **1969(c)**.
- 1974(i)** Selbstbefreiung durch das Wissen. In *Der Sinn der Geschichte*, edited by Leonhard Reinisch, pp. 100-116. C.H. Beck, Munich. Fourth impression of **1961(d)**.
- 1974(j)** Die Logik der Sozialwissenschaften. In *Der Positivismusstreit in der deutschen Soziologie (Soziologische Texte 58)*, edited by H. Maus & F. Fürstenberg, pp. 103-123. Luchterhand Verlag, Darmstadt & Neuwied. Fifth impression of **1969(m)**.
- 1974(k)** Selbstbefreiung durch Wissen. *Sonntagsbeilage zur Stuttgarter Zeitung, Stuttgarter Zeitung*, No. 92, 20th April 1974, p. 49. An abridged version, with new introduction, of **1954(b)**.

- 1974(l)** Immanuel Kant: Der Philosoph der Aufklärung. In *Immanuel Kant zu Ehren*, edited by J. Kopper & R. Malter. Suhrkamp Taschenbuch Wissenschaft **61**, Frankfurt am Main. A reprint of **1954(b)**.
- 1974(m)** *A sociedade aberta e seus inimigos*. Itatiaia, Belo Horizonte, and Editora da Universidade de São Paulo. São Paulo. New edition of **1959(c)**.
- 1974(n)** *The Logic of Scientific Discovery*. Hutchinson, London. Seventh impression of **1959(a)**.
- 1974(o)** Causality, Explanation, and the Deduction of Predictions. In *Philosophical Problems of Causation*, edited by Tom L. Beauchamp, pp. 39f. Dickenson Publishing Company, Encino & Belmont, California. A reprint of section 12 of **1972(c)**.
- 1974(p)** A Note on Natural Laws and so-called 'Contrary to Fact Conditionals'. *Ibid.*, pp. 42f. A reprint of **1949(a)**.
- 1974(q)** Kneale and Natural Laws. *Ibid.*, pp. 50f. A reprint of sections (14)-(16) of *Appendix *x* of **1972(c)**.
- 1974(r)** A Revised Definition of Natural Necessity. *Ibid.*, pp. 66f. A reprint of **1968(j)**.
- 1974(s)** Addendum 1968. *Ibid.*, pp. 72f. A reprint of p. 441 of *Appendix *x* of **1972(c)**.
- 1974(t)** Falsifiability and Freedom. In *Reflexive Water*, edited by Fons Elders, pp. 69-131. Souvenir Press, London.
- 1974(u)** Kagaku-Suisoku To Ronpaku. In *Hihanteki Gōrishugi [Critical Rationalism]*, Modern Thought Series, volume 6, pp. 5-52. Diamond Inc., Tokyo. A Japanese translation by Junichi Aomi et al. of chapter 1 of **1969(h)**.
- 1974(v)** Benshōhō To Wa Nani Ka. *Ibid.*, pp. 57-90. Japanese translation by Junichi Aomi et al. of chapter 15 of **1969(h)**.
- 1974(w)** Bemerkungen zu Roehles Arbeit und zur Axiomatik. *Conceptus* **8**, No. 24, 1974, pp. 53-56.
- 1974(x)** Truth, Rationality, and the Growth of Knowledge. In *Philosophical Problems of Science and Technology*, edited by Alexander C. Michalos, pp. 76-117. Allyn & Bacon Inc., Boston, Mass. A reprint of chapter 10 of **1972(d)**.
- 1974(y)** Epistemologia fără subiect cunoscător. In *Epistemologie orientări contemporane*, Materialismul dialectic și științele moderne, volume XV, edited by Ilie Pârvu, pp. 69-120. Editura politică, Bucharest. Romanian translation of **1968(s)**.

- 1974(z)** *Das Elend des Historizismus*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Fourth German edition of **1965(g)**.
- 1974(z₁)** *Kyakkanteki Chishiki*. Mokutakusha, Tokyo. Japanese translation by H. Mori of **1972(a)**.
- 1974(z₂)** Scientific Reduction and the Essential Incompleteness of All Science. In *Studies in the Philosophy of Biology*, edited by F.J. Ayala & T. Dobzhansky, pp. 259-284. Macmillan, London. Contained in **1982(a)**. German translation contained in **1994(y)**.
- 1974(z₃)** Normal Science and Its Dangers. In *Criticism and the Growth of Knowledge*, Proceedings of the International Colloquium in the Philosophy of Science, London, 1965, volume 4, edited by I. Lakatos & A.E. Musgrave, pp. 51-58. Cambridge University Press, Cambridge. A reprint, with corrections, of **1970(w)**.
- 1974(z₄)** *Conjectures and Refutations*. Routledge & Kegan Paul, London. Second impression of **1972(d)**.
- 1974(z₅)** *Avoin Yhteiskunta Ja Sen Viholliset [The Open Society and Its Enemies]*. Kustannusosakeyhtiö Otava, Helsinki. Finnish translation by Paavo Löppönen of **1966(a)**.
- 1974(z₆)** Conjecture and Refutation; The Nature of Observations; and The Myth of Induction. In *Readings in Science Education*, Science Teacher Education Project, edited by Edgar Jenkins & Richard Whitfield, pp. 16f. McGraw-Hill, London & New York. Excerpts from pp. 35-37, 43, and 53 respectively of **1969(h)**.
- 1974(z₇)** *The Poverty of Historicism*. Routledge & Kegan Paul, London. Eighth impression of **1961(b)**.
- 1974(z₈)** *The Open Society and Its Enemies*. Volumes I and II. Routledge & Kegan Paul, London. A reprint of **1966(a)**.
- 1974(z₉)** Some Philosophical Comments on Tarski's Theory of Truth. In *Proceedings of the Tarski Symposium, Proceedings of Symposia in Pure Mathematics XXV*, 1974, edited by Leon Henkin, John Addison, C.C. Chang, William Craig, Dana Scott & Robert Vaught, pp. 397-408. American Mathematical Society, Providenc, R.I. First published as chapter 9 of **1972(a)**.
- 1974(z₁₀)** Die Normalwissenschaft und ihre Gefahren. In *Kritik und Erkenntnisfortschritt*, edited by I. Lakatos & A.E. Musgrave, translated by P.K. Feyerabend & A Szabó, pp. 51-57. Vieweg, Braunschweig. German translation by A. Szabó of **1970(w)**.
- 1974(z₁₁)** Scientific Reduction and the Essential Incompleteness of All Science. In *Studies in the Philosophy of Biology*, edited by F.J. Ayala & T. Dobzhansky,

pp. 259-284. University of California Press, Berkeley, California. American edition of **1974(z₂)**.

- 1974(z₁₂)** Ti Einai Dialektikē. *Ēridanos* (Athens), 1974, pp. 56-66. Greek translation by E. Kastrinos of part of chapter 15 of **1963(a)**.
- 1974(z₁₃)** Science: Conjectures and Refutations. In *Compilation of Papers for the Use of Post-Graduate Students of Psychiatry*, fourth edition, edited by B. Barraclough, B. Heine & A. Smith. John Wyeth, Maidenhead. Reproduction of pp. 33-39 and 42-59 of chapter 1 of **1972(d)**.
- 1975(a)** *Miseria dello storicismo*. Feltrinelli Editore, Milan. Italian translation by Carlo Montaleone of **1961(b)**.
- 1975(b)** *Conoscenza oggettiva. Un punto di vista evoluzionistico*. Armando Armando Editore, Rome. Italian translation by Arcangelo Rossi of **1972(a)**.
- 1975(c)** *Conhecimento objetivo. Uma abordagem evolucionária*. Editora da Universidade de São Paulo and Editora Itatiaia limitada, Belo Horizonte. Portuguese translation by Milton Amado of **1973(e)**.
- 1975(d)** The Yugoslav Philosophers. *New Statesman*, 21st March 1975. A letter to the Editor.
- 1975(e)** Contributions to the Discussion. In *The Creative Process in Science and Medicine*, edited by Hans A. Krebs & Julian Shelley. Excerpta Medica, American Eisevier Publishing Co., New York.
- 1975(f)** Die Logik der Sozialwissenschaften. In *Der Positivismusstreit in der deutschen Soziologie (Soziologische Texte 58)*, edited by H. Maus & F. Fürstenberg, pp. 103-123. Luchterhand Verlag, Darmstadt & Neuwied. Sixth impression of **1969(m)**.
- 1975(g)** Kritischer Rationalismus – Eine Unterhaltung mit Karl Popper. In *Kritischer Rationalismus und Sozialdemokratie*, edited by Georg Lührs, Thilo Sarrazin, Frithjof Spreer, & Manfred Tietzel, pp. 55-72. J.H.W. Dietz, Berlin & Bonn/Bad Godesberg. German translation of **1971(l)**.
- 1975(g₁)** Die Aufgabe der Wissenschaft. *Ibid.*, pp. 89-102. A reprint of chapter 5 of **1974(e)**.
- 1975(h)** Was ist Dialektik?. *Ibid.*, pp. 167-199. A revised version of **1965(n)**.
- 1975(i)** Utopie und Gewalt. *Ibid.*, pp. 303-315. A new translation of chapter 18 of **1963(a)**.
- 1975(j)** *Die offene Gesellschaft und ihre Feinde*. UTB 472 and 473, Francke Verlag, Munich. Fourth edition **1957(k)** and **1958(i)**.

- 1975(k)** Bylting eða umbartur?. *Stefnir* **26**, 1.tbl., pp. 15-17. Icelandic translation of part of **1972(f)**.
- 1975(l)** Kritischer Rationalismus – Eine Unterhaltung mit Karl Popper. In *Kritischer Rationalismus und Sozialdemokratie*, edited by Georg Lührs, Thilo Sarrazin, Frithjof Spreer, & Manfred Tietzel. J.H.W. Dietz, pp. 55-72. Berlin & Bonn/Bad Godesberg. Second edition of **1975(g)**.
- 1975(l₁)** Die Aufgabe der Wissenschaft. *Ibid.*, pp. 89-102. Second edition of **1975(g₁)**.
- 1975(m)** Was ist Dialektik?. *Ibid.*, pp. 167-199. Second edition of **1975(h)**.
- 1975(n)** Utopie und Gewalt. *Ibid.*, pp. 303-315. Second edition of **1975(i)**.
- 1975(o)** How I See Philosophy. In *The Owl of Minerva. Philosophers on Philosophy*, edited by Charles J. Bontempo & S. Jack Odell, pp. 41-55. McGraw-Hill Book Company, New York. Contained in **1992(t)**. Serbo-Croat translation: **1978(d)**; Polish: **1988(l)**.
- 1975(p)** The Rationality of Scientific Revolutions. In *Problems of Scientific Revolution. Scientific Progress and Obstacles to Progress in the Sciences, The Herbert Spencer Lectures 1973*, edited by Rom Harré, 1975, pp. 72-101. Clarendon Press, Oxford. Contained in **1994(r)**. Portuguese translation: **1976(p)**; Italian **1977(f)**; Romanian: **1981(z₁₆)**; Spanish: **1983(u)**; French: **1991(d)**.
- 1975(q)** Die moralische Verantwortlichkeit des Wissenschaftlers. In *Universitas, Zeitschrift für Wissenschaft, Kunst und Literatur*, edited by H.W. Bähr, Wissenschaftliche Verlagsanstalt, Stuttgart **30**, No. 7, 1975, pp. 689-699. German translation **1970(z)**.
- 1975(r)** *Objective Knowledge: An Evolutionary Approach*. Clarendon Press, Oxford. Fourth impression, with corrections, of **1972(a)**.
- 1975(s)** Schlüssel zur Kontrolle der Dämonen. *Die Presse*, Wochenendbeilage, 6th/7th September 1975, pp. 17-19. An interview.
- 1975(t)** Wissenschaft und Kritik. In *Idee und Wirklichkeit: 30 Jahre Europäisches Forum Alpbach*, pp. 65-75. Springer-Verlag, Vienna & New York. English translation contained in **1992(t)**.
- 1975(u)** *The Logic of Scientific Discovery*. Hutchinson, London. Eighth impression (in paperback and hardback) of **1959(a)**.
- 1975(v)** Von den Quellen unseres Wissens und unserer Unwissenheit. In *Mannheimer Forum*, 75/76, edited by H. von Ditfurth, pp. 9-52. Boehringer Mannheim GmbH, Mannheim, 1975. German translation (without footnotes) of the introduction to **1974(z₄)**.

- 1975(w)** Gendrivelseskriteriet. In *Temaer i nutidens tænkning – Filosofiske, etiske og teologiske tekster i udvalg*, edited by Peter Kemp, pp. 23-26. Gyldendalske Boghandel, Copenhagen. An excerpt from **1973(k)** [chapter 1 of **1972(d)**].
- 1975(x)** Undvik Harklyveri! *Radikal-Liberalen*, No. 4, 1975, p. 12. An interview, by letter, with a Swedish journal.
- 1975(y)** *A lógica da pesquisa científica*. Editora Cultrix, São Paulo and Editora da Universidade de São Paulo. Portuguese translation of **1972(c)** by Leonidas Hegenberg & Octanny Silveira da Mota.
- 1975(z)** Law, Legislation and Liberty. *The Times Literary Supplement*, 5th December 1975, p. 1462. A letter to the Editor.
- 1975(z₁)** Der Baum der Erkenntnis – wissenschaftliche Evolution. In *Wissenschaftstheorie*, edited by R. Bensch & W. Trutwin, pp. 29f. Patmos Verlag, Düsseldorf.
- 1975(z₂)** Alle Erkenntnisse sind theoriegetränkt, auch unsere Beobachtungen. *Ibid.*, pp. 83f.
- 1975(z₃)** Grenzen der Induktion. *Ibid.*, pp. 86f.
- 1975(z₄)** Das Falsifikationsprinzip. *Ibid.*, pp. 87f.
- 1975(z₅)** La ciencia normal y sus peligros. In *La crítica y el desarrollo del conocimiento*, edited by I. Lakatos & A. Musgrave, pp. 149-157. Ediciones Grijalbo, Barcelona, Buenos Aires, & Mexico. Spanish translation by Francisco Hernán of **1972(y)**.
- 1975(z₆)** Sociale revolutie of Sociale hervorming?. In *Herbert Marcuse/Karl Popper, Sociale revolutie of sociale hervorming – een confrontatie*, edited by Franz Stark, pp. 5, 9-11, 22-29, 34-38, and 40. Het Werelvenster, Baarn, The Netherlands. Second impression of **1971(w)**.
- 1976(a)** *Logik der Forschung*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Sixth edition, revised, of **1934(b)** and **1966(e)**.
- 1976(b)** The Logic of the Social Sciences. In *The Positivist Dispute in German Sociology*, pp. 87-104. Heinemann Educational Books Ltd, London, 1976. English translation by Glyn Adey & David Frisby of **1974(j)**. Contained in **1992(t)**.
- 1976(c)** Reason or Revolution?. In *The Positivist Dispute in German Sociology*, Heinemann Educational Books Ltd, London, pp. 288-300. A revised version of **1970(z₄)**.
- 1976(d)** Die Theorien sollen sterben – nicht wir. Die Demokratie lebt von der Korrektur ihrer Fehler. *Deutsche Zeitung* (Bonn), No. 1, 2nd January 1976, p. 9 (Kultur).

- 1976(e)** Some Fundamental Problems in the Logic of Scientific Discovery; and Background Knowledge and Scientific Growth. In *Can Theories be Refuted? Essays on the Duhem-Quine Thesis*, edited by Sandra G. Harding, pp. 89-112 and 113-115. D. Reidel Publishing Company, Dordrecht. Excerpts from **1974(n)** and **1969(h)**.
- 1976(f)** *Misericordia dello storicismo*. Feltrinelli Editore, Milan. Second impression of **1975(a)**.
- 1976(g)** *Unended Quest: An Intellectual Autobiography*, Fontana/Collins, London. A revised version of **1974(b)**.
- Further impressions, editions, and translations of **1976(g)** into other languages are indicated in the year in which they are published.
- 1976(h)** A Note on Verisimilitude. *The British Journal for the Philosophy of Science* **27**, 2, June 1976, pp. 147-159.
- 1976(i)** Hat die Weltgeschichte einen Sinn?. *Seminar: Geschichte und Theorie*, edited by H.M. Baumgartner & J. Rüsen, pp. 305-336. Suhrkamp Verlag, Frankfurt am Main. An abridged version of pp. 320-347 of **1958(i)**.
- 1976(j)** Ész vagy forradalom?. In *Tény, Érték, Ideológia*, pp. 147-163. Gondolat, Budapest. Hungarian translation of **1970(z₄)**.
- 1976(k)** A társadalomtudományok logikája. *Ibid.*, pp. 279-301. Hungarian translation of **1969(m)**.
- 1976(l)** Revolución o reforma?. In *A la búsqueda del sentido*, pp. 31, 34-36, 47-54, 57-63. Ediciones Sigueme, Salamanca. Spanish translation of **1971(g)**.
- 1976(m)** *Unended Quest: An Intellectual Autobiography*. Open Court Publishing Company, La Salle, Illinois. American edition of **1976(g)**.
- 1976(n)** Normal Science and Its Dangers. In *Criticism and the Growth of Knowledge*, Proceedings of the International Colloquium in the Philosophy of Science, London, 1965, volume 4, edited by I. Lakatos & A.E. Musgrave, pp. 51-58. Cambridge University Press, Cambridge. A reprint of **1974(z₃)**.
- 1976(o)** The Myth of the Framework. In *The Abdication of Philosophy: Philosophy and the Public Good. Essays in Honor of Paul Arthur Schilpp*, edited by Eugene Freeman, pp. 23-48. Open Court Publishing Co., La Salle, Illinois. Contained in **1994(r)**. Japanese translation: **1979(o)**; Italian: **1985(m)**; French: **1989(n)**.
- 1976(p)** A racionalidade das revoluções científicas. In *Problemas da revolução científica, incentivos e obstáculos ao progresso das ciências*, edited by R. Harré, pp. 91-122. Editora Itatiaia, Belo Horizonte, and Editora da Universidade de São Paulo, São Paulo, pp. 91-122. Portuguese translation by Leonidas Hegenberg & Octanny Silveira da Mota of **1975(p)**.

- 1976(q)** The Logic of the Social Sciences. In *The Positivist Dispute in German Sociology*, pp. 87-104. Harper Torchbooks, Harper & Row, New York. American edition of **1976(b)**.
- 1976(r)** Reason or Revolution?. *Ibid.*, pp. 288-300. American edition of **1976(c)**.
- 1976(s)** *The Poverty of Historicism*. Routledge & Kegan Paul, London. Ninth impression of **1961(b)**.
- 1976(t)** *Conjectures and Refutations*. Routledge & Kegan Paul, London. Third impression of **1972(d)**.
- 1976(u)** *Unended Quest: An Intellectual Autobiography*. Fontana/Collins, London. Second impression of **1976(g)**.
- 1976(v)** Über Wahrheit und Sicherheit: Eine Richtigstellung. *Rechtstheorie: Zeitschrift für Logik, Methodenlehre, Kybernetik und Soziologie des Rechts* **7**, No. 1, 1976, pp. 65f.
- 1976(w)** *La ricerca non ha fine: Autobiografia intellettuale*. Armando Armando Editore, Rome. Italian translation by Dario Antiseri of **1976(g)**.
- 1976(x)** *Congetture e confutazioni: Lo sviluppo della conoscenza scientifica*. Universale Paperbacks, volumes 50* and 50**, Il Mulino, Bologna. A paperback edition of **1972(e)**.
- 1976(y)** Darwinism as a Metaphysical Research Programme. *Methodology and Science* **9**, No. 3, 1976, pp. 103-117. An excerpt from **1976(g)**.
- 1976(z)** Induction: Deduction; Objective Truth. *Methodology and Science* **9**, No. 4, 1976, pp. 163-171. An excerpt from **1976(g)**.
- 1976(z₁)** Freiheit vor Gleichheit. *Frankfurter Allgemeine Zeitung*, No. 291, 24th December 1976, Supplement: 'Bilder und Zeiten', p. 1.
- 1976(z₂)** Come io vedo la filosofia. *La Cultura* **XIV**, No. 4, 1976, pp. 389-403. Italian translation of **1975(o)**.
- 1976(z₃)** Μία γενικὴ ματιά σὲ μερικά θεμελιώδη προβλήματα. *Deucalion (Athens)* **3**, No. 16, 1976, pp. 364-389. Greek translation of chapter 1 of **1972(c)**.
- 1976(z₄)** The Social Revolution. In *The Open University 1976, Supplementary Material, Social Sciences: A Third Level Course, D302 Patterns of Inequality, Units 32-33*, pp. 60-69. An excerpt from chapter 19 of **1966(a)**.
- 1976(z₅)** La scienza normale e i suoi pericoli. In *Critica e crescita della conoscenza*, edited by I. Lakatos & A.E. Musgrave, pp. 121-128. Feltrinelli Editore, Milan. Italian translation by Giulio Giorello of **1974(z₃)**.

- 1976(z₆)** On Reason and The Open Society]. In *Revolution or Reform? A Confrontation*, edited by A.T. Ferguson, pp. 55, 61-63, 78-87, 94-100, and 102f. New University Press, Chicago (New University Press Studies in Ethics and Society, volume II). English translation of **1971(g)**. Revised English version, **1972(f)**.
- 1976(z₇)** *The Open Society and Its Enemies*. Princeton University Press, Princeton, N.J. A reprint in hardback and paperback of **1966(p)**.
- 1976(z₈)** Revolution oder Reform?. In *Revolution oder Reform? Herbert Marcuse und Karl Popper – Eine Konfrontation*, edited by Franz Stark, pp. 3, 9f., 22-29, 34-39, and 11. Kösel-Verlag, Munich. Fourth impression of **1971(g)**.
- 1976(z₉)** Revolution or Reform?. In *Revolution or Reform?*, New University Press, Inc., Chicago. English translation of **1971(g)**.
- 1977(a)** Utopia and Violence. In *The Essayist*, edited by Sheridan Barker, third edition, pp. 57-68. Thomas Y. Crowell Company, New York.
- 1977(b)** Karl Popper: Kritik am utopischen Denken. In *Menschliche Zukunft* (Kurs 11/2), edited by H. Buchner & Valentin Klingel, pp. 26f. Katechetisches Institut des Bistums Trier.
- 1977(c)** (with H. Marcuse) *Rivoluzione o riforme?*, edited by Franz Stark. Collana di metodologia delle scienze e di filosofia del linguaggio 1, Diretta da Dario Antiseri. Armando Armando Editore, Rome. Italian translation by Paolo Massimi of **1971(g)**.
- 1977(d)** *Autobiografia Intelectual*. Editora Cultrix and Editora da Universidade de São Paulo. Portuguese translation by Leonidas Hegenberg & Octanny Silveira da Mota of **1976(g)**.
- 1977(e)** *Unended Quest: An Intellectual Autobiography*, Fontana/Collins, London. Third impression of **1976(g)**.
- 1977(f)** La razionalità delle rivoluzioni scientifiche. In *Rivoluzioni scientifiche e rivoluzioni ideologiche*, edited by R. Harré, pp. 89-122. Armando Armando Editore, Rome. Italian translation by Enrico Petrozzi of **1975(p)**.
- 1977(g)** The Death of Theories and of Ideologies. In *La réflexion sur la mort*, 2e Symposium International de Philosophie, pp. 296-328. École libre de philosophie 'PLETHON', Athens.
- 1977(h)** Ho Thanatos tōn Theōriōn kai tōn Ideologiōn. Ibid., pp. 297-329. Greek translation of **1977(g)**, printed on alternate pages with **1977(g)**.
- 1977(i)** How I See Philosophy. In *Philosophers on Their Own Work*, edited by André Mercier & Maja Svilar, pp. 125-148. Fédération Internationale des Sociétés de

Philosophie, Peter Lang, Berne, Frankfurt am Main, & Las Vegas. A revised version of **1975(o)**

- 1977(j)** Mes vues sur la philosophie. *Ibid.*, pp. 149-173. French translation of **1977(i)**.
- 1977(k)** Wie ich Philosophie nicht sehe. *Süddeutsche Zeitung*, No. 167, 23rd/24th July 1977, p. 100. German translation of excerpt from **1975(o)**.
- 1977(l)** Viewpoint [The Meaning of History]. In *Living Philosophy*, edited by Philip Lawton & Marie-Louise Bishop, p. 173. Canfield Press, San Francisco. An excerpt from chapter 25, p. 278, of **1966(a)**.
- 1977(m)** Normal Science and Its Dangers. In *Criticism and the Growth of Knowledge*, Proceedings of the International Colloquium in the Philosophy of Science, London, 1965, volume 4, edited by I. Lakatos & A.E. Musgrave, pp. 51-58. Cambridge University Press, Cambridge. A reprint of **1974(z₃)**.
- 1977(n)** Some Remarks on Panpsychism and Epiphenomenalism. *Dialectica* **31**, Nos 1-2, 1977, pp. 177-186.
- 1977(o)** Die moralische Verantwortlichkeit des Wissenschaftlers. In *Probleme der Erklärung sozialen Verhaltens*, edited by Klaus Eichner & Werner Habermehl, pp. 294-304. Verlag Anton Hain, Meisenheim. German translation of **1969(a)**.
- 1977(p)** *La lógica de la investigación científica*. Editorial Tecnos, Madrid. Fifth impression of **1962(r)**.
- 1977(q)** *The Poverty of Historicism*. Harper Torchbooks. Harper & Row, New York. A reprint of **1964(a)**.
- 1977(r)** *The Logic of Scientific Discovery*. Hutchinson, London. Ninth impression of **1959(a)**.
- 1977(s)** L'apriori dans la connaissance. In *Recueil de Textes Philosophiques à l'usage des classes de première, sections B et C*, pp. 17f. (texte No. 11). Ministère de l'éducation nationale, Grand-Duché de Luxembourg, Luxembourg. French translation of an excerpt from pp. 46-48 of **1974(z₄)**.
- 1977(s₁)** Das Induktionsproblem. *Ibid.*, pp. 31f. (texte No. 17). An excerpt from pp. 3-8 of **1971(o)**.
- 1977(s₂)** Die Interpretation der Geschichte. *Ibid.*, pp. 49f. (texte No. 26). An excerpt from pp. 328f., 331-333, and 344, of volume II of **1975(j)**.
- 1977(t)** Grundprobleme der Erkenntnislogik, zum Problem der Methodenlehre. In *Wahrheitstheorien*, edited by Gunnar Skirbekk, pp. 109-139. Suhrkamp Taschenbuch Wissenschaft **210**, Suhrkamp Verlag, Frankfurt am Main. A reprint of Part I of **1966(e)**.

- 1977(u)** (with John C. Eccles) *The Self and Its Brain: An Argument for Interactionism*. Springer International. Springer Verlag, Berlin, Heidelberg, London, & New York.
- Further impressions, editions, and translations of **1977(u)** into other languages are indicated in the year in which they are published.
- 1977(v)** *Busqueda sin termino: Una autobiografía intelectual*. Editorial Tecnos, Madrid. Spanish translation by Carmen García Trevijano of **1976(g)**.
- 1977(w)** De logica der sociale wetenschappen. In *Tekstboek algemene literatuurwetenschap*, edited by W.J.M. Bronzwaer, D.W. Fokkema, & Elrud Kunne-Ibsch, pp. 14-24. Basisboeken, Ambo, Baarn. Dutch translation of an excerpt from **1974(j)**.
- 1977(x)** *Logika odkrycia naukowego*. Państwowe wydawnictwo naukowe, Warsaw. Polish translation by Urszula Niklas of **1974(n)**.
- 1977(y)** *The Open Society and Its Enemies*. Volumes I and II. Routledge & Kegan Paul, London. A reprint of **1966(a)**.
- 1977(z)** Utopie und Gewalt. In *Logik, Mathematik und Philosophie des Transzendenten* (Festgabe für Uno Saarnio), edited by Ahti Hakamies, pp. 97-108. Verlag Ferdinand Schöningh, Munich, Paderborn, & Vienna. A slightly revised version of **1975(i)**.
- 1977(z₁)** Li-hsing-chu-i tui Chuan-t'ung-fen-hsi chih T'an-t'ao [The Rationalists' Inquiries Concerning the Analysis of Tradition]. *East Wind* (Tunghai University, Taiwan) **46**, May 1977, pp. 54-61. Chinese translation by Huang Po-chi of chapter 4 of **1968(b)**.
- 1977(z₂)** (with John C. Eccles) *The Self and Its Brain: An Argument for Interactionism*. Springer International, Berlin, Heidelberg, New York, & London. American edition of **1977(u)**, with a new Preface to Part II by John C. Eccles, dated January 1978.
- 1977(z₃)** *Die offene Gesellschaft und ihre Feinde*. Francke Verlag, Munich. Fifth edition, in paperback, of **1957(k)** and **1958(i)**.
- 1977(z₄)** [Utopia and Violence]. *Grani* (Frankfurt am Main) **XXXII**, No. 106, 1977, pp. 234-250. Russian translation of **1948(a)**.
- 1978(a)** On the Possibility of an Infinite Past: A Reply to Whitrow. *British Journal for the Philosophy of Science* **29**, March 1978, pp. 47f. Abstract: **1978(x)**.
- 1978(b)** Universals and Dispositions. In *Dispositions*, edited by R. Tuomela, pp. 147-153. D. Reidel, Dordrecht, & Hingham, Mass. A reprint of sections 1-3 and the latter half of 17 of *Appendix *x* of **1959(a)**.

- 1978(c)** The Propensity Interpretation of Probability. *Ibid.*, pp. 247-265. A reprint of **1959(e)**.
- 1978(d)** Kako gledam na filozofiju [How I See Philosophy]. In *Filozofija Karla Poppera*, filozofske studije **VIII**, 1976, pp. 7-21. Filozofsko Drustvo Srbije, Belgrade. Serbo-Croat translation by P. Lazar Stojanović of **1975(o)**.
- 1978(e)** Wie ich die Philosophie sehe. In *Theorie und Politik aus kritisch-rationaler Sicht*, edited by Georg Lührs, Thilo Sarrazin, Frithjof Spreer, & Manfred Tietzel, pp. 1-16. J.H.W. Dietz, Berlin & Bonn. German translation of **1975(o)**.
- 1978(f)** Gespräch mit Karl Popper. *Ibid.*, pp. 17-30. A reprint of **1975(s)**.
- 1978(g)** *Unended Quest*, Fontana/Collins, London. Fourth impression, with corrections, of **1976(g)**.
- 1978(h)** *Der armoece van het historicisme*. Uitgeverij Het Spectrum, Utrecht. Third impression of **1974(d)**.
- 1978(i)** *De groei van kennis*. Boom Meppel, Amsterdam. Dutch translation by Zeno Swijtkink of the Introduction and chapters 1, 3, 10, 15, and 16 of **1963(a)**. Contains 'Foreword to the Netherlands Edition, February 1978'.
- 1978(j)** Wie ich die Philosophie sehe. *Der Report* **2**, No. 24, 15th June 1978, pp. 12 and 14. An abbreviated version of **1978(e)**.
- 1978(k)** Wie ich die Philosophie sehe. In *Die Welt*, No. 150, 1st July 1978, Geistige Welt, p. 1.
- 1978(l)** *Miseria dello storicismo*. Feltrinelli Editore, Milan. Third impression of **1975(a)**.
- 1978(m)** Sociale revolutie of Sociale hervorming?. In *Herbert Marcuse/Karl Popper, Sociale revolutie of sociale hervorming – een confrontatie*, edited by Franz Stark, pp. 5, 9-11, 22-29, 34-38, and 40. Het Werelvenster, Baarn, The Netherlands. Third impression of **1971(w)**.
- 1978(n)** Die Logik der Sozialwissenschaften. In *Der Positivismusstreit in der deutschen Soziologie (Soziologische Texte 58)*, edited by H. Maus & F. Fürstenberg, pp. 103-123. Luchterhand Verlag, Darmstadt & Neuwied. Seventh impression of **1969(m)**.
- 1978(o)** *The Open Society and Its Enemies*. Princeton University Press, Princeton, N.J. A reprint of **1966(p)**.
- 1978(p)** Observaciones sobre el panpsiquismo y el epifenomenalismo. *Teorema* **8**, No. 1, 1978, Valencia, pp. 5-18. Spanish translation of **1977(n)**.

- 1978(q)** *La connaissance objective*. Editions complexe, Brussels. French translation by Catherine Bastyns of chapters 1-3 of **1972(a)**, with a Preface by the author.
- 1978(r)** Wie ich die Philosophie sehe. *Mitteilungen (Gesellschaft der Freunde der Universität Mannheim e.V.)* **27**, No. 2, October 1978, pp. 1-5. A reprint of **1978(j)**.
- 1978(s)** Wie ich die Philosophie sehe. In *Österreichische Philosophen und ihr Einfluss auf die analytische Philosophie der Gegenwart*, Band 1, edited by J.C. Marek et al., pp. 11-20. *Conceptus*, Innsbruck & elsewhere, 1977. A reprint of **1978(e)**.
- 1978(t)** *Conjectures and Refutations*. Routledge & Kegan Paul, London. Fourth impression of **1972(d)**.
- 1978(u)** Eine Diskussion war gar nicht geplant. Richtigstellung zu einem Vortrag in Alpbach. *Die Presse* (Vienna). Tribüne der Leser. Samstag/Sonntag, 16th/17th September 1978.
- 1978(v)** Aestheticism. Perfectionism. Utopianism. In *The Philosophy of Society*, edited by R. Bechler & A.R. Drengson, pp. 212-234. Methuen, London. A reprint of chapter 5 of **1966(a)**.
- 1978(w)** *Hatashinari tanjiku. dutoku jiden*. Iwanami Shoten, Tokyo. Japanese translation by Hiroshi Mori of **1976(u)**.
- 1978(x)** Abstract of 'On The Possibility of An Infinite Past: A Reply to Whitrow'. *The Philosopher's Index* **12**, No. 2, Summer 1978. Abstract of **1978(a)**.
- 1978(y)** *La logique de la découverte scientifique*. Payot, Paris. Second impression of **1973(d)**, containing a partial correction to the translation of the author's Preface, 1934.
- 1978(z)** Natural Selection and the Emergence of Mind. *Dialectica* **32**, Fasc. 3-4, 1978, pp. 339-355. Chinese translation: **1980(h)**; Spanish: **1980(z₁)**.
- 1978(z₁)** *Autobiografie*. Uitgeverij Het Spectrum, Utrecht/Antwerp. Dutch translation by Jack Birner & Rob de Vries of **1976(g)**.
- 1978(z₂)** Critical Rationalism as a Philosophy of Scientific Humanism. In *Secular Alternatives to Religion*, pp. 71f. The Open University Press, Milton Keynes.
- 1978(z₃)** L'objectif de la science. In *Le pouvoir et la science en France*, edited by Pierre Papon, pp. 305-307. Éditions du Centurion, Paris. French translation of excerpts from chapter 5 of **1972(a)**.
- 1978(z₄)** *The Poverty of Historicism*. pirated edition, photographically reproduced in Hong Kong, at a date and by persons unknown, from the Routledge edition.

- 1978(z₅)** Bemerkungen über die Frankfurter Schule. In *Gespräche mit Herbert Marcuse*, Suhrkamp Verlag, Frankfurt am Main, pp. 130-134. German translation of comments prepared for an English broadcast, used by the organizers only in an abbreviated form.
- 1978(z₆)** Une science requiert des points de vue et des problèmes théoriques. In *La Science: Epistemologie générale*, edited by Marie-Claude Bartholy, Jean-Pierre Despin, & Gerald Grandpierre, pp. 146f. Éditions Magnard, Paris. An excerpt from p. 106 of **1973(d)**. The title is the editors'.
- 1978(z₇)** La falsifiabilité comme critère de démarcation. *Ibid.*, pp. 176-178. An excerpt from pp. 36-39 of **1973(d)**. The title is the editors'.
- 1978(z₈)** Les quatre procédures de vérification d'une théorie. *Ibid.*, pp. 179f. An excerpt from pp. 29-30 of **1973(d)**. The title is the editors'.
- 1978(z₉)** (with John C. Eccles) *The Self and Its Brain: An Argument for Indeterminism*. F. Hoffmann-La Roche & Co. Ltd, Basle. Licensed reprint of **1977(u)**.
- 1978(z₁₀)** *La ricerca non ha fine: Autobiografia intellettuale*. Armando Armando Editore, Rome. Second edition, revised, of **1976(w)**.
- 1979(a)** *Objective Knowledge: An Evolutionary Approach*. Clarendon Press, Oxford. Second, revised, edition of **1972(a)**, containing a new *Appendix* ('Supplementary Remarks (1978)').
- 1979(b)** Is It True What She Says About Tarski?. *Philosophy* **54**, No. 207, 1979, p. 98.
- 1979(c)** Die Wechselwirkung und die Wirklichkeit der Welt 3. In *Wissen und Macht*, edited by O. Molden. Verlag Fritz Molden, Vienna, Munich, Zürich, & Innsbruck, pp. 108-116.
- 1979(d)** The Philosopher King. In *Introductory Philosophy*, edited by R.P. Wolff, pp. 435-446. Prentice-Hall, Englewood Cliffs, N.J. An excerpt from chapter 8 of **1966(a)**.
- 1979(e)** *Scienza e filosofia: Problemi e scopi della scienza*. Giulio Einaudi Editore, Turin. Fourth impression of **1969(c)**.
- 1979(f)** Three Worlds. *Michigan Quarterly Review*, No. 1, Winter 1979, pp. 1-23. The Obert C. Tanner Lecture, delivered on 7th April 1978 at the University of Michigan.
- 1979(g)** Epistemology and Industrialisation. Remarks on the Influence of Philosophical Ideas on the History of Europe. *Ordo* **30** (Festgabe für F.A. von Hayek), 1979, pp. 3-20. Gustav Fischer Verlag, Stuttgart & New York. Contained in **1994(r)**. Spanish translation: **1981(l)**.
- 1979(h)** Letter to the Editor. *Eton College Chronicle* **748**, No. 3, 25th May 1979, p. 3.

- 1979(i)** *La Société ouverte et ses ennemis. Volume I: L'Ascendant de Platon; volume II. Hegel et Marx.* Éditions du Seuil, Paris. French translation (abridged) by Jacqueline Bernard & Philippe Monod of **1973(f)**.
- 1979(j)** Die Logik der Sozialwissenschaften. In *Der Positivismusstreit in der deutschen Soziologie (Soziologische Texte 58)*, edited by H. Mäus & F. Fürstenberg, pp. 103-123. Luchterhand Verlag, Darmstadt & Neuwied. Eighth impression of **1969(m)**.
- 1979(k)** The Unity of Method. In *Social Research: Principles and Procedures*, edited by John Bynner & Keith M. Stribley. Longmans, London, in association with The Open University Press, Milton Keynes, pp. 17-24. A reprint of section 29 of **1957(g)**.
- 1979(l)** Ich weiss, dass ich fast nichts weiss, und kaum das. *Frankfurter Rundschau*, Dokumentation. 19th June 1979, No. 139, p. 14.
- 1979(m)** Ich weiss, dass ich fast nichts weiss, und kaum das. *Uni-Report* (Johann Wolfgang Goethe Universität) **12**, No. 9, 13th June 1979, pp. 3f.
- 1979(n)** *Das Elend des Historizismus*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Fifth edition, revised, of **1965(g)**.
- 1979(o)** Junkyowaka to in Shinwa. *Shiso* **8**, No. 662, 1979, pp. 109-133. Japanese translation by Tadashi Kobayashi of **1976(o)**.
- 1979(p)** Schöpferische Selbstkritik in Wissenschaft und Kunst (Rede zur Eröffnung der Salzburger Festspiele, 1979). In *Salzburger Festspiele 1979, Offizielles Programm*, Residenz Verlag, Salzburg and Vienna, pp. 25-31. English translation: **1979(q)**; French: **1989(e)**; Chinese: **1990(v)**.
- 1979(q)** Creative Self-Criticism in Science and in Art (Opening Address of The Salzburg Festival, 1979). In *Salzburg Festival 1979, Offizielles Programm*, Residenz Verlag, Salzburg & Vienna, 1979, pp. 32-38. Contained in **1992(t)**.
- 1979(r)** Massenkultur – warum denn nicht?. *Geistige Welt*, 4th August 1979, p. 1. Abbreviated version of **1979(p)**. The title was not approved by the author.
- 1979(s)** Die Kulturpessimisten sind blind und machen andere blind. *Integral* **4**, No. 8, 1979, pp. 27 and 33. Highly abbreviated version of **1979(p)**.
- 1979(t)** Schöpferische Selbstkritik in Wissenschaft und Kunst. *Salzburger Nachrichten* **35**, No. 172, 27th July 1979, p. 3. A reprint of **1979(p)**, with the beginning omitted.
- 1979(u)** Normal Science and Its Dangers. In *Criticism and the Growth of Knowledge*, Proceedings of the International Colloquium in the Philosophy of Science, London, 1965, volume 4, edited by I. Lakatos & A.E. Musgrave, pp. 51-58. Cambridge University Press, Cambridge. A reprint of **1974(z₃)**.

- 1979(v)** [Tribute to Hayek]. *The Mont Pèlerin Society Newsletter*, August 1979, p. 10.
- 1979(w)** *Ausgangspunkte. Meine Intellektuelle Entwicklung*. Hoffman & Campe, Hamburg, 1979. German translation of **1978(g)** by Friedrich Griesse & the author. The German text deviates slightly from the English original.
- 1979(x)** Creative Self-Criticism in Science and in Art. *Encounter* **53**, No. 5, November 1979, pp. 10-14. A slightly corrected version of **1979(q)**.
- 1979(y)** Erkenntnisse eines unbelehrbaren Kulturoptimisten. *Wien aktuell*, No. 9, September 1979, pp.viii-xi.
- 1979(z)** [Brief opinion of the Salam-Weinberg Model]. *Omni* **2**, No. 1, October 1979, p. 144, column 3. Publications International Ltd, New York.
- 1979(z₁)** Due generi di musica: and Progressismo nell'arte, specialmente nella musica. In *Mitologie, convivenze di musica e mitologie testi e studi*, edited by Giovanni Morelli. La Biennale di Venezia, September/October 1979. Excerpts from **1976(w)**.
- 1979(z₂)** *Die beiden Grundprobleme der Erkenntnistheorie*, aufgrund von Manuskripten aus den Jahren 1930-1933 [on the basis of manuscripts from the years 1930-1933], edited by Troels Eggers Hansen. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Contains also Vorwort, 1978: pp. XIII-XIV, and Einleitung, 1978: pp. XV-XXXIII.
- Further impressions, editions, and translations of **1979(z₂)** into other languages are indicated in the year in which they are published.
- 1979(z₃)** [Brief Review of E.J. Steele. *Somatic Selection and Adaptive Evolution*]. *The Times Literary Supplement*, No. 4001, 23rd November 1979, p. 5. Appears under the heading 'In absentia. Some books of the year'.
- 1979(z₄)** Die moralische Verantwortlichkeit des Wissenschaftlers. *Forum Heute* **2**, 1979, Bibliographisches Institut, Mannheim, Vienna, & Zürich, pp. 209-214. Based on **1969(a)**.
- 1979(z₅)** *The Growth of Scientific Knowledge*. Vittorio Klosterman, Frankfurt am Main, 1979. A reprint of chapter 10 of **1963(a)**. Contains 'Preface, Summer 1979'.
- 1979(z₆)** *The Poverty of Historicism*. Routledge & Kegan Paul, London. 10th impression of **1961(b)**.
- 1979(z₇)** Gibt es eine philosophische Wissenschaft?. In *Artibus Ingenuis, Jahresquerschnitt 1979*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. An excerpt from pp. 341-344 of **1979(z₂)**.
- 1979(z₈)** The Principle of Leadership. In *Werkboek Staatskunde*, pp. 610-616. Leiden: Commissie Staatskunde. An excerpt from chapter 7 of **1969(q)**.

- 1979(z₉)** Über die sogenannten Quellen der menschlichen Erkenntnis. *Das Fenster*, No. 25. Innsbruck. Winter 1979/1980. pp. 2527-2529. English translation contained in **1992(t)**.
- 1979(z₁₀)** La logique des sciences sociales. In *De Vienne à Francfort, la querelle allemande des sciences sociales*. Éditions complexe, Brussels, pp. 75-90. French translation by J. Dewitte of **1962(k)**.
- 1979(z₁₁)** Raison ou révolution?. *Ibid.*, pp. 237-247. French translation by C. Bastyns of **1970(z₄)**.
- 1980(a)** *The Logic of Scientific Discovery*. Hutchinson, London. Tenth impression, revised, of **1959(a)**.
- 1980(b)** *La lógica de la investigación científica*. Editorial Tecnos, Madrid. Sixth impression of **1962(r)**.
- 1980(c)** Wie ich die Philosophie sehe. In *Reclams Literatur – Kalender XXVI*, 1980, pp. 106-110. Philipp Reclam Jun., Stuttgart. An excerpt from **1978(e)**.
- 1980(d)** La démarcation entre la science et la métaphysique. In *De Vienne à Cambridge: L'héritage du positivisme logique de 1950 à nos jours*, edited by Pierre Jacob, pp. 121-176. Éditions Gallimard, Bibliothèque des Sciences Humaines, Paris. French translation of **1964(b)**.
- 1980(e)** Science: Conjectures and Refutations. In *The Search for the Human Sciences*, pp. 95f. Deakin University. Open Campus Program, School of Humanities, Belmont, Victoria, Australia. An excerpt from pp. 38f. of **1978(t)**.
- 1980(f)** Epistemology Without a Knowing Subject. *Ibid.*, pp. 97-102. An excerpt from pp. 106-122 of **1979(a)**.
- 1980(g)** Worlds 1, 2, 3. *Ziran Kerue Zhezue Wenti Conkan [Philosophical Problems in the Natural Sciences]*. No. 1, 1980, pp. 2-10. Chinese translation of chapter P2 of **1977(u)**.
- 1980(h)** Natural Selection and the Emergence of Mind. *Ibid.*, pp. 11-18. Chinese translation of **1978(z)**.
- 1980(i)** *Suisoku To Hanbaku*. Hosei University Press, Tokyo. Japanese translation by Fujimoto, Ishigaki, & Mori of **1972(d)**.
- 1980(j)** Schöpferische Selbstkritik in Wissenschaft und Kunst. *Ergo*, January 1980, pp. 4-8. An abbreviated version of **1979(p)**.
- 1980(k)** *Miseria dello storicismo*. Feltrinelli Editore, Milan. Fourth impression of **1975(a)**.

- 1980(l) *Peri Tes Epistemes Kai Tes Metaphysikēs* [On the Status of Science and of Metaphysics]. In *Metaphysikē: Apopseis Panō Stē Synchronē Metaphysikē*, edited by Konstantinos I. Boudoures. Athēnaikos Kyklos Philosophias, Athens. Greek translation of **1958(f)**.
- 1980(m) *A miséria do historicismo*. Editora Cultrix and Editora da Universidade de São Paulo, São Paulo. Portuguese translation by Octanny Silveira da Mota & Leonidas Hegenberg of **1977(q)**.
- 1980(n) Hat die Weltgeschichte einen Sinn?. In *Diskurs: Geschichte, Philosophische Arbeitsbücher*, edited by Willi Oelmüller, Ruth Dölle, & Rainer Piepmeier, pp. 290-301. Verlag Ferdinand Schöningh, Paderborn, Munich, Vienna, & Zürich.
- 1980(o) Wissen und Nichtwissen. In *Verleihung der Ehrendoktorwürde an Sir Karl Popper*, 8th June 1979. Pressestelle der Johann Wolfgang Goethe Universität, Frankfurt am Main. English translation contained in **1992(t)**.
- 1980(p) Three Worlds. In *The Tanner Lectures on Human Values*, volume 1, edited by Sterling M. McMurrin. pp. 141-167. University of Utah Press, Salt Lake City, Utah, and Cambridge University Press, Cambridge. An earlier version of **1979(f)**.
- 1980(q) *Unended Quest: An Intellectual Autobiography*. Fontana/Collins, London. Fifth impression of **1976(g)**.
- 1980(r) (with John C. Eccles) *El yo y su cerebro*. Éditiones <Roche>, Basel. Spanish translation by C. Solis Santos of **1977(u)**.
- 1980(s) (with John C. Eccles) *El yo y su cerebro*. Editorial Labor Universitaria, S.A., Barcelona. A paperback edition of **1980(r)**.
- 1980(t) Several contributions to *Structure in Science and Art*, Proceedings of the Third C.H. Boehringer Sohn Symposium, 1979, edited by Peter Medawar & Julian Shelley. Excerpta Medica, Amsterdam, Oxford, & Princeton.
- 1980(u) Wie ich die Philosophie sehe. In *Was ist Philosophie?*, edited by Kurt Salamun, pp. 172-188. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen.
- 1980(v) 'Vorwort' to *Grundfragen der Lebensraumgestaltung* by Winfried Moewes. Walter de Gruyter, Berlin & New York, pp. v-viii.
- 1980(w) A lógica da investigação científica; Tres concepções acerca do conhecimento humano; and A sociedade aberta e seus inimigos. In *Os Pensadores* (also entitled *Popper*), edited by Victor Civita. Abril Cultural, São Paulo. An anthology containing Portuguese translations by Pablo Ruben Mariconda & Paulo de Almeida of excerpts from **1980(a)**, **1956(f)**, **1966(p)**. Contains an Introduction *not* by the author.

- 1980(x)** *Die offene Gesellschaft und ihre Feinde*. UTB, Francke Verlag, Munich. Sixth impression, in paperback, of **1957(k)** and **1958(i)**.
- 1980(y)** *The Open Society and Its Enemies*. Volumes I and II. Routledge & Kegan Paul, London. A reprint of **1966(a)**.
- 1980(z)** *Hē anoichtē koinonía kai hoi echthroi tes*. Volume I: *Hē goēteia tou Platona*. Dodoni, Athens & Ioannina. Greek translation by Irene Papadaki of volume I of **1966(a)**. Continued in **1982(z)**.
- 1980(z₁)** Selección natural y la emergencia de la mente. *Teorema* (Madrid) **10**, 1980, pp. 191-213. Spanish translation of **1978(z)**.
- 1980(z₂)** Ein Interview mit Sir Karl Popper über Kurt Gödel und den Wiener Kreis. *Wissenschaft aktuell* (Vienna), No. 4, 1980, pp. 50f.
- 1980(z₃)** Science: Conjectures and Refutations. In *Challenges to Empiricism*, edited by Harold Morick, pp. 128-160. Hackett Publishing Co. Inc., Indianapolis, Indiana, and Methuen & Co. Ltd, London. A reprint of **1972(q)**.
- 1980(z₄)** Foreword to *Foundations of Quantum Physics* by Toyoki Koga. Wood & Jones, Pasadena, California.
- 1980(z₅)** Science: Conjectures and Refutations. In *Introductory Readings in the Philosophy of Science*, edited by E.D. Klemke, Robert Hollinger, & A. David Kline, pp. 19-35. Prometheus Books, Buffalo, New York. A shortened version of chapter I of **1978(t)**.
- 1980(z₆)** Normalventskap och dess faror [Normal Science and Its Dangers]. In *Filosofin Genom Tiderna. Filosofiska Strömningar efter 1950*, edited by A.V. Konrad Marc-Wogau, pp. 39-47. Alvert Bonniers Förlag AB, Stockholm. Swedish translation of **1980(z₃)**.
- 1980(z₇)** *Det öppna samhället och dess fiender*. Volume I: *Platon*. Förlaget Akademilitteratur AB, Gotab, Stockholm. Swedish translation by Andrew Casson of volume I of **1966(a)**. Continued in **1981(z₂₂)**.
- 1980(z₈)** Letter to the Editor. *New Scientist*, 8th August 1980, p. 611. In response to an article by Dr Beverley Halstead.
- 1980(z₉)** *Faghre Tarichigari*. Kharazmi, Tehran. Second edition of **1971(z₃)**.
- 1980(z₁₀)** Das Induktionsproblem. Reformulierung und Lösung. In *Lehrbuch Philosophie, Teil 1, Begriffe, Methoden, Disziplinen*, edited by Dietrich Gondosch, Martha Helmle, & Gregor Paul, pp. 67-69. Verlag Moritz Diesterweg, Frankfurt. An excerpt from **1973(e)**.
- 1980(z₁₁)** *Congetture e confutazioni: Lo sviluppo della conoscenza scientifica*. Universale Paperbacks, volumes 50* and 50**, Il Mulino, Bologna. Second impression: of

1976(x).

- 1981(a)** Against methods of discovery. In *Conceptions of Inquiry*, edited by Stuart Brown, John Fauvel, & Ruth Finnegan, pp. 82-88. Methuen & Co. Ltd, in association with The Open University Press, London & New York. An excerpt from **1968(a)**.
- 1981(b)** Conjectures and Refutations. *Ibid.*, pp. 100-107. An excerpt from **1957(a)**.
- 1981(c)** The rationality of scientific revolutions. *Ibid.*, pp. 138-140. An excerpt from **1975(p)**.
- 1981(d)** Letter to the Editor. *New Scientist*, 12th February 1981, p. 431. In response to an article by Professor John Sparkes.
- 1981(e)** *Miseria dello storicismo*. Feltrinelli Editore, Milan. Fifth impression of **1975(a)**.
- 1981(f)** The Myth of Induction. In *Concise Readings in Philosophy*, edited by William H. Halverson, pp. 70-76. Random House, New York. An excerpt from sections 1-3 of **1959(a)**, with some deletions. The title is the editor's.
- 1981(g)** 'Vorwort' to *Es kam ganz anders* by Fritz Kolb, pp. 7-9. Österreichischer Bundesverlag GmbH, Vienna.
- 1981(h)** Diritto d'errore, conversazione con Karl Raimund Popper. *Panorama XIX*, No. 779, 23rd March 1981, pp. 162-172.. Italian translation of an interview with Corrado Augias.
- 1981(i)** 'Sir Karl Popper Criticizes B.F. Skinner'. *Free Inquiry* 1, No. 2, Spring 1981, p. 3. A letter to the Editor. Neither the title, nor the signature, 'Sir Karl Popper', is Popper's.
- 1981(j)** (with J.C. Eccles) *L'io e il suo cervello*. Volume I: K.R. Popper: *Materia, coscienza e cultura*; volume II: J.C. Eccles: *Strutture e funzioni cerebrali*; volume III: *Dialoghi aperti tra Popper ed Eccles*, edited by Francesco Barone and Vittorio Somenzi. Armando Armando Editore, Rome. Italian translations by Giuseppe Mininni of Part I of **1977(u)**, and by Barbara Continenza of Part II and Part III. of **1977(u)**.
- 1981(k)** A Dialogue Between Giants. *Science Digest*, June 1981, p. 102. A short recommendation of the book *The Born-Einstein Letters*.
- 1981(l)** La Influencia de las Ideas Filosóficas en la Historia de Europa. *Economía y Ciencia*, volume 2 of *Estudios Públicos*, March 1981. Spanish translation of **1979(g)**.
- 1981(m)** (with Augusto Garuccio & Jean-Pierre Vigier) An Experiment to Interpret E.P.R. Action-at-a-Distance: The Possible Detection of Real De Broglie Waves.

Epistemological Letters, 30, July 1981.

- 1981(n)** A Survey of Some Fundamental Problems. *Ziran Kexue Zhexue Wenti Conkan* [*Philosophical Problems in the Natural Sciences*], No. 1, 1981, pp. 1-6. Chinese translation of chapter 1 of **1977(r)**.
- 1981(o)** (with John C. Eccles) *The Self and Its Brain: An Argument for Interactionism*. Springer International, Springer Verlag, Berlin, Heidelberg, London, & New York. A reprint, with corrections, of **1977(u)**.
- 1981(p)** Tutta la conoscenza scientifico e congetturale. *LEGGERE LA REALTÁ*, Antologia Interdisciplinare, edited by Dario Antiseri, pp. 14f. Editrice la Scuola, Brescia. An excerpt from **1969(c)**.
- 1981(q)** Contro Kuhn. *Ibid.*, pp. 41-45. An excerpt from **1976(z₅)**.
- 1981(r)** La 'metafisica' dell' 'economismo' marxista. *Ibid.*, pp. 320-322. An excerpt from **1973(m)**.
- 1981(s)** L'epistemologia contro l'holismo. *Ibid.*, pp. 322-325. An excerpt from **1954(e)**.
- 1981(t)** Le regole della democrazia. *Ibid.*, pp. 356-358. An excerpt from **1973(m)**.
- 1981(u)** Il mondo 3. *Ibid.*, pp. 451-453. An excerpt from **1976(w)**.
- 1981(v)** La funzione della tradizione. *Ibid.*, pp. 456f. An excerpt from **1972(e)**.
- 1981(w)** Il linguaggio, la critica, e il mondo 3. *Ibid.*, pp. 464-466. An excerpt from **1975(b)**.
- 1981(x)** *Logica Cercetarii*. Editura științifică și enciclopedică, Bucharest. Romanian translation by M. Flonta, A. Surdu & E. Tivig of **1975(u)**.
- 1981(y)** *Ausgangspunkte: Meine Intellektuelle Entwicklung*. Lizenzausgabe für den Buchklub Ex Libris, Zürich. A reprint of **1979(w)**.
- 1981(z)** Wissenschaft: Wissen und Nichtwissen. *Weltwoche Magazin*, No. 33, 12th August 1981, pp. 10-12.
- 1981(z₁)** Über den Zusammenprall von Kulturen. In *25 JAHRE Staatsvertrag*, herausgegeben von der Österreichischen Bundesregierung, pp. 118-122. Österreichischer Bundesverlag, Vienna. Contained in **1992(t)**.
- 1981(z₂)** Lo scopo della scienza. In *Razionalismo critico e socialdemocrazia*, pp. 110-126. Università Cattolica, Milan. Italian translation by Herta Ament & Maria Baiocchi Frattini of **1975(g₁)**.

- 1981(z₃)** Che cos'è la dialettica?. *Ibid.*, pp. 205-243. Italian translation by Herta Ament & Maria Baiocchi Frattini of **1975(h)**.
- 1981(z₄)** Utopia e violenza. *Ibid.*, pp. 372-386. Italian translation by Herta Ament & Maria Baiocchi Frattini of **1975(i)**.
- 1981(z₅)** *Fornuft og rimelighet som tenkemåte: Utvalgte essays*, edited by Jan M. Døderlein. Dreyers Forlag, Oslo. An anthology containing Norwegian translations by the editor of **1957(a)**, **1957(l)**, **1948(d)**, and of excerpts from **1973(f)** (one excerpt), **1974(c)** (three excerpts), and **1974(a)** (seven excerpts).
- 1981(z₆)** Contro il determinismo storicista. In *Epistemologia e storiografia*, edited by Paola Masat Lucchette, pp. 137-141. Editrice la Scuola, Brescia. An excerpt from **1972(e)**.
- 1981(z₇)** 'Geschichtliche Offenbarung als Gotteslästerung'. In *Der nahe und der ferne Gott: Nichttheologische Texte zur Gottesfrage im 20. Jahrhundert*, edited by Leszek Kolakowski, pp. 204-208. Severin & Siedler, Berlin. An excerpt from **1977(z₃)**.
- 1981(z₈)** *Conjecturas e refutações (O progresso do conhecimento científico)*. Coleção Pensamento Científico. Editora Universidade de Brasília, Brasília. Portuguese translation by Sérgio Bath of **1972(d)**. Contains a new Author's Preface ('Contra a Indução' [= Against Induction]).
- 1981(z₉)** La Signification Actuelle de Deux Arguments d'Henri Poincaré. In *Relations Actuelles entre les Sciences, les Arts et la Philosophie*, pp. 91-94. La Maison de l'UNESCO. French translation by Renée Bouveresse of **1981(z₁₂)**.
- 1981(z₁₀)** Joseph Henry Woodger. *The British Journal for the Philosophy of Science* **32**, No. 3, September 1981, pp. 328-330. An obituary notice.
- 1981(z₁₁)** *La quête machévrée*. Calmann-Lévy, Paris. French translation by Renée Bouveresse in collaboration with Michelle Bouin-Naudin of **1976(g)**. Contains an unauthorized Preface by Christian Schmidt.
- 1981(z₁₂)** The Present Significance of Two Arguments of Henri Poincaré. *Methodology and Science* **14**, No. 4, pp. 260-264. French translation: **1981(z₉)**; Italian: **1983(k)**.
- 1981(z₁₃)** *Objective Knowledge. An Evolutionary Approach*. Clarendon Press, Oxford. Second impression of **1979(a)**.
- 1981(z₁₄)** Die Zielsetzung der Erfahrungswissenschaft. In *Geschichte der Philosophie in Text und Darstellung* **8**, 20. Jahrhundert, edited by Reiner Wiehl, pp. 349-367. Philipp Reclam jun., Stuttgart. A reprint of **1957(j)**.
- 1981(z₁₅)** The Importance of Critical Discussion: An Argument for Human Rights and Democracy. *Free Inquiry* **2**, No. 1, Winter 1981/1982, pp. 7-11.

- 1981(z₁₆)** Raționalitatea Revoluțiilor Științifice. In *Istoria științei și reconstrucția ei conceptuală*, edited, with an introduction, by Ilie Părvu, pp. 275-304. Editura științifică și enciclopedică, Bucharest. Romanian translation of **1975(p)**.
- 1981(z₁₇)** Foreword to *Cosmology, Physics and Philosophy* by Benjamin Gal-Or. Springer Verlag N.Y. Inc., New York.
- 1981(z₁₈)** The Present Significance of Two Arguments of Henri Poincaré. In *Essays in Scientific Philosophy, Philosophie als Wissenschaft* (Festschrift für Paul Weingartner), edited by E. Morscher, D. Neumaier, & G. Zecha, pp. 19-24. C. G. Böhrer Verlag, Bad Reichenhall. A reprint of **1981(z₁₂)**.
- 1981(z₁₉)** (with Augusto Garbaccio & Jean-Pierre Vigié) Possible Direct Physical Detection of De Broglie Waves. *Physics Letters* **86A**, No. 8, 7th December 1981, pp. 397-400.
- 1981(z₂₀)** The Rationality of Scientific Revolutions. In *Scientific Revolutions*, edited by Ian Hacking, pp. 80-106. Oxford University Press, Oxford & New York. A reprint, with corrections, of **1975(p)**.
- 1981(z₂₁)** *La miseria del historicismo*. Taurus Ediciones, Alianza Editorial, Madrid. Second edition of **1973 (j)**.
- 1981(z₂₂)** *Det öppna samhället och dess fiender*. Volume II: *Förkunnandets tidevarv: Hegel och Marx*. Förlaget Akademilitteratur AB, Go'rab, Stockholm. Swedish translation by Bo Ekström of volume I of **1966(a)**. Continuation of **1980(z₇)**.
- 1981(z₂₃)** *Il Pensiero Politico. Critiche e proposte del grande teorico liberale in una ricostruzione antologica*. Biblioteca della libertà, No. 22 Armando Armando Editore, Rome. An anthology containing Italian translations of **1971(g)**, **1975(b)**, and **1976(w)**; and excerpts from **1972(e)**, **1973(m)**, **1973(v)**, and **1976(f)**. Contains an introduction by Angelo M. Petroni.
- 1981(z₂₄)** *Conjectures and Refutations*. Routledge & Kegan Paul, London. Fifth impression of **1972(d)**.
- 1981(z₂₅)** The Present Significance of Two Arguments of Henri Poincaré. *Fundamenta Scientiae* **2**, Nos 3/4, 1981, pp. 263-266. A reprint of **1981(z₁₂)**.
- 1981(z₂₆)** La Signification Actuelle de Deux Arguments d'Henri Poincaré. *Fundamenta Scientiae* **2**, Nos 3/4, 1981, pp. 259-262. A reprint of **1981(z₉)**.
- 1981(z₂₇)** *La sociedad abierta y sus enemigos*. Ediciones Paidós. Barcelona & Buenos Aires. A new edition of **1957(c)**.
- 1981(z₂₈)** A Survey of Some Fundamental Problems. In *Supplementary Course Reader*, No. A313, pp. 21-35. The Open University Press, Milton Keynes. A reprint of chapter 1 of **1959(a)**.

- 1981(z₂₉)** Conjectures and Refutations. *Ibid.*, pp. 36-57. A reprint of chapter 1 of **1978(t)**.
- 1981(z₃₀)** The Aldine Aristophanes. Venice, 1499. In *Fine Books and Book Collecting* (Festschrift for Alan G. Thomas), edited by Christopher de Hamel & Richard A. Lindenthal. James Hall, Leamington Spa.
- 1981(z₃₁)** *Li Shih-Feng Lun Chu-I Ti Ch'ung K'un*. Lien Ching Publishing Co., Taiwan. Chinese translation by Feng-pin Li of **1979(z₆)**. Unauthorized.
- 1981(z₃₂)** Pien Cheng Fa Shih Shên Mo. *Ibid.*, appendix 1. Chinese translation by Feng-pin Li of **1940(a)**. Unauthorized.
- 1981(z₃₃)** Shê Hui K'io Hsüeh Chung Ti Yü T'sé Yü Yü Yen. *Ibid.*, appendix 2. Chinese translation by Feng-pin Li of **1948(d)**. Unauthorized.
- 1982(a)** *The Open Universe: An Argument for Indeterminism*, edited by W.W. Bartley III. Hutchinson & Company, London. Volume II of the *Postscript to The Logic of Scientific Discovery*.
- Further impressions, editions, and translations of **1982(a)** into other languages are indicated in the year in which they are published.
- 1982(b)** *Quantum Theory and the Schism in Physics*, edited by W.W. Bartley III. Hutchinson & Company, London. Volume III of the *Postscript to The Logic of Scientific Discovery*.
- Further impressions, editions, and translations of **1982(b)** into other languages are indicated in the year in which they are published.
- 1982(c)** *The Open Universe: An Argument for Indeterminism*, edited by W.W. Bartley III. Rowman & Littlefield, Totowa, N.J. American edition of **1982(a)**.
- 1982(d)** *Quantum Theory and the Schism in Physics*, edited by W.W. Bartley III. Rowman & Littlefield, Totowa, N.J. American edition of **1982(b)**.
- 1982(e)** *Logik der Forschung*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Seventh German edition, revised, with a new Preface and six new *Addenda*, of **1934(b)** and **1966(e)**.
- 1982(f)** Les chemins de la vérité: L'Express va plus loin avec Karl Popper. *L'Express*, No. 1598, 26th February 1982, pp. 82-88. French translation of an interview with Sophie Lannes & Alain Boyer.
- 1982(g)** *Unended Quest: An Intellectual Autobiography*, Open Court Publishing Company, La Salle, Illinois. Second edition, revised and enlarged, of **1976(m)**.

- 1982(h)** Popper's Psychologism: A Reply to Ball. *Philosophy of the Social Sciences* **12**, No. 1, March 1982, p. 69.
- 1982(i)** *Ausgangspunkte: Meine Intellektuelle Entwicklung*. Hoffmann und Campe, Hamburg. Second edition, revised, with augmented bibliography, of **1979(w)**.
- 1982(j)** *Objektive Erkenntnis: Ein Evolutionärer Entwurf*. Hoffmann und Campe, Hamburg. Third impression of **1973(i)**.
- 1982(k)** *Unended Quest: An Intellectual Autobiography*. Fontana/Collins, London. Sixth impression, with extended bibliography and some further corrections, of **1976(g)**.
- 1982(l)** Self Reference and Meaning in Ordinary Language. A corrected reprint (privately printed) of chapter 14 of **1972(d)**.
- 1982(m)** Proposal for a Simplified New Variant of the Experiment of Einstein, Podolsky and Rosen. In *Physik, Philosophie und Politik (Festschrift für Carl Friedrich von Weizsäcker zum 70. Geburtstag)*, edited by Klaus Michael Meyer-Abich, pp. 310-313. Carl Hanser Verlag, Munich.
- 1982(n)** Science: Conjectures and Refutations. In *Philosophy of Science and the Occult*, edited by Patrick Grim, pp. 87-93. State University of New York Press, Albany, New York. An excerpt from chapter 1 of **1963(a)**.
- 1982(o)** *Conocimiento objetivo: Un enfoque evolucionista*. Editorial Tecnos, Madrid. Second edition of **1974(f)**.
- 1982(p)** *La logique de la découverte scientifique*. Payot, Paris. Third impression of **1978(y)**.
- 1982(q)** (with John C. Eccles) *Das Ich und sein Gehirn*. R. Piper & Co. Verlag, Munich & Zürich. German translations by Willy Hochkeppel (Popper's contributions) and by Angela Hartung (Eccles's contributions) of **1981(o)**.
- 1982(r)** *Offene Gesellschaft – offenes Universum. Franz Kreuzer im Gespräch mit Karl R. Popper*, edited by Franz Kreuzer. Franz Deuticke, Vienna. Contains discussions from 1979, 1981, and 1982, and a lecture delivered by Popper in Tübingen in May 1981.
- Further impressions, editions, and translations of **1982(r)** into other languages are indicated in the year in which they are published.
- 1982(s)** The Place of Mind in Nature. In *Mind in Nature* (Nobel Conference XVIII, Gustavus Adolphus College), edited by Richard Q. Elvee, pp. 31-59. Harper & Row, San Francisco. Italian translation (unauthorized): **1992(z₁₆)**.
- 1982(t)** *La lógica de la investigación científica*. Editorial Tecnos, Madrid. Seventh impression of **1962(r)**.

- 1982(u)** Duldsamkeit und Intellektuelle Verantwortlichkeit. In *Toleranz*, edited by Peter Stuhlmacher & Luise Abramowski, pp. 173-185. Attempto Verlag, Tübingen University Press, 1982.
- 1982(v)** Introduction to *Evoluzione e selezione somatica*, by E.J. Steele, translated into Italian by Paola Manzoni, p. 7. Il Saggiatore, Milan.
- 1982(w)** (with John C. Eccles) *El yo y su cerebro*. Editorial Labor Universitaria S.A., Barcelona. Second impression of **1980(r)**.
- 1982(x)** (with John C. Eccles) *Das Ich und sein Gehirn*. R. Piper & Co. Verlag, Munich. Second edition of **1982(q)**.
- 1982(y)** A Critical Note on the Greatest Days of Quantum Theory. *Foundations of Physics* **12**, No. 10, 1982, pp. 971-976.
- 1982(z)** *Hē anoichtē koivonía kai hoi echthroí tēs*. Volume II: *Hegel, Marx kai ta epakolontha*. Dodoni, Athens & Ionnina. Greek translation by Irene Papadaki of volume II of **1966(a)**. Continuation of **1980(z)**.
- 1982(z₁)** Chapters 8, 10, and 40 of *Unended Quest: An Intellectual Autobiography*. *Grani* (Frankfurt am Main) **XXXVI**, No. 126, 1982, pp. 242-271. Russian translation of excerpts from **1976(g)**.
- 1982(z₂)** *Yulín Suhui Wa Guh Jukduel*. Mineumsa Publishing Company, Seoul. Korean translation by Han-koo Lee & Myung-hyun Lee of **1980(y)**.
- 1982(z₃)** *La sociedad abierta y sus enemigos*. Ediciones Paidós, Barcelona & Buenos Aires. Second impression of **1981(z₂₇)**.
- 1983(a)** *The Logic of Scientific Discovery*. Hutchinson Publishing Company Ltd, London. Eleventh impression of **1959(a)**.
- 1983(b)** *Realism and the Aim of Science*, edited by W.W. Bartley III. Hutchinson & Company, London. Volume III of the *Postscript to The Logic of Scientific Discovery*.
- Further impressions, editions, and translations of **1983(b)** into other languages are indicated in the year in which they are published.
- 1983(c)** *Realism and the Aim of Science*, edited by W.W. Bartley III. Rowman & Littlefield, Totowa, N.J. American edition of **1983(b)**.
- 1983(d)** Is Determinism Self-Refuting?. *Mind* **92**, N.S., No. 365, 1983, pp. 103f.
- 1983(e)** Bücher und Gedanken. *Anzeiger des Österreichischen Buchhandels* **118**, 1983. Contained in **1992(t)**. Greek translation: **1995(k)**.

- 1983(f)** (with David Miller) A proof of the impossibility of inductive probability. *Nature* **302**, 5910, 21st April 1983, pp. 687f. Continued in **1984(d)**, **1985(h)**, **1985(j)**, and **1987(c)**. Italian translation: **1983(j)**; Chinese: **1986(f)**; French: **1990(u)**.
- 1983(g)** *Conjeturas y refutaciones: El desarrollo del conocimiento científico*. Ediciones Paidós, Barcelona & Buenos Aires. Second edition, revised, of **1967(n)**. Spanish translation by Néstor Míguez & Rafael Grasa of **1972(d)**.
- 1983(h)** *Offene Gesellschaft – offenes Universum. Franz Kreuzer im Gespräch mit Karl R. Popper*, edited by Franz Kreuzer. Franz Deuticke Verlagsgesellschaft m.b.H., Vienna. Second edition, revised, of **1982(r)**.
- 1983(i)** *A Pocket Popper*, edited, with an introduction, by David Miller. Fontana Pocket Readers, Fontana Paperbacks, London & Glasgow. An anthology containing **1957(e)**, the English version of **1967(d)**, and excerpts from **1959(a)** (three excerpts), **1961(b)** (two excerpts), **1966(a)** (seven excerpts), **1972(d)** (seven excerpts), **1974(c)** (two excerpts), **1975(p)** (one excerpt), **1977(u)** (two excerpts), **1978(z)**, and **1979(a)** (four excerpts).
- 1983(j)** (with David Miller) L'impossibilità della probabilità induttiva. *Scienza duemila* **5**, July–August 1983, pp. 20f. Italian translation by Marcello Pera of **1983(f)**.
- 1983(k)** Due argomenti di Poincaré. *Ibid.*, pp. 21–23. Italian translation by Marcello Pera of **1981(z₁₂)**.
- 1983(l)** L'induzione e la verità. *Ibid.*, pp. 22f. Italian translation by Marcello Pera of a letter from Popper to Pera.
- 1983(m)** *Logika i rost nauchnogo znaniya [Logic and the Growth of Scientific Knowledge]*, edited, with an introduction, by V.N. Sadovsky. Progress Publishers, Moscow. An anthology containing Russian translations by L.V. Blinnikov, V.N. Briushinkin, S.L. Nappelbaum, & A.L. Nikiforov of excerpts from **1959(a)**, **1972(a)**, and **1979(a)**, and from *Addendum I* to volume II of **1966(a)**.
- 1983(n)** *Scienza e filosofia: Problemi e scopi della scienza*. Einaudi Editore, Turin. Fifth impression of **1969(c)**.
- 1983(o)** *Logica della scoperta scientifica*. Einaudi Editore, Turin. Third impression of **1970(e)**.
- 1983(p)** Dedication – Remembering Wolfgang Yourgrau. In *Essays in Honor of Wolfgang Yourgrau*, edited by Alwyn van der Merwe, pp. vii–viii. Plenum Press, New York & London.
- 1983(q)** (with Neil McIntyre) The critical attitude in medicine: the need for a new ethics. *British Medical Journal* **287**, 24th/31st December 1983, pp. 1919–1923. French translation: **1992(z₂₀)**.

- 1983(r)** *Objective Knowledge. An Evolutionary Approach*. Clarendon Press, Oxford. Third impression of **1979(a)**.
- 1983(s)** Freeman on Peirce's anticipations of Popper. In *The Relevance of Charles Peirce*. Monist Library of Philosophy, La Salle, Illinois. A reprint of pp. 1065f. of **1974(c)**.
- 1983(t)** *Wu Qiong De Tan Shou*. Fujian Province People's Press. Chinese translation by Duan Juan & Qiu Renzong of **1982(g)**.
- 1983(u)** La racionalidad de las revoluciones científicas. *Teorema* (Valencia) **13**, 1983, pp. 109-110. Spanish translation of **1975(p)**.
- 1983(v)** *Offene Gesellschaft – offenes Universum. Franz Kreuzer im Gespräch mit Karl R. Popper*, edited by Franz Kreuzer. Franz Deuticke Verlagsgesellschaft m.b.H., Vienna. Third edition, revised, of **1982(r)**.
- 1983(w)** *De groei van kennis*. Boom Meppel, Amsterdam. A paperback reprint of **1978(i)**.
- 1983(x)** De volta aos Pré-Socráticos. *Humanidades* (Brasília) **1**, No. 2, January/March 1983, pp. 125-137. Portuguese translation of chapter 5 of **1981(z24)**.
- 1984(a)** *Logik der Forschung*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Eighth German edition (50 Jahre *Logik der Forschung*), revised, with a new *Addendum*, of **1934(b)** and **1966(e)**.
- 1984(b)** *Auf der Suche nach einer besseren Welt: Vorträge und Aufsätze aus dreißig Jahren*. R. Piper & Co. Verlag, Munich.
- Further impressions, editions, and translations of **1984(b)** into other languages are indicated in the year in which they are published.
- 1984(c)** *Objektive Erkenntnis: Ein Evolutionärer Entwurf*. Hoffmann und Campe, Hamburg. Fourth edition, revised, of **1973(i)**. German translation by Herman Vetter and Ingeborg & Cord Fleischmann of **1983(r)**.
- 1984(d)** (with David Miller) The Impossibility of Inductive Probability. Popper and Miller Reply. *Nature* **310**, No. 5976, 2nd August 1984, pp. 433f. A continuation of **1983(f)**.
- 1984(e)** (with John C. Eccles) *The Self and Its Brain*. Routledge & Kegan Paul, London. First paperback reprint, with new corrections, of **1977(u)**.
- 1984(f)** *The Open Society and Its Enemies*. Volumes I and II. Routledge & Kegan Paul, London. A reprint of **1966(a)**.
- 1984(g)** *Ausgangspunkte: Meine Intellektuelle Entwicklung*. Hoffmann und Campe, Hamburg. Third edition, with further augmented bibliography, of **1979(w)**.

- 1984(h)** Against Induction: One of Many Arguments. In *Rationality in Science and Politics*, edited by Gunnar Andersson, pp. 245-247. D. Reidel Publishing Co., Dordrecht.
- 1984(i)** *Poscritto alla logica della scoperta scientifica*, edited by W.W. Bartley III. Volume I: *Il realismo e lo scopo della scienza*. Il Saggiatore, Milan. Italian translation by Margherita Benzi & Sabrina Mancini of **1983(b)**.
- 1984(j)** *Poscritto alla logica della scoperta scientifica*, edited by W.W. Bartley III. Volume II: *L'universo aperto. Un argomento per l'indeterminismo*. Il Saggiatore, Milan. Italian translation by Roberto Festa of **1982(a)**.
- 1984(k)** *Poscritto alla logica della scoperta scientifica*, edited by W.W. Bartley III. Volume III. *La teoria dei quanti e lo scisma nella fisica*. Il Saggiatore, Milan. Italian translation, edited by Alberto Artosi, of **1982(b)**.
- 1984(l)** Bücher und Gedanken. In *Ökonomisches Denken und soziale Ordnung* (Festschrift for Erik Boettcher), edited by Hans Albert. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. A reprint of **1983(e)**.
- 1984(m)** *La société ouverte et ses ennemis*. Volume I: *L'ascendant de Platon*. Édition du Seuil, Paris. A reprint of volume I of **1979(i)**.
- 1984(n)** *La société ouverte et ses ennemis*. Volume II: *Hegel et Marx*. Édition du Seuil, Paris. A reprint of volume 2 of **1979(i)**.
- 1984(o)** *Miseria dello storicismo*. Giangiacomo Feltrinelli Editore, Milan. Sixth impression of **1975(a)**.
- 1984(p)** *Nedza Historyczmu [z dodaniem fragmentów Autobiografii]*. Warszawa, 1984. Polish translation by Stanisław Zerski of **1979(z₆)** and of excerpts from **1980(q)** and **1984(f)**.
- 1984(q)** Evolutionary Epistemology. In *Evolutionary Theory: Paths into the Future*, edited by J.W. Pollard, pp. 239-255. John Wiley & Sons Ltd, Chichester, New York, Brisbane, Toronto, & Singapore. Italian translation: **1984(z₆)**.
- 1984(r)** *El universo abierto. Un argumento en favor del indeterminismo. Post Scriptum a La lógica de la investigación científica*, volume II, edited by W.W. Bartley III. Editorial Tecnos, Madrid. Spanish translation by Sansigre Vidal of **1982(a)**.
- 1984(s)** *L'Univers incertain. Plaidoyer pour l'indéterminisme*. Hermann éditeurs des sciences et des arts, Paris. French translation by Renée Bouveresse of **1982(a)**.
- 1984(t)** (with John C. Eccles) *Das Ich und sein Gehirn*. R. Piper & Co. Verlag, Munich. Third edition of **1982(q)**.

- 1984(u)** 'Oui, nous sommes libres!'. *Le Nouvel Observateur*, 14th/20th September 1984, pp. 95-97. A discussion edited by Didier Eribon.
- 1984(v)** A Critical Note on the Greatest Days of Quantum Theory. In *Quantum, Space and Time – The Quest continues*. Studies and Essays in Honour of Louis de Broglie, Paul Dirac, and Eugene Wigner, edited by Asim O. Barut, Alwyn van der Merwe, & Jean-Pierre Vigié, pp. 49-54. Cambridge University Press, Cambridge. A reprint of **1982(y)**.
- 1984(w)** Selbstkritik und Offenheit als Überlebensstrategien. *Start und Aufstieg*, Zeitschrift des Berufsförderungsinstituts, No. 5, September/October 1984. A written interview with Bernard Ingrisch. The title is not Popper's.
- 1984(x)** *The Open Universe: An Argument for Indeterminism*, edited by W.W. Bartley III. Rowman & Littlefield, Totowa, N.J. A paperback edition of **1982(c)**.
- 1984(y)** *Quantum Theory and the Schism in Physics*, edited by W.W. Bartley III. Rowman & Littlefield, Totowa, N.J. A paperback edition of **1982(d)**.
- 1984(z)** *Società aperta, universo aperto*. Borla Edizioni, Rome. Italian translation by Dario Antiseri of **1982(r)**.
- 1984(z₁)** *The Poverty of Historicism*. Routledge & Kegan Paul, London. 11th impression of **1961(b)**. In paperback.
- 1984(z₂)** The Myth of the Framework. *LACE* [London Association of Comparative Educationists], London. A reprint of **1976(o)**.
- 1984(z₃)** Critical Remarks on the Knowledge of Lower and Higher Organisms, the So-Called Sensory Motor Systems. In *Sensory-Motor Integration in the Nervous System*, edited by O. Creutzfeldt, R.F. Schmidt, & W.D. Willis, pp. 19-31. Springer Verlag, Berlin, Heidelberg, New York, & Tokyo.
- 1984(z₄)** *Sociedad abierta, universo abierto*. Editorial Tecnos, Madrid. Spanish translation by Salvador Mas Torres of **1983(v)**. Contains also 'Tolerancia y responsabilidad intelectual', a Spanish translation by Angeles Jiménez Perona of **1987(z₂)**.
- 1984(z₅)** *Conjectures and Refutations*. Routledge & Kegan Paul, London. Sixth impression of **1972(d)**.
- 1984(z₆)** Epistemologia Evolutiva. In *Paradigmi, Rivista di Critica Filosofica* (Brindisi) 2, No. 4, 1984. Italian translation by Silvana Corato of **1984(q)**.
- 1984(z₇)** Reason or Revolution?. In *Foundations of the Frankfurt School of Social Research*, edited by Judith Marcus & Soltán Tar, pp. 155-165. Transaction Books, New Brunswick, N.J., & London, pp. 155-165. A reprint of **1970(z₄)**.

- 1984(z₈)** The Frankfurt School: An Autobiographical Note. *Ibid.*, pp. 167-170. English translation of **1978(z₅)**.
- 1984(z₉)** (with John C. Eccles) *Das Ich und sein Gehirn*. R. Piper & Co. Verlag, Munich. Fourth edition of **1982(q)**.
- 1984(z₁₀)** *Keradmāndān Dar Khedmat-e Khodkāmegān ya Jame-ye Azad va Doshmānānash* [*Sages at the Service of Despots or The Open Society and Its Enemies*], volume I. Melli Publishing House Inc., Beverly Hills, California. Persian translation by A.A. Mohajer of volume I of **1980(y)**. Continued in **1985(p)**.
- 1984(z₁₁)** Über unser Wissen und unsere Unwissenheit. In *Jahresquerschnitt 1984*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. An excerpt from pp. 449-452 of **1984(a)**.
- 1984(z₁₂)** *K'ai Fung Shé Hui Ch'í Ch'í Tí Jen*. Volume I: *Pai La T'u Ti Fu Chou*. Kuei Kuan Books, Taiwan. Chinese translation by Wen Jui Chuang & Ying Ming Li of volume I of **1980(y)**. Unauthorized.
- 1984(z₁₃)** *K'ai Fung Shé Hui Ch'í Ch'í Tí Jen*. Volume I: *Yü Yen Tí Kao Chao*. Kuei Kuan Books, Taiwan. Chinese translation by Wen Jui Chuang & Ying Ming Li of volume II of **1980(y)**. Unauthorized.
- 1984(z₁₄)** A Proof of the Impossibility of Inductive Probability. *Epistemologia* **7**, 2, July/December 1984, pp. 183-188. A reprint of **1983(f)**. Included as part of Giovanni Blandino S.J. Critical Remarks on an Argumentation by K. Popper & D. Miller: Discussion about Induction. *Ibid.*, pp. 183-206. Unauthorized.
- 1985(a)** Realism in Quantum Mechanics and a New Version of the EPR experiment. In *Open Questions in Quantum Physics*, edited by G. Tarozzi & A. van der Merwe, pp. 3-25. D. Reidel Publishing Company, Dordrecht. Contains a discussion on pp. 26-32.
- 1985(b)** Evolutionary Epistemology. *Ibid.*, pp. 395-413. A reprint of **1984(q)**.
- 1985(c)** (with John C. Eccles) *The Self and Its Brain*. Springer International, London. A hardback reprint, with new corrections, of **1977(u)**.
- 1985(d)** *Poscritto alla logica della scoperta scientifica*, edited by W.W. Bartley III. Volume I: *Il realismo e lo scopo della scienza*. Il Saggiatore, Milan. A reprint of **1984(i)**.
- 1985(e)** *Popper Selections*, edited, with an introduction, by David Miller. Princeton University Press, Princeton, N.J. American edition of **1983(i)**.
- 1985(f)** *Hadsha Va Ebtalha: Rosha Shenakht Elni*. Enteshar Co. Ltd, Tehran. Persian translation by Ahmad Aram of **1984(z₅)**.

- 1985(g)** (with Konrad Lorenz) *Die Zukunft ist offen. Das Altenberger Gespräch. Mit den Texten des Wiener Popper Symposiums*, edited by Franz Kreuzer. Serie Piper, R. Piper & Co. Verlag, Munich & Zürich. Contains contributions from Roman Sexl, Rupert Riedl, Friedrich Wallner, Paul Weingartner, Irene Papadaki, Franz Seitelberger, Marianne Fillenz, Gerhard Vollmer, W.W. Bartley III, Gerard Radnitzky, Ivan Slade, Alexandre Petrovic, Peter Michael Lingers, and Norbert Leser.
- Further impressions, editions, and translations of **1985(g)** into other languages are indicated in the year in which they are published.
- 1985(h)** The Non-Existence of Probabilistic Inductive Support. In *Foundations of Logic and Linguistics: Problems and their Solutions*, edited by Georg Dorn & Paul Weingartner, pp. 303-318. Plenum Press, New York & London. A continuation of **1983(f)** and **1984(d)**.
- 1985(i)** *Objektive Erkenntnis: Ein Evolutionärer Entwurf*. Buchclub Ex Libris, Zürich. A licensed reprint of **1984(c)**.
- 1985(j)** Has Inductive Probability been Proved Impossible?. Popper and Miller Reply. *Nature* **315**, 6019, 6th June 1985, p. 461. A continuation of **1983(f)**, **1984(d)**, and **1985(h)**.
- 1984(j₁)** (with John C. Eccles) *Das Ich und sein Gehirn*. R. Piper & Co. Verlag, Munich. Fifth edition of **1982(q)**.
- 1985(k)** *Realism and the Aim of Science*, edited by W.W. Bartley III. Rowman & Littlefield, Totowa, N.J. A paperback edition of **1983(c)**.
- 1985(l)** *Realism and the Aim of Science*, edited by W.W. Bartley III. Hutchinson Publishing Company Ltd. London. A paperback edition of **1983(b)**.
- 1985(m)** Il mito della cornice. In *I Modi del Progresso: Teorie e episodi della razionalità*, edited by Marcello Pera & Joseph Pitt. Il Saggiatore, Milan. Italian translation of **1976(o)**.
- 1985(n)** Elogio del Disaccordo. *Corriere della Sera*, 16th June 1985. The title was not approved by Popper.
- 1985(o)** Frühe Erinnerungen. In *SOS Kinderdorf, Jahrbuch 1985*, Herausgeber SOS Kinderdorf-Verlag, Innsbruck.
- 1985(p)** *Jame-ye Bāz Doshmananash [The Open Society and Its Enemies]*, volume II. Melli Publishing House Inc., Beverley Hills, California. Persian translation by A.A. Mohajer of volume II of **1980(y)**. Continuation of **1984(z₁₀)**.
- 1985(q)** *Gagalnya Historisisme*. Hak terjemahan pada LP3ES, Jakarta. Indonesian translation by Nena Suprpto of **1984(z₁)**.

- 1985(r)** *Conjectures et réfutations – La croissance du savoir scientifique*. Éditions Payot, Paris, 1985. French translation by Michelle-Irène B. de Launay & Marc B. de Launay of **1984(z₅)**.
- 1985(s)** *El coneixement objectiu. Un enfocament evolutiu*. Edicions 62, Barcelona. Catalan translation by Núria Roig of **1983(r)**.
- 1985(t)** (with Thomas D. Angelidis) Towards a Local Explanatory Theory of the Einstein-Podolsky-Rosen-Bohm Experiment. In *Symposium on the Foundations of Modern Physics*, edited by P. Lahti & P. Mittelstaedt, pp. 37-49. World Scientific Publishing Co.
- 1985(u)** *Realismo y el objetivo de la ciencia. Post Scriptum a La lógica de la investigación científica*, volume I, edited by W.W. Bartley III. Editorial Tecnos, Madrid. Spanish translation by Marta Sansigre Vidal of **1983(b)**.
- 1985(v)** *Teoría cuántica y el esma en física. Post Scriptum a La lógica de la investigación científica*, volume III, edited by W.W. Bartley III. Editorial Tecnos, Madrid. Spanish translation by Marta Sansigre Vidal of **1982(b)**.
- 1985(w)** (with John C. Eccles) *El yo y su cerebro*. Editorial Labor Universitaria, S.A., Barcelona. Third impression of **1980(r)**. A paperback.
- 1985(x)** Realism and Quantum Theory. In *Determinism in Physics. Proceedings of the Second International Meeting on Epistemology*, edited by Eftichios Bitsakis & Nikos Tambakis, pp. 11-29. Gutenberg Publishing Company, Athens.
- 1985(y)** Falsifizierbarkeit. In *Christlicher Glaube in Moderner Gesellschaft*. Quellenband 2, *Im Bann der Natur*, edited by Hoimar von Ditfurth & Rudolf Walter, pp. 214f. Herder Verlag, Freiburg im Breisgau. An excerpt from **1984(a)**. The title is not Popper's.
- 1985(z)** Grenzenlose Unwissenheit. *Ibid.*, p. 221. An excerpt from **1975(v)**. The title is not Popper's.
- 1985(z₁)** Einstein und die Amöbe. *Ibid.*, pp. 227-229. An excerpt from **1984(c)**. The title is not Popper's.
- 1985(z₂)** Festvortrag: 40 Jahre Naturwissenschaft. In *Der Beitrag: Europas Erbe und Auftrag. Europäisches Forum Alpbach 1984*, edited by Otto Molden. Österreichisches College, Vienna, 1985.
- 1985(z₃)** Foreword to *Treatise on Critical Reason* by Hans Albert. Princeton University Press, Princeton, N.J.
- 1985(z₄)** *Conjectures and Refutations*. Routledge & Kegan Paul, London. Seventh impression of **1972(d)**.

- 1985(z₅)** *La lògica de la investigació científica*. Editorial Tecnos, Madrid. Eighth impression of **1962(r)**.
- 1985(z₆)** *Uncended Quest: An Intellectual Autobiography*. Open Court Publishing Company, La Salle, Illinois. First paperback edition of **1982(g)**.
- 1985(z₇)** *Misericordia dello storicismo*. Giangiacomo Feltrinelli Editore, Milan. Seventh impression of **1975(a)**.
- 1985(z₈)** *Búsqueda sin término: Una autobiografía intelectual*. Editorial Tecnos, Madrid, 1985. Second impression of **1977(v)**.
- 1985(z₉)** *La lògica de la investigació científica*. Editorial Laia, Barcelona, 1985. Catalan translation by Ronald Puppo i-Bunds, with some *Appendices* omitted, of **1983(a)**. Edited and introduced by Magí Cadevall i Soler.
- 1986(a)** *Poscritto alla logica della scoperta scientifica*, edited by W.W. Bartley III. Volume II: *L'universo aperto. Un argomento per l'indeterminismo*. Il Saggiatore, Milan. Second impression of **1984(j)**.
- 1986(b)** *El universo abierto. Un argumento en favor del indeterminismo. Post Scriptum a La lógica de la investigación científica*, volume. II, edited by W.W. Bartley III. Editorial Tecnos, Madrid. Second impression of **1984(r)**.
- 1986(c)** *Popper: Autobiografia Intelectual*. Editoria Cultrix, São Paulo. Portuguese translation by Leonidas Hegenberg & Octanny Silveira da Motta of **1976(g)**.
- 1986(d)** *Uncended Quest: An Intellectual Autobiography*. Flamingo Series of Fontana Paperbacks, Collins Publishing Group, London. A reprint, with Postscript and updated bibliography, of **1982(k)**.
- 1986(e)** *Offene Gesellschaft – offenes Universum. Ein Gespräch über das Lebenswerk des Philosophen*. Serie Piper, R. Piper GmbH & Co. KG, Munich. A shortened edition of **1983(h)**.
- 1986(f)** (with David Miller) Yin-nai gezui bu ko nang seng ti cheng ming. *Tze-yen huo-hsueh yü tze-hsueh wen ti*, 2, 1986, pp. 2-4. Chinese translation by Qiu Renzong of **1983(f)**.
- 1986(g)** (with David Miller) Deductive Dependence. In *Actes IV Congrès Català de Lògica*. Universitat Politècnica de Catalunya & Universitat de Barcelona, pp. 21-29.
- 1986(h)** *The Poverty of Historicism*. Ark Paperbacks, Routledge & Kegan Paul, London & New York. Twelfth impression of **1961(b)**.
- 1986(i)** Die Erkenntnistheorie und das Problem des Friedens. In *Denken über die Zukunft* (Vorträge und Diskussionen des Symposiums vom 12./13. September

1985 im Kongresshaus Zürich), edited by Hannes Keller. Schweizer Illustrierte, Zürich. Contained in **1994(y)**.

- 1986(j)** (with Konrad Lorenz) *Die Zukunft ist offen. Das Altenberger Gespräch. Mit den Texten des Wiener Popper Symposiums*, edited by Franz Kreuzer. Serie Piper. R. Piper & Co. Verlag, Munich. Second impression of **1985(g)**.
- 1986(k)** *The Open Society and Its Enemies*. Volumes I and II. Routledge & Kegan Paul, London. A reprint of volume II of 1973(m).
- 1986(l)** *La società aperta e i suoi nemici*. Volume II: *Hegel e Marx falsi profeti*. Armando Armando Editore, Rome. Italian translation by Renato Pavetto of **1969(q)**.
- 1986(m)** Realism and a Proposal for a Simplified New Variant of the EPR-Experiment. In *Foundations of Physics. A Selection of Papers Contributed to the Physics Section of the 7th International Congress of Logic, Methodology and Philosophy of Science*, edited by Paul Weingartner & Georg Dorn, pp. 227-249. Verlag Holder-Pichler-Tempsky, Vienna.
- 1986(n)** Preface to *El debate de la teoria cuántica* by Franco Selleri. Alianza Editorial, S.A., Madrid. Spanish translation (from the Italian) by Miguel Ferrero Meiga.
- 1986(o)** Preface to *He Diamache gia ten Kuantike Theoria* by Franco Selleri. Seira Epistemologie 1. Gutenberg Solonos 103, Athens. Greek translation by Nixos Tampaxes.
- 1986(p)** (with Konrad Lorenz) [Japanese translation of **1985(g)**]. Orion Press, Shisaku-sha Publishing Co., Tokyo. Japanese translation of **1985(g)** by Hikaru Teruji.
- 1986(q)** *Bída Historicismu*. Preklad Edice Prameny, Brno. Czech translation of **1984(z₁)**.
- 1986(r)** Cel Nauki. In *Fakt i teoria. Teksty filozoficzne*, edited by Dariusz Aleksandrowicz. University of Wrocław. Polish translation by Leszek Koczanowicz of chapter 5 of **1979(a)**.
- 1986(s)** Prawda, racjonalność i rozwój wiedzy naukowej. *Ibid.* Polish translation by Dariusz Aleksandrowicz of chapter 10 of **1972(d)**.
- 1986(t)** *Cai Xiang Yu Fan Be*. The Shanghai Translation Publishing House, Shanghai. Chinese translation by Fu Ji-zhong of **1984(z₅)**.
- 1986(u)** Predicting overt behavior versus predicting hidden states. *Behavior and Brain Sciences* 9, No. 2, June 1986, pp. 254f.

- 1986(v)** (with John C. Eccles) Dialog über den selbstbewussten Geist, das Gehirn und die Grenzen der Evolutions-Theorie. In *Lust an der Erkenntnis: Die Philosophie des 20. Jahrhunderts*, edited by Volker Spierling, pp. 344-357. Serie Piper, Band 547. Munich, 1986. An excerpt from pp. 649-659 of **1985(j)**. The title of the excerpt is the publishers'.
- 1986(w)** Naturgesetze und theoretische Systeme. *Ibid.*, pp. 379-391. A reprint of **1972(j)**.
- 1986(x)** Utopia and Violence. *World Affairs* **149**, 1, 1986. A reprint of chapter 18 of **1985(z₄)**.
- 1986(y)** *K'ui Fang Shé Hui Ch'í Ch'í Ti Jen*, volume I: *Pai La T'u Ti Fu Chou*. Kuei Kuan Books, Taiwan. Second impression of **1984(z₁₂)**. Unauthorized.
- 1986(z)** *K'ui Fang Shé Hui Ch'í Ch'í Ti Jen*, Volume II: *Yü Yen Ti Kao Chao*. Kuei Kuan Books, Taiwan. Second impression of **1984(z₁₃)**. Unauthorized.
- 1986(z₁)** (with John C. Eccles) *The Self and its Brain*. Routledge, London. A reprint of **1977(u)**.
- 1986(z₂)** *La ricerca non ha fine: Autobiografia intellettuale*. Armando Armando Editore, Rome. A reprint of **1978(z₁₀)**.
- 1986(z₃)** A new interpretation of darwinism. The first Medawar Lecture, given 12 June 1986, at the Royal Society, London.
- 1987(a)** Bell's Theorem: A Note on Locality. In *Microphysical Reality and Quantum Formalism*, edited by G. Tarozzi & A. van der Merwe, pp. 413-417. D. Reidel Publishing Co., Dordrecht.
- 1987(b)** (with John C. Eccles) *Das Ich und sein Gehirn*. R. Piper & Co. Verlag, Munich. Sixth edition, in paperback, of **1982(q)**.
- 1987(c)** (with David Miller) Why Probabilistic Support Is Not Inductive. *Philosophical Transactions of the Royal Society*, A **321**, No. 1562, 30th April 1987, pp. 569-596. A continuation of **1983(f)**, **1984(d)**, **1985(h)**, and **1985(j)**.
- 1987(d)** *Offene Gesellschaft – offenes Universum. Franz Kreuzer im Gespräch mit Karl R. Popper*, edited by Franz Kreuzer. Franz Deuticke Verlagsgesellschaft m.b.H., Vienna. Fourth edition, revised, of **1982(r)**.
- 1987(e)** *Sociedade aberta Universo aberto. Entrevista conduzida por Franz Kreuzer*. Publicações Dom Quixote, Lisbon. Portuguese translation by Maria Helena Rodrigues de Carvalho of **1982(r)**.
- 1987(f)** The Myth of the Framework. In *Rational Changes in Science: Essays on Scientific Reasoning* (BSPS, volume 98), edited by Joseph C. Pitt & Marcello

Pera. D. Reidel Publishing Company, Dordrecht. A reprint of **1976(o)**.
English version of **1985(m)**.

- 1987(g)** *Objektive Erkenntnis – Ein Evolutionärer Entwurf*. Hoffman & Campe, Hamburg. Contains a Postscript by Helmut Schmidt. Edited by Joachim Fest & Wolf Jobst Siedler. A licensed reprint of **1984(c)**.
- 1987(h)** Capitalism and its Fate. In *Marr Refuted*, edited by Ronald Duncan & Colin Wilson. Ashgrove Press, Bath. An excerpt (without footnotes) from chapter 20 of **1966(a)**.
- 1987(i)** **absent**
- 1987(j)** Reden in verwickelter Weise – die Erbsünde deutscher Philosophie. *Die Welt*, 6th July 1987, No. 153. An interview with Albert Menne & Günter Zehmin. The title is not Popper's. Continued in **1987(k)** and **1987(l)**. Czech translation: **1993(z₃)**.
- 1987(k)** Nicht nur Physiker, auch Philosophen können schlechte Philosophen sein. *Die Welt*, 8th July 1987, No. 155. An interview with Albert Menne & Günter Zehmin. The title is not Popper's. A continuation of **1987(j)**.
- 1987(l)** Wir spielen auf unserem Gehirn wie ein Pianist auf dem Flügel. *Die Welt*, 11th July 1987, No. 158. An interview with Albert Menne & Günter Zehmin. The title is not Popper's. A continuation of **1987(j)** and **1987(k)**.
- 1987(m)** *Sociedade aberta Universo aberto. Entrevista conduzida por Franz Kreuzer*. Publicações Dom Quixote, Lisbon. A reprint of **1987(e)** in the series OPUS Biblioteca de Filosofia.
- 1987(n)** Ninguém sabe o suficiente para ser intolerante. *Expresso*, 28th March 1987. Portuguese translation of an interview with João Carlos Espada. The title is not Popper's.
- 1987(o)** Die erkenntnistheoretische Position der Evolutionären Erkenntnistheorie. In *Die Evolutionäre Erkenntnistheorie: Bedingungen. Lösungen. Kontroversen*, edited by Rupert Riedl & Franz M. Wuketits. Verlag Paul Parey, Berlin & Hamburg. Contained in **1994(y)**. Italian translation (unauthorized): **1992(z₁₆)**.
- 1987(p)** *O realismo e o objectivo da ciência*. Publicações Dom Quixote, Lisbon. Portuguese translation by Nuno Ferreira da Fonseca of **1983(b)**.
- 1987(q)** Letter to the Editor about Hegel and Hitler. *The Salisbury Review*, July 1987, p. 54.
- 1987(r)** Zur Theorie der Demokratie. *Der Spiegel* (Hamburg), 32, 3rd August 1987, pp. 54f. Continued in **1994(y)**.

- 1987(s)** Apuntes a la teoría de la democracia. *El País*, 8th August 1987. Spanish translation of **1987(r)**
- 1987(t)** Popper versus Copenhagen. *Nature* **328**, 6132, 20th August 1987, p. 675. A letter in reply to M.J. Collett & R. London.
- 1987(u)** (with John C. Eccles) *Jigu to Noh*. Shisaki Sha Publishers, Tokyo. Japanese translation by Utaoka Ohmura & Nobushige Sawada of **1984(e)**.
- 1987(v)** *Das Elend des Historizismus*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Sixth edition, revised, of **1965(g)**.
- 1987(w)** *Ausgangspunkte: Meine Intellektuelle Entwicklung*. Hoffmann und Campe, Hamburg. Fourth edition of **1979(w)**. This edition wrongly carries the imprint of **1984(g)**.
- 1987(x)** *Objektive Erkenntnis: Ein Evolutionärer Entwurf*. Hoffmann und Campe, Hamburg. In the series *Klassiker des Denkens*. Contains a Postscript by Helmut Schmidt. A licensed reprint of **1984(c)**.
- 1987(y)** Zur Theorie der Demokratie. In *Aus erster Hand: Meinungen, Informationen und Perspektiven*. Die Erste Österreichische Spar-Casse-Bank, No. 3, Vienna.
- 1987(z)** *Li Shi Jue Ding Lun De Pin Kun*. Hua Xia Press, Beijing. Chinese translation by Du Ruji of **1986(h)**.
- 1987(z₁)** Philosophie et physique: L'influence de la physique sur certaines spéculations métaphysiques touchant la structure de la matière. *Revue de Métaphysique et de Morale*, No. 2, 1987, pp. 230-237. French translation by Michelle-Irène B. de Launay of **1960(e)**.
- 1987(z₂)** Toleration and Intellectual Responsibility. In *On Toleration*, edited by Susan Mendus & David Edwards. Clarendon Press, Oxford. Contained in **1992(t)**. Italian translation: **1990(l)**.
- 1987(z₃)** *The Logic of Scientific Discovery*. Hutchinson Publishing Company Ltd, London. Twelfth impression of **1959(a)**.
- 1987(z₄)** *The Logic of Scientific Discovery*. Hutchinson Publishing Company Ltd, London. Thirteenth impression of **1959(a)**.
- 1987(z₅)** *Kexue Zhishi Jinhualun* [*The Evolution of Scientific Knowledge*]. Joint Pub. Cop., 166 Chaonei Dajie, Beijing. Chinese translation by Shu-Li Ji of fifteen essays taken from Popper's scientific works.
- 1987(z₆)** Conjectures and Refutations: The Growth of Scientific Knowledge. In *Congetture e confutazioni di Popper e il dibattito epistemologico post-popperiano*, edited by Giorgio Brianese. A reprint of chapter 1 of **1984(z₅)**.

- 1987(z₇)** *Auf der Suche nach einer besseren Welt: Vorträge und Aufsätze aus dreißig Jahren.* Serie Piper. R. Piper & Co. Verlag, Munich. Second edition of **1984(b)**.
- 1987(z₈)** *Ke Guan Zhu Shu.* The Shanghai Translation Publishing House, Shanghai. Chinese translation by Shu Wei-gang of **1983(r)**.
- 1987(z₉)** Campbell on the Evolutionary Theory of Knowledge. In *Evolutionary Epistemology, Rationality, and the Sociology of Knowledge*, edited by Gerard Radnitzky & W.W. Bartley III, pp. 115-120. Open Court Publishing Company, La Salle, Illinois. In hardback and paperback.
- 1987(z₁₀)** Natural Selection and the Emergence of Mind. *Ibid.*, pp. 139-155. A reprint of **1978(z)**.
- 1987(z₁₁)** Excerpts from volume II of **1958(i)**, **1982(q)**, **1982(r)**, and **1984(c)**. In *Philosophie zum Lesen*. Verlegergemeinschaft Neues Schulbuch. Verlag Herder, Vienna. Unauthorized.
- 1987(z₁₂)** (with John C. Eccles) *Das Ich und sein Gehirn.* R. Piper & Co. Verlag, Munich. Seventh edition, in paperback, of **1982(q)**.
- 1987(z₁₃)** *Spoleczeństwo Otwarte i Jego Wrogowie. Urok Platona.* Kwartalnik Polityczny 'Krytyka', Warsaw. Polish translation by Tadeusz Korczyński of volume I of **1986(k)**. Unauthorized. Contains an unauthorized Preface by L. Kolakowski.
- 1987(z₁₄)** *Spoleczeństwo Otwarte i Jego Wrogowie. Wysoka Fala Proroctw: Hegel, Marks i Następców.* Kwartalnik Polityczny 'Krytyka', Warsaw. Polish translation by Tadeusz Korczyński of volume II of **1986(k)**. Unauthorized.
- 1987(z₁₅)** *I due problemi della teoria della conoscenza.* Il Saggiatore, Milan. Italian translation by Mario Trinchero of **1979(z₂)**.
- 1988(a)** Karl Popper: Un génie dans la banlieue de Londres. *Le Figaro* (Paris), No. 13517, 13th February 1988. French translation of an interview with Guy Sorman. The title is the interviewer's.
- 1988(b)** Es gibt keine Eliten. *INNOVATIO – Internationales Forum für Wirtschaft und Gesellschaft*, 1/2, 1988. Innovatio Verlag AG, Fribourg.
- 1988(c)** *O universo aberto.* Publicações Dom Quixote, Lisbon. Portuguese translation by Nuno Ferreira da Fonseca of **1982(a)**.
- 1988(d)** Sur la théorie de la démocratie. *MEDIASPOUVOIR*, No. 10, April/June 1988, Bayard-Press, Paris. French translation by Philippe Thureau-Dangin of revised version of **1987(r)**.

- 1988(e)** *Realism and the Aim of Science*, edited by W.W. Bartley III. Hutchinson Publishing Company Ltd. London. Second impression of **1985(l)**.
- 1988(f)** *A miséria do historicismo*. Editora Cultrix, São Paulo. Second impression of **1980(m)**.
- 1988(g)** *Auf der Suche nach einer besseren Welt: Vorträge und Aufsätze aus dreißig Jahren*. R. Piper & Co. Verlag, Munich. Third edition, corrected, of **1984(b)**.
- 1988(h)** La psicología debería formar parte de la biología. *La Nación*, 23rd June 1988. Spanish translation of an interview with Sara Gallardo.
- 1988(i)** *Misère de l'historicisme*. Éditions Presses Pocket AGORA, Paris. French translation by Hervé Rousseau of **1969(o)**, revised and augmented by Renée Bouveresse.
- 1988(j)** Popper on Democracy: The open society and its enemies revisited. *The Economist*, April 23rd 1988, pp. 25-28. The title is not Popper's. Chinese translation (unauthorized): **1989(zg)**.
- 1988(k)** *Misericordia dello storicismo*. Giangiacomo Feltrinelli Editore, Milan. Eighth impression of **1975(a)**.
- 1988(l)** Jak widzę filozofię [How I See Philosophy]. *Przegląd Powszechny*, 7/7/803-804, 1988. Jesuizi, Warszawa. Polish translation by Tadeusz Szubka of **1975(o)**.
- 1988(m)** *The Open Universe: An Argument for Indeterminism*, edited by W.W. Bartley III. Hutchinson & Company, London. A paperback reprint of **1982(a)**.
- 1988(n)** (with Konrad Lorenz) *Die Zukunft ist offen. Das Altenberger Gespräch. Mit den Texten des Wiener Popper Symposiums*, edited by Franz Kreuzer. Serie Piper, R. Piper GmbH & Co. KG, Munich. Third impression of **1985(g)**.
- 1988(o)** The allure of the open future. *The Guardian*, 29th August 1988. An excerpt from the lecture, entitled 'A World of Propensities: Two New Views of Causality', given at the XVIIIth World Congress of Philosophy, Brighton on 24th August 1988. Unauthorized. First authorized English publication: **1990(f)**.
- 1988(p)** Il futuro ha i dadi truccati. *Corriere della Sera* (Milan), 4th September 1988. Italian translation by Monica Levy of 'A World of Propensities: Two New Views of Causality' (**1988(o)**, **1990(f)**).
- 1988(q)** Karl Popper y la verdad – Un mundo de propensiones (Un nuevo aspecto de la causalidad). *El País* (Madrid), 15th September 1988. Spanish translation by Carmen Viamonte of 'A World of Propensities: Two New Views of Causality' (**1988(o)**, **1990(f)**).

- 1988(r)** El encanto de un futuro abierto. *La Vanguardia* (Barcelona), 6th September 1988. Spanish translation by Marita Oses of **1988(o)**, **1988(o)**, **1990(f)**).
- 1988(s)** Um mundo de propensões – Uma nova perspectiva a causalidade. *Semanário* (Lisbon), 1st October 1988. Portuguese translation by João Carlos Espada of 'A World of Propensities: Two New Views of Causality' (**1988(o)**, **1990(f)**).
- 1988(t)** Socrate nel 2000. *L'ora*, 14th October 1988. Italian translation of an interview with Giacchino Lavanco. The title is the interviewer's.
- 1988(u)** Odio le mode e i media. *Sagra dell'Uva – Canicutti, Giornale di Sicilia*, 15th October 1988. Italian translation of an interview with Gianfranco Marrone. The title is the interviewer's.
- 1988(v)** Społeczeństwo Otwarte. *Tygodnik Demokratyczny* (Warsaw), No. 45, 6th November 1988. Polish translation by Jerzy Gochnio of **1988(j)**.
- 1988(w)** *Bemerkungen zur Theorie und Praxis des demokratischen Staates*. Bank Hofmann AG, Zürich. Contained in **1994(y)**. English version contained in **1996(z₂)**. Portuguese translation: **1990(i)**; Spanish: **1990(z)**, **1990(z₁)**; Italian: contained in **1992(z₃)**; French: contained in **1993(y)**; **1997(j)**.
- 1988(x)** Campbell on the Evolutionary Theory of Knowledge. In *Evolutionary Epistemology, Rationality, and the Sociology of Knowledge*, edited by Gerard Radnitzky & W.W. Bartley III, pp. 115-120. Open Court Publishing Company, La Salle, Illinois. A reprint, with some alterations, of the paperback edition of **1987(z₉)**.
- 1988(y)** Natural Selection and the Emergence of Mind. *Ibid.*, pp. 139-155. A reprint, with some alterations, of **1987(z₁₀)**.
- 1988(z)** *En Intellektuell Sjåle Biografi*. Bokförlaget Doxa AB, Lund. Swedish translation by Svante Hansson of **1986(d)**.
- 1988(z₁)** [Excerpts from volume 2 of **1958(i)**, **1982(q)**, **1982(r)**, and **1984(c)**]. In *Philosophie zum Lesen*, Verlegergemeinschaft Neues Schulbuch. Verlag Herder, Vienna. A reprint of **1987(z₁₁)**. Unauthorized.
- 1988(z₂)** Um mundo de propensões – Uma nova perspectiva sobre a causalidade. *Risco* (Lisbon), No. 10, Autumn/Winter 1988. A revised reprint of **1988(s)**.
- 1988(z₃)** *Conocimiento objetivo: Un enfoque evolucionista*. Editorial Tecnos, Madrid. Third edition of **1974(f)**.
- 1988(z₄)** *Bo Pu Er Si Xiang Zi Shu*. The Shanghai Translation Publishing House, Shanghai. Chinese translation by Yue-Se Zhao of **1986(d)**. Unauthorized.
- 1989(a)** *Conjeturas y refutaciones: El desarrollo del conocimiento científico*. Ediciones Paidós, Barcelona & Buenos Aires. Third edition of **1967(n)**.

- 1989(b)** *La quête machéenne*. Éditions Presses Pocket AGORA, Calmann-Lévy, Paris. French translation by Renée Bouveresse in collaboration with Michelle Bouin-Naudin of **1976(g)**, revised and augmented by Renée Bouveresse. Second edition of **1981(z₁₁)**.
- 1989(c)** *Auf der Suche nach einer besseren Welt: Vorträge und Aufsätze aus dreißig Jahren*. R. Piper & Co. Verlag, Munich. Fourth edition, revised, of **1984(b)**.
- 1989(d)** *Em busca de um mundo melhor*. Editorial Fragmentos Lda, Lisbon. Portuguese translation by Teresa Curvelo of **1988(g)**. First volume of a series entitled *Colecção Problemas*, edited by João Carlos Espada.
- 1989(e)** Rôle de l'auto-critique dans la création. *Diogène* (Paris) **145**, January/February 1989, pp. 38-48. French translation by Marc-André Béra of **1979(p)**.
- 1989(f)** Über das Problem der Demokratie. In *Leidenschaft zur praktischen Vernunft* (Festschrift in honour of Helmut Schmidt's 70th birthday), edited by Manfred Lahnstein & Hans Matthöfer, pp. 391-398. Siedler Verlag GmbH, Berlin, 1989.
- 1989(g)** (with Konrad Lorenz) *Il futuro è aperto*. Rusconi Libri S.p.A., Milan. Italian translation by Dario Antiseri of **1988(n)**. Contains an introduction by the translator.
- 1989(h)** (with John C. Eccles) *Das Ich und sein Gehirn*. R. Piper & Co. Verlag, Munich. Eighth edition (first edition in Serie Piper paperback) of **1982(q)**.
- 1989(i)** *Quantum Theory and the Schism in Physics*, edited by W.W. Bartley III. Unwin Hyman Ltd, London. Volume III of the *Postscript to The Logic of Scientific Discovery*. First paperback edition of **1982(b)**.
- 1989(j)** *Logik der Forschung*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Ninth German edition, revised, of **1934(b)** and **1966(e)**.
- 1989(k)** Meglio liberi che verdi. *Panorama* **XXVII**, No. 1213, 16th July 1989, pp. 112f. Italian translation of an interview with Sebastiano Maffettone.
- 1989(l)** *A lógica da pesquisa científica*. Editora Cultrix, São Paulo and Editora da Universidade de São Paulo. First paperback edition of **1975(y)**.
- 1989(m)** *A teoria dos quanta e o cisma na física*. Publicações Dom Quixote, Lisbon. Portuguese translation by Nuno Ferreira da Fonseca of **1982(b)**.
- 1989(n)** Le mythe du cadre de référence. In *Karl Popper et la science d'aujourd'hui* (Colloque de Cerisy), pp. 13-44. Aubier, Paris. French translation by Renée Bouveresse of **1976(o)**.
- 1989(o)** Gedankenskizzen über das, was wichtig ist. In *Geist und Natur*, edited by Hans-Peter Dürr & Walther Ch. Zimmerli, pp. 381-388. Scherz Verlag Berne, Munich, & Vienna. German translation by Markus Würmli of 'Outline of my

Views of What Matters', a lecture given before an International Congress, 'Welt und Wirklichkeit im Wandel der Erfahrung', Hanover on 27th May 1988. The title is the translator's.

- 1989(p)** *Changing Our View of Causality: A World of Propensities*. In *The Philosophy of Science Lecture Series*, sponsored by the Vera & Walter Morris Foundation, Little Rock, Arkansas. The text of the lecture, given on 27th April 1988, is followed by the text of a Round Table Discussion held on the same day.
- 1989(q)** *A Historizmus Nyomorúsága*. Akadémiai Kiadó, Budapest. Hungarian translation by Kelemen Tamás of **1986(h)**.
- 1989(r)** Zwei Bedeutungen von Falsifizierbarkeit. In *Handlexikon zur Wissenschaftstheorie*, edited by Helmut Seiffert & Gerard Radnitzky, pp. 82-86. Ehrenwirt GmbH Verlag, Munich.
- 1989(s)** *Alla ricerca di un mondo migliore*. Armando Editore, Rome. Italian translation by Barbara di Noi of **1984(b)**.
- 1989(t)** Foreword to *Evolution of the Brain: Creation of the Self* by John C. Eccles. Routledge, London.
- 1989(u)** *Conjectures and Refutations*. Routledge, London. Fifth edition, revised and corrected, of **1963(a)**. Incorporates **1982(i)**.
- 1989(v)** Popper on Schrödinger. *Nature* **342**, 6248, 23rd November 1989, p.337.
- 1989(w)** *Otevřená Společnost a Její Nepřátelé. Příliv Věstectví: Hegel, Marx a Co Následovalo*. Překlad Edice Prameny, Brno. Czech translation of volume II of **1986(k)**. Unauthorized.
- 1989(x)** On a Little Known Chapter of Mediterranean History. *Catalònia*, No. 15, September 1989, pp. 31-37. Centre Unesco de Catalunya, Barcelona. This is the full text of an address given at the Palace of the Generalitat de Catalunya in Barcelona on 24th May 1989, on being awarded the Catalonia International Prize. The title has been omitted. Contained in **1992(t)**. Catalan translation: **1989(y)**.
- 1989(y)** Sobre un capítol poc conegut de la història mediterrània. Programme of the presentation ceremony of the Catalonia International Prize, pp. 11-15. Institut Català d'Estudis Mediterranis, Barcelona. Catalan translation of **1989(x)**.
- 1989(y₁)** Sobre un capítol poc conegut de la història mediterrània. *Diari de Barcelona [Mirador]*, 25th May 1989, pp. 26f. A reprint of **1989(y)**.
- 1989(y₂)** Sobre un capítol poc conegut de la història mediterrània. *Catalònia*, No. 15, September 1989, [no page numbers]. Centre Unesco de Catalunya, Barcelona. A reprint of **1989(y)**.

- 1989(z)** Creative Self-Criticism in Science and in Art. *Diogenes* (Casalini Libri, Fiesole) **145**. Spring 1989; pp. 36-45. English translation of **1979(p)**.
- 1989(z₁)** *La sociedad abierta y sus enemigos*. Ediciones Paidós, Barcelona, & Buenos Aires. Third impression of **1981(z₂₇)**.
- 1989(z₂)** *Nedza historyczna... z dodaniem fragmentów Autobiografii*. Wydawnictwo Krag, Warsaw. Polish translation by Stefan Amsterdamski of **1986(h)** and excerpts from **1986(d)**. Unauthorized.
- 1989(z₃)** (with John C. Eccles) *Das Ich und sein Gehirn*. R. Piper & Co. Verlag, Munich. Ninth edition (second edition in Serie Piper paperback) of **1982(q)**.
- 1989(z₄)** *The Open Society and Its Enemies*. Volume I. Routledge London. A reprint of volume I of **1966(a)**.
- 1989(z₅)** *The Poverty of Historicism*. Ark Paperbacks, Routledge & Kegan Paul, London & New York. Thirteenth impression of **1961(b)**.
- 1989(z₆)** Entrevista con Karl Popper. *Arbor* (Madrid), No. 522, June 1989, pp. 9-35. Spanish translation of an interview with Jesús Mosterín.
- 1989(z₇)** *K'ai Fang Shé Hui Chi Ch'i Ti Jen*. Volume I: *Pai La T'u Ti Fu Chou*. Kuei Kuan Books, Taiwan. Third impression of **1984(z₁₂)**. Unauthorized.
- 1989(z₈)** *K'ai Fang Shé Hui Chi Ch'i Ti Jen*. Volume II: *Yü Yen Ti Kao Chao*. Kuei Kuan Books, Taiwan. Third impression of **1984(z₁₃)**. Unauthorized.
- 1989(z₉)** K'ai Fang Shé Hui Chi Ch'i Ti Jen Ti Tsai Shêng Szu. *Contemporary* (Taiwan), **42**. Chinese translation by Chiün-Hsiung Hua of **1988(j)**. Unauthorized.
- 1989(z₁₀)** Interview with Karl Popper. In *Jüdische Portraits. Photographien und Interviews* by Herlinde Koelbl. S. Fischer Verlag GmbH, Frankfurt am Main, pp. 189-191.
- 1989(z₁₁)** *Congetture e confutazioni: Lo sviluppo della conoscenza scientifica*. Universale Paperbacks, Il Mulino, Bologna. Third impression of **1976(x)**.
- 1990(a)** (with Konrad Lorenz) *L'avenir est ouvert*. Flammarion, Paris. French translation by Jeanne Etoré of **1985(g)**.
- 1990(b)** 'Die Deutschen sollten ihre Wohltäter nicht beißen'. *Die Welt*, No. 42, 19th February 1990. Part I of an interview with Manfred Schell. Spanish translations: **1990(h)**, **1990(m)**; Portuguese: **1990(z₇)**.
- 1990(c)** 'Die Technik hat uns befreit. Was soll diese Hetze?'. *Die Welt*, No. 44, 21st February 1990. Part II of an interview with Manfred Schell. Spanish translation: **1990(h)**, **1990(n)**; Portuguese: **1990(z₇)**.

- 1990(d)** 'Intellektuelle sind anmaßend und bestechlich'. *Die Welt*, No. 46, 23rd February 1990. Part III of an interview with Manfred Schell. Spanish translations: **1990(h)**, **1990(o)**; Portuguese: **1990(z7)**.
- 1990(e)** *Die Welt* im Gespräch mit dem Philosophen Sir Karl Popper – Sonderdruck des dreiteiligen Interviews. Februar 1990. Special edition of **1990(b)**, **1990(c)**, and **1990(d)**. Czech translation: **1993(z4)**.
- 1990(f)** *A World of Propensities*. Thoemmes, Bristol. Chinese translation: **1991(r)**; Portuguese: **1991(s)**; Italian: **1991(z3)**; French: **1992(b)**; Spanish: **1992(c)**; Catalan: **1993(q)**; German: **1995(o)**; Japanese: **1995(p)**; Polish: **1996(e)**; Swedish: **1996(n)**; Danish (in part): **1994(z10)**.
- 1990(g)** Pyrite and the origin of life. *Nature* **344**, No. 6265, 29th March 1990, p. 387.
- 1990(h)** Alemania y la revolución europea. *El Independiente* (Madrid), 1st April 1990. Spanish translation of **1990(b)**, **1990(c)**, and **1990(d)**.
- 1990(i)** Algumas observações sobre a teoria e a praxis do Estado democrático. *Risco* (Lisbon). No. 13, Spring 1990, pp. 5-21. Portuguese translation by Teresa Curvelo of **1988(w)**.
- 1990(j)** (with John C. Eccles) *Das Ich und sein Gehirn*. R. Piper & Co. Verlag, Munich. Tenth edition (third edition in Serie Piper paperback) of **1982(q)**.
- 1990(k)** Mille anni di pace. *Il Mondo*, 14th/21st May 1990, pp. 91-94. Italian translation of an interview with Giuseppe Leuzzi.
- 1990(l)** Tolleranza e responsabilità intellettuale. In *Saggi sulla tolleranza*, Arnoldo Mondadori Editore, Milan. Italian translation by Daniela Gobetti of **1987(z2)**.
- 1990(m)** La causa de la paz. *La Nación* (Buenos Aires), 24th June 1990. Spanish translation by Rodolfo Modern of **1990(b)**.
- 1990(n)** La técnica nos ha liberado. *La Nación* (Buenos Aires), 1st July 1990. Spanish translation by Rodolfo Modern of **1990(c)**.
- 1990(o)** De las teorías a las ideologías. *La Nación* (Buenos Aires), 8th July 1990. Spanish translation by Rodolfo Modern of **1990(d)**.
- 1990(p)** (with Konrad Lorenz) *O futuro esta aberto*. Editorial Fragmentos, Lisbon. Portuguese translation by Teresa Curvelo of **1985(g)**.
- 1990(q)** Epistemology Without a Knowing Subject; and What is Dialectic?. Editorial Vaniias Semelides, Thessaloniki. Greek translation of chapter 3 of **1972(a)** and chapter 15 of **1972(d)**. Unauthorized.

- 1990(r)** Det åbne samfund vandt. *Berlingske Magasin*, Copenhagen, 29th July 1990. Danish translation of an interview with Peter Wivel.
- 1990(s)** Science: Conjectures and Refutations. In *Philosophy of Science and the Occult*, edited by Patrick Grim, pp. 104-110. State University of New York Press, Albany, N.Y. Second edition of **1982(n)**.
- 1990(t)** Alguns Problemas Práticos da Democracia. In *Balanço do Século, Ciclo de Conferências promovido pelo Presidente da República*, edited by Fernando Gil, pp. 73-85. Imprensa Nacional – Casa da Moeda, Lisbon. Portuguese translation by João Carlos Espada of a lecture entitled ‘On the Theory of Democracy’, given at Lisbon on 3rd October 1987.
- 1990(u)** *Le réalisme et la science. Post-scriptum à la Logique de la découverte scientifique*. I. Hermann éditeurs des sciences et des arts, Paris. French translation by Daniel Andler & Alain Boyer of **1983(b)**. Contains French translation of **1983(f)**.
- 1990(v)** Chinese translation by Feng Yunwen of **1979(p)**. *Diogenes* (Paris), No. 145, pp. 1-8.
- 1990(w)** *La lógica de la investigación científica*. Editorial Tecnos, Madrid. Ninth impression of **1962(r)**.
- 1990(x)** Eine wunderschöne Welt. In *Woran ich glaube*, p. 215. Gütersloher Verlagshaus Gerd Mohn.
- 1990(y)** Attenti ai pericoli della democrazia. *Europeo*, No. 44, 2nd November 1990, pp. 112-116. Italian translation of an interview with Giuliano Ferrieri.
- 1990(z)** Teoría y práctica del Estado democrático: Mercado cultural y democracia. *La Nación* (Buenos Aires), 23rd September 1990. Spanish translation by Rudolfo Modern of sections I-IV of **1988(w)**.
- 1990(z₁)** Teoría y práctica del Estado democrático: El Estado mínimo. *La Nación* (Buenos Aires), 30th September 1990. Spanish translation by Rudolfo Modern of sections V-XII of **1988(w)**.
- 1990(z₂)** *Freiheit und intellektuelle Verantwortung*. Liberales Forum, St Gallen. Unrevised text of a lecture given at St Gallen on 27th June 1989. Not for general distribution. Contained in **1994(y)**. Italian translation contained in **1992(z₃)**; French: contained in **1993(y)**.
- 1990(z₃)** *Alla ricerca di un mondo migliore*. Armando Editore, Rome. Second impression of **1989(s)**.
- 1990(z₄)** *Ti ine dialectiki? – Gnosioteoria choris gnostico hypokiemenon*. Philosophiki Erevna [Philosophical Enquiry Series], Vani Publishing House, Thessaloniki.

Greek translation by Nicolaos Avgelis of chapter 15 of **1963(a)** and of chapter 3 of **1972(a)**.

- 1990(z₅)** (with Günter Wächtershäuser) Progenote or Protogenote?. *Science* **250**, No. 4984, 23rd November 1990, p. 1070. A letter.
- 1990(z₆)** Preface to *O Poder – Uma nova análise social*, a Portuguese translation by Isabel Belchior of Bertrand Russell's *Power. A New Social Analysis*. Editorial Fragmentos Lda, Lisbon.
- 1990(z₇)** *Die Welt* em diálogo com o filósofo Sir Karl Popper. *Risco* (Lisbon), No. 15, Winter 1990–1991, pp. 9–29. Portuguese translation of **1990(b)**, **1990(c)**, and **1990(d)**.
- 1990(z₈)** 'I was a Marxist until the age of 17'. *Moscow News*, No. 46, 25th November / 2nd December 1990. A conversation with Andranik Migranyan, published in the form of an interview. Contains also an excerpt from **1966(a)**. Publication prepared by Vladimir Orlov. Unauthorized.
- 1990(z₉)** *La scena e la storia sul filo dei ricordi*. Edizione Casagrande, Bellinzona and Editoriale Jaca Book, Milan. Italian translation of an interview with Guido Ferrari in April 1984 at Locarno. Unauthorized.
- 1990(z₁₀)** L'orologio di Cartesio. *La Repubblica Cultura* (Rome), 6th October 1990. Italian translation by Ottavio Fatica of an address, entitled 'Mechanisms versus Creative Inventions – A Brief Survey of a Wide Open Problem', given in Venice on 5th October 1990. Another Italian translation: **1991(t)**.
- 1990(z₁₁)** *Auf der Suche nach einer besseren Welt: Vorträge und Aufsätze aus dreißig Jahren*. R. Piper & Co. Verlag, Munich. Fifth edition, corrected, of **1984(b)**.
- 1990(z₁₂)** *Unended Quest: An Intellectual Autobiography*. Open Court Publishing Company, La Salle, Illinois. Second paperback edition of **1982(g)**.
- 1990(z₁₃)** *The Logic of Scientific Discovery*. Unwin Hyman, London. Fourteenth impression of **1959(a)**.
- 1990(z₁₄)** (with John C. Eccles) *The Self and its Brain*. Routledge, London. A paperback reprint of **1977(u)**.
- 1990(z₁₅)** *The Open Society and Its Enemies*. Volume II. Routledge, London. A reprint of volume II of **1966(a)**.
- 1991(a)** Schöpferische Selbstkritik: Die Arbeit am Kunstwerk. *Österreichische Musikzeitschrift* (Vienna), JG. **46**, 1/2, 1991. A shortened and revised version of **1979(p)**.
- 1991(b)** Kepler: Seine Metaphysik des Sonnensystems und seine empirische Kritik. In *Wege der Vernunft. Festschrift zum 70. Geburtstag von Hans Albert*, edited by

- A. Bohnen & A.E. Musgrave, pp. 11-16. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Contained in **1994(y)**.
- 1991(c)** The Logic of Scientific Discovery. In *The World Treasury of Physics, Astronomy and Mathematics*, edited by Timothy Ferris, pp. 795-800. Little, Brown & Company, Boston. A reprint of section 85 of **1987(z₄)**.
- 1991(d)** Révolutions scientifiques et révolutions idéologiques. Introduction to *Karl Popper: Science et Philosophie*, edited by Renée Bouveresse & Hervé Barreau. Librairie Philosophique J. Vrin, Paris. French translation by Renée Bouveresse of **1975(p)**.
- 1991(e)** *La connaissance objective*. Aubier bibliothèque philosophique, Paris. French translation by Jean-Jacques Rosat of **1979(a)**. Contains a Preface by the translator.
- 1991(f)** The Non-Existence of Probabilistic Inductive Support. In *Logical Foundations* (Essays in honour of D.J. O'Connor), edited by Indira Mahalingam & Brian Carr, pp. 31-41. The Macmillan Press Ltd, London. A reprint of **1985(h)**.
- 1991(g)** Creative Self-Criticism in Science and in Art. In *Österreichische Musikzeitschrift* (Mozart Special 1991). A shortened and revised version of **1979(p)**.
- 1991(h)** Freiheit und intellektuelle Verantwortung. *Atel* (Hauszeitung der Aare-Tessin AG für Elektrizität), No. 138, Olten (Switzerland), May 1991, pp. 34-40. A reprint of **1990(z₂)**.
- 1991(i)** Ich bin ein Optimist, der nichts über die Zukunft weiß. Feuilleton Beilage der *Süddeutschen Zeitung*, 15th-16th June 1991, No. 136. Abbreviated version of **1992(e)**.
- 1991(j)** La libertad contra el cinismo. *El País*, Madrid, 3rd July 1991. Spanish translation by Luis Meana of **1992(e)**.
- 1991(k)** *A miséria do historicismo*. Editora Cultrix, São Paulo. Third impression of **1980(m)**.
- 1991(l)** *The Open Society and Its Enemies*. Volume I. Routledge, London. A reprint of volume I of **1966(a)**.
- 1991(m)** Two Theories on Democracy. *Al-Democrati*, 25th July 1991. Arabic translation by Samir S. Mahmood of **1988(j)**. Published by The Association of Iraqi Democrats, London.
- 1991(n)** *The Poverty of Historicism*. Routledge, London, & New York. Fourteenth impression of **1961(b)**.

- 1991(o)** *Ich weiss, dass ich nichts weiss – und kaum das? Karl Popper im Gespräch über Politik, Physik und Philosophie.* Verlag Ullstein GmbH, Berlin. A reprint of **1987(j)**, **1987(k)**, **1987(l)**, and **1990(e)**.
- 1991(p)** In URSS bancarotta intellettuale. *Corriere della Sera* (Milan), 28th August 1991. Italian translation of an interview with Riccardo Chiaberge.
- 1991(q)** *Conjeturas y refutaciones: El desarrollo del conocimiento científico.* Ediciones Paidós, Barcelona. Fourth edition of **1967(n)**.
- 1991(r)** *Chü Hsiang Ti Shih Chieh.* IES, Hong Kong. Chinese translation by Yue-Ching Ho of **1990(f)**. Contains an Author's Preface to the Chinese Edition.
- 1991(s)** *Um mundo de propensões.* Editorial Fragmentos, Lisbon. Portuguese translation by Teresa Barreiros & Rui G. Feijó of **1990(f)**.
- 1991(t)** Meccanismi contro invenzione creativa: brevi considerazioni su un problema aperto. In *L'Automa Spirituale*. Biblioteca di Cultura Moderna Laterza, Rome, pp. 7–17. Italian translation by Gabriella Rojatti & Raffaella Rojatti of an address, entitled 'Mechanisms versus Creative Inventions – A Brief Survey of a Wide Open Problem', given in Venice on 5th October 1990. Translated into Italian also in **1990(z₁₀)**.
- 1991(u)** A World of Propensities: Two New Views of Causality. In *Advances in Scientific Philosophy. Essays in Honour of Paul Weingartner on the Occasion of the 60th Anniversary of his Birthday*, edited by Gerhard Schurz & Georg J.W. Dorn. Rodopi, Amsterdam. A reprint of **1990(f)**.
- 1991(v)** *Muntq-i Ektehaf-i Ebni.* Scientific & Cultural Publications Company (Shirkat-i Intsharat-i 'Ilmi wa Farhangi), Tehran. Persian translation by S. Hossein Kamali of **1991(z₁₃)**. Edited by Abdul-Karim Sorūsh.
- 1991(w)** 'Alles stürzte sodann in ein geistiges schwarzes Loch'. *Die Weltwoche* (Zürich), No. 37, 12th September 1991, p. 61–63. An interview with Felix Müller & Hanspeter Born.
- 1991(x)** *Auf der Suche nach einer besseren Welt: Vorträge und Aufsätze aus dreißig Jahren.* R. Piper & Co. Verlag, Munich. Sixth edition, revised, of **1984(b)**.
- 1991(y)** *Objective Knowledge. An Evolutionary Approach.* Clarendon Press, Oxford. Fourth impression of **1979(a)**.
- 1991(z)** *Conjectures and Refutations.* Routledge, London. Second impression of **1989(u)**.
- 1991(z₁)** *The Open Universe: An Argument for Indeterminism*, edited by W.W. Bartley III. Routledge, London. A reprint of **1988(m)**.

- 1991(z₂)** Selections from *The Logic of Scientific Discovery*. In *The Philosophy of Science*, edited by Richard Boyd, Philip Gasper, & J.D. Trout, pp. 99-119. MIT Press (a Bradford Book), Cambridge, Mass. A reprint of sections 9-11 of **1985(e)**.
- 1991(z₃)** *Un universo di propensioni*. Vallecchi Editore, Florence. Italian translation by Arnaldo Benini of **1990(f)**.
- 1992(a)** On a Little Known Chapter of Mediterranean History. In *Transition to Modernity* (dedicated to Ernest Gellner), edited by John A. Hall & I.C. Jarvie, pp. 113-119. Cambridge University Press, Cambridge. A slightly shortened version of **1989(x)**.
- 1992(b)** *Un univers de propensions: deux études sur la causalité et l'évolution*. Éditions de l'Éclat, Combas. French translation by Alain Boyer of **1990(f)**. Contains an introduction by the translator.
- 1992(c)** *Un mundo de propensiones*. Editorial Tecnos S.A., Madrid. Spanish translation by José Miguel Esteban C'loquell of **1990(f)**.
- 1992(d)** Ideologien machen die Menschen blind gegenüber der Wirklichkeit. In *Der Traum aber bleibt*, edited by Norbert Sommer, pp. 84-92. Wichern-Verlag, Berlin. An interview with Ingolf Karnahl, originally for Saarländischer Rundfunk.
- 1992(e)** Gegen den Zynismus in der Interpretation der Geschichte. In *Eichstätt Materialien*. Verlag Friedrich Pustet, Regensburg. Lecture given at the Catholic University of Eichstätt on 27th May 1991, on the occasion of being awarded an honorary doctorate. Contained in **1994(y)**. Czech translation (unauthorized): **1992(g)**; Portuguese: **1992(u)**.
- 1992(f)** Popper, lo, il Papa e Gorbaciov. *La Stampa* (Turin), 9th April 1992, 'Società & Cultura', p. 17. Italian translation of an interview with Barbara Spinelli.
- 1992(g)** Karl Raimund Popper: Jsem optimista, který o budoucnosti nic neví. *Literární Noviny* (Prague), 6th-12th February 1992. Czech translation by Jaromír Povejsil of **1992(e)**. Unauthorized.
- 1992(h)** *Objective Knowledge. An Evolutionary Approach*. Clarendon Press, Oxford. Fifth impression of **1979(a)**.
- 1992(i)** *Die offene Gesellschaft und ihre Feinde*. Volume I: *Der Zauber Platons*, Volume II: *Hegel, Marx und die Folgen*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Seventh German edition. An entirely new edition, with eight new *Appendices* and many changes in the text. Hardback and paperback. Contains also a brief survey of Europe from 1914 to 1992.
- 1992(j)** Broadcast Review of [A] *History of Western Philosophy. Russell: the Journal of the Bertrand Russell Archives* **12**, N.S., No. 1, Summer 1992, pp. 11-13 (German) and 19-21 (English). English translation by I. Grattan-Guinness.

Originally broadcast on Austrian Radio (ÖRF) on 19th January 1947.
Contains also four letters from Popper to Russell. A part of
I. Grattan Guinness, Russell and Karl Popper: Their personal contacts, *Ibid.*,
pp. 3-21.

- 1992(k)** How the Moon Might Throw Some of Her Light Upon the Two Ways of
Parmenides. *Classical Quarterly* **XLII**, No. 1, 1992, pp. 12-19.
- 1992(l)** Russian translation by Enga M. Dvin of **1992(f)**. *Literaturnaya Gazeta*
(Moscow), No. 21, 5398, 5th May 1992.
- 1992(m)** *Realism and the Aim of Science*, edited by W.W. Bartley III. Routledge,
London. Third impression of **1985(l)**.
- 1992(n)** The Communist Road to Self-Enslavement. *Cato Policy Report* (Washington)
XIV, No. 3, May/June 1992. Text of an address given at the International
Congress of the American Economic Association at New Orleans on 4th
January 1992.
- 1992(o)** *Ba's al-Ethologia*. Dar al Saqi, London. New edition, with a changed title, of
1959(b)
- 1992(p)** *Unended Quest: An Intellectual Autobiography*. Routledge, London. A
reprint, with second Postscript ('Postscript to Marxism, 1992') and updated
bibliography, of **1982(k)**. Contains also a foreword, included without the
author's knowledge, by Helmut Kohl.
- 1992(q)** A giant mind winks wisdom from the small print of history. *The Sunday*
Times, 12th July 1992, News Review section, p. 8. An interview with Lesley
White. Italian translation: **1992(s)**.
- 1992(r)** Il Novecento che abbiamo attraversato. *Corriere della Sera* (Milan), 16th July
1992, p. 7. Italian translation of an interview with Riccardo Chiaberge.
- 1992(s)** Buon compleanno Sir Karl. *La Repubblica*, 28th July 1992, p. 27. Italian
translation by Ottavio Fatica of **1992(q)**.
- 1992(t)** *In Search of a Better World: Lectures and Essays from Thirty Years*.
Routledge, London. Translation by Laura J. Bennett, with additional
material translated by Melitta Mew, of **1984(b)**. Translation revised by the
author & Melitta Mew.
- 1992(u)** Contra o cinismo na interpretação da história. *Risco* (Lisbon), Nos 17/18, 1992.
Portuguese translation by Teresa Curvelo of **1992(e)**.
- 1992(v)** *Objektive Erkenntnis: Ein Evolutionärer Entwurf*. Hoffmann und Campe,
Hamburg. In the series *Klassiker des Denkens*. Second impression of
1987(x).

- 1992(w)** *Teoría cuántica y el espin en física. Post Scriptum a La lógica de la investigación científica*, volume III, edited by W.W. Bartley III. Editorial Tecnos, Madrid. Second edition of **1985(v)**.
- 1992(x)** *Ausgangspunkte: Meine Intellektuelle Entwicklung*. Hoffmann und Campe, Hamburg. Fifth edition of **1979(w)**.
- 1992(y)** (with Konrad Lorenz) *El porvenir está claro*. Tusquets Editores, Barcelona. Spanish translation by Teófilo de Lozoya of **1988(n)**.
- 1992(z)** Der Liberalismus – Einige Thesen. In *Kleines Lesebuch über den Liberalismus*, edited by Detmar Doering. Fridrich-Naumann-Stiftung, Comdok Verlagsabteilung, Königswinter. An excerpt from chapter 11 of **1991(x)**.
- 1992(z₁)** (with John C. Eccles) *O eu e seu cérebro, O cérebro e o pensamento*. Editora Universidade de Brasília, Brasília. Portuguese translation by Silvio Meneses Garcia, Helena Cristina Fontenelle Arantes, & Aurélio Osmar Cardoso de Oliveira of **1984(t)**. In two volumes.
- 1992(z₂)** *Objektive Erkenntnis: Ein Evolutionärer Entwurf*. Hoffmann und Campe, Hamburg. In the series *Klassiker des Denkens*. Third impression of **1987(x)**.
- 1992(z₃)** *La lezione di questo secolo. Intervista di Giancarlo Bosetti*. Marsilio Editori S.p.A., Venice, May 1992. Contains also Italian translations (from German) by Nicola Paoli of **1988(w)** and **1990(z)**.
- Further impressions, editions, and translations of **1992(z₃)** into other languages are indicated in the year in which they are published.
- 1992(z₄)** *La lezione di questo secolo. Intervista di Giancarlo Bosetti*. Marsilio Editori S.p.A., Venice. Second edition, September 1992, of **1992(z₃)**.
- 1992(z₅)** Letter on the conflict in the former Yugoslavia. *The Times*, 26th August 1992.
- 1992(z₆)** Popper questa Europa sbagliata. *La Stampa Società & Cultura*, 6th October 1992. Italian translation of an interview with Marcello Pera.
- 1992(z₇)** *The Open Society and Its Enemies*. Volume II. Routledge, London. A reprint of volume II of **1966(a)**.
- 1992(z₈)** *Quantum Theory and the Schism in Physics*, edited by W.W. Bartley III. Routledge, London. Volume III of the *Postscript to The Logic of Scientific Discovery*. Second impression of **1989(i)**.
- 1992(z₉)** *The Logic of Scientific Discovery*. Routledge, London & New York. Fifteenth impression of **1959(a)**.

- 1992(z₁₀)** Russian translation of **1991(n)** [*The Poverty of Historicism*]. *Voprosi filosofii* (Moscow) **8**, 1992, pp. 49-79; **9**, 1992, pp. 22-48; **10**, 1992, pp. 29-58. Russian translation by Sonya A. Kudrina, edited by N.A. Ovchinnikov.
- 1992(z₁₁)** Russian translation of **1970(z₇)** [Die Logik der Sozialwissenschaften]. *Voprosi filosofii* (Moscow), 1992, pp. 65-75.
- 1992(z₁₂)** About the EPR Controversy. *Foundations of Physics* **22**, No. 10, October 1992, pp. 1303-1316. An interview with Marie-Christine Combourieu. Followed on pp. 1316-1322 by an English translation and the German original in facsimile of a letter dated 11.IX.35 by Einstein to Popper (an excerpt from Appendix *xi of **1959(a)**).
- 1992(z₁₃)** *Wiedza obiektywna: Ewolucyjna teoria epistemologiczna*. Wydawnictwo Naukowe PWN, Warsaw. Polish translation by Adam Chmielewski of **1992(h)**.
- 1992(z₁₄)** *La lezione di questo secolo. Intervista di Giancarlo Bosetti*. Marsilio Editori S.p.A., Venice. Third edition, November 1992, of **1992(z₃)**.
- 1992(z₁₅)** Eine Frage der Verantwortung. *Süddeutsche Zeitung* (Munich), 23rd October 1992. In Serie 'Zeitbombe Mensch'.
- 1992(z₁₆)** *Tre saggi sulla mente umana*. Vallecchi Editore S.p.A., Florence. Italian translation by Arnolfo Benini of **1982(s)** and of **1987(o)**, and of a commentary by W.W. Bartley III on **1987(o)**. Unauthorized.
- 1992(z₁₇)** Kriege führen für den Frieden: Der Philosoph Karl R. Popper über den Kollaps des Kommunismus und die neuen Aufgaben der Demokratie. *Der Spiegel*, No. 13, April 1992, pp. 202-211. An interview with Olaf Ihlau. Contained in **1994(y)**.
- 1992(z₁₈)** *Otkrytoie Obschestvo i ego Vragi* [ОТКРЫТОЕ ОБЩЕСТВО И ЕГО ВРАГИ]. Soros Foundation, Moscow. Russian translation by Vadim Sadovsky et al. of **1966(a)**.
- 1992(z₁₉)** Ideologien machen die Menschen blind gegenüber der Wirklichkeit. In *Der Traum aber bleibt: Sozialismus und christliche Hoffnung*, edited by Norbert Sommer, pp. 84-92. Wichern-Verlag GmbH, Berlin. An interview with Ingolf Karnahl. The title is not the author's.
- 1992(z₂₀)** Necessité d'une nouvelle éthique professionnelle en science et en médecine. In *Éthique de l'intervention*, Les Cahiers du CERFEE (Centre en Recherche Formation Enfance Éducation), Nos 8-9, Université Paul Valéry, Montpellier III. French translation by Michel Ramos of **1983(q)**.
- 1992(z₂₁)** [Contribution to programme note for organ recital by Dr Gillian Weir, 5th August 1992.] Paisley International Organ Festival.

- 1992(z22)** *Conocimiento objetivo: Un enfoque evolucionista*, Editorial Tecnos, Madrid. Fourth edition of **1974(f)**.
- 1993(a)** Universals, Dispositions, and Natural or Physical Necessity. In *Foundations of Philosophy of Science*, edited by James H. Fetzer. Paragon House Publishers, New York. A reprint of *Appendix *x* of **1987(z3)**.
- 1993(b)** Science: Conjectures and Refutations. *Ibid.* A reprint of chapter 1 of **1991(z)**, without the *Appendix*.
- 1993(c)** Three Views Concerning Human Knowledge. *Ibid.* A reprint of chapter 3 of **1991(z)**.
- 1993(d)** Letter on Churchill's refusal to negotiate peace with Hitler in 1940. *The Times*, 12th January 1993.
- 1993(e)** *Miscria dello storicismo*. Giangiacomo Feltrinelli Editore, Milan. Ninth impression of **1975(a)**.
- 1993(f)** (with Konrad Lorenz) *Die Zukunft ist offen. Das Altenberger Gespräch. Mit den Texten des Wiener Popper Symposiums*, edited by Franz Kreuzer. Serie Piper. R. Piper GmbH & Co. KG, Munich. Fourth impression of **1985(g)**.
- 1993(g)** The Rationality Principle. In *The Philosophy and Methodology of Economics* (The International Library of Critical Writings in Economics 23), volume III, edited by Bruce J. Caldwell. Edward Elgar Publishing Limited, Aldershot. A reprint of section 29 of **1985(e)**.
- 1993(h)** (with John C. Eccles) *The Self and its Brain*. Routledge, London. Seventh impression of **1977(u)**. Paperback.
- 1993(i)** Eine Frage der Verantwortung. In *Zeitbombe Mensch - Überbevölkerung und Überlebenschance*, pp. 182f. Deutscher Taschenbuch Verlag, Munich. A reprint of **1992(z15)**.
- 1993(j)** In Bosnien nichts zu tun wäre selbstmörderische Dummheit. *Berliner Morgenpost*, 21st April 1993, p. 3.
- 1993(k)** Der Aufschrei des Verstandes. *Hamburger Abendblatt*, No. 92, 21st April 1993, p. 3. A reprint of **1993(j)**. The title is not the author's.
- 1993(l)** *Spółeczeństwo Otwarte i Jego Wrogowie*. Volume I: *Platon*. Wydawnictwo Naukowe PWN, Warsaw. Polish translation by Halina Krahelska (revised by Adam Chmielewski) of volume I of **1966(a)**. Continued in **1993(m)**.
- 1993(m)** *Spółeczeństwo Otwarte i Jego Wrogowie*. Volume II: *Hegel, Marks*. Wydawnictwo Naukowe PWN, Warsaw. Polish translation by Halina Krahelska (revised by Adam Chmielewski) of volume II of **1966(a)**. A continuation of **1993(m)**.

- 1993(n)** Political guru urges Balkan air attacks. Title of article in *The Independent*, 18th March 1993, p. 6, containing a statement by Popper on the crisis in the former Yugoslavia.
- 1993(o)** Entschlossenheit zeigen. *Donaukurier*, 18th May 1993, p. 2. A reprint of **1993(j)**.
- 1993(p)** *Societatea deschisă și dușmanii ei: Vraja lui Platon*. Humanitas, Bucharest. Romanian translation by D. Stoianovici of volume I of **1966(a)**. Continued in **1993(z₂)**.
- 1993(q)** *Un món de propensions*. Edicions 62, Barcelona. Catalan translation by Ferran Meler i Orti of **1990(f)**. Contains an introduction by Baltasar Porcel.
- 1993(r)** *Objektive Erkenntnis: Ein evolutionärer Entwurf*. Hoffmann und Campe, Hamburg. First 'campe paper-pack' edition of **1987(x)**.
- 1993(s)** Russian translation of **1991(n)** [*The Poverty of Historicism*]. Progress Publishing Group Corporation, Moscow. A reprint of **1992(z₁₀)**.
- 1993(t)** *Otvoreno Društvo i Njegovi Neprijatelji*. Volume I: *Čar Platona*; volume II: *Plina Proročanstva: Hegel, Marks i Posledice*. Bigz, Belgrade. Serbian translation by Branimir Gligorich of **1966(a)**.
- 1993(u)** Ludwig Boltzmann und die Richtung des Zeitablaufs: Der Pfeil der Zeit. In *Klassiker der modernen Zeitphilosophie*, edited by Walther Ch. Zimmerli & Mike Sandbothe, pp. 172-181. Wissenschaftliche Buchgesellschaft, Darmstadt. An excerpt from pp. 227-236 of **1992(x)**.
- 1993(v)** *Búsqueda sin término: Una autobiografía intelectual*. Editorial Tecnos, Madrid. Second impression of **1985(z₈)**.
- 1993(w)** Discussion of the Mind-Brain Problem, by K.R. Popper, B.I.B. Lindahl, and P. Århem. *Theoretical Medicine* **14**, No. 2, June 1993. Kluwer Academic Publishers, Dordrecht, Boston, & London.
- 1993(x)** Karl Popper in conversation with Professor George Wedell, Governor of the European Institute for the Media. In *The European Citizen and the Media. European Media Policy: Should it Exist?*, pp. 9-12. Proceedings of the Colloquium, The Council Chamber, BBC Broadcasting House, London, 4th June 1993. Mediafact Series.
- 1993(y)** *La leçon de ce siècle*. Anatolia Éditions, Paris. French translation by Jacqueline Henry & Claude Orsoni of **1992(z₃)**.
- 1993(z)** *Otvorenoto obshtestvo i negovite vragove*. Volume I: *Platon*. Soros Foundation, Sofia. Bulgarian translation by Kamen Lozev of volume I of **1966(a)**. Continued in **1996(y)**.

- 1993(z₁)** Tolerancia y responsabilidad intelectual. In *Ciencia, Filosofía y Religión en relación con el Cáncer*, pp. 23f. Librería AKADIA Editorial, Buenos Aires. An excerpt from **1984(z₄)**.
- 1993(z₂)** *Societatea deschisă și dușmanii ei: Epoca Marilor Profeții: Hegel și Marx*. Humanitas, Bucharest. Romanian translation by D. Stoianovici of volume II of **1966(a)**. A continuation of **1993(p)**.
- 1993(z₃)** Resumé filosofického myšlení. Úvodem: Etapy na cestě k světové slávě. *Reflexe* (Prague), Nos 7-8, pp. 1-20, 1993. Czech translation of **1987(j)**.
- 1993(z₄)** Resumé filosofického myšlení. *Reflexe* (Prague), No. 9, 1993, pp. 1-23. Czech translation of **1990(e)**.
- 1993(z₅)** *A sociedade aberta e os seus inimigos*. Volume I: *O Fascínio de Platão*. Editorial Fragmentos, Lisbon. Portuguese translation by Anabela Sottomayore & Catarina Labisa, revised by João Carlos Espada, of volume I of **1966(a)**. Continued in **1993(z₆)**.
- 1993(z₆)** *A sociedade aberta e os seus inimigos*. Volume II: *A mare alta da profecia: (Hegel, Marx e as sequelas)*. Editorial Fragmentos, Lisbon. Portuguese translation by Teresa Carvelo, revised by João Carlos Espada, of volume II of **1966(a)**. A continuation of **1993(z₅)**.
- 1993(z₇)** *Offene Gesellschaft – offenes Universum*. Serie Piper, R. Piper GmbH & Co. KG, Munich. Second impression of **1986(e)**.
- 1993(z₈)** Warum ist der Staat notwendig?. In *Handreichung für den Sozialkundenunterricht im Gymnasium*, p. 37. Staatsinstitut für Schulpädagogik und Bildungsforschung, Munich. An excerpt from pp. 169f. of **1991(x)**. The title is not the author's.
- 1993(z₉)** *Unended Quest: An Intellectual Autobiography*. Routledge, London. Second impression of **1992(p)**.
- 1993(z₁₀)** *The Open Society and Its Enemies*. Volume I. Routledge, London. Tenth reprint of volume I of **1966(a)**.
- 1994(a)** *Auf der Suche nach einer besseren Welt: Vorträge und Aufsätze aus dreißig Jahren*. R. Piper & Co. Verlag, Munich. Seventh edition, revised, of **1984(b)**.
- 1994(b)** Science: Conjectures and Refutations. In *Philosophy and the Natural Sciences*, Compton Phil 244 #1, Vanderbilt University Class Pak/ Spring 1994, Nashville, Tennessee. A reprint of chapter 1 of **1989(u)**.
- 1994(c)** *Objektive Erkenntnis: Ein evolutionärer Entwurf*. Hoffmann und Campe, Hamburg. Second 'campe paper-pack' edition of **1987(x)**.

- 1994(d)** *En busca de un mundo mejor*. Ediciones Paidós Iberica, S.A., Barcelona. Spanish translation by Jorge Vigil Rubio of **1984(b)**.
- 1994(e)** *Logik der Forschung*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Tenth German edition, revised, with new addendum, of **1934(b)** and **1966(e)**.
- 1994(f)** Science: Conjectures and Refutations. In *Positivist Sociology and its Critics* (Schools of Thought in Sociology 11), volume I, edited by Peter Halfpenny & Peter McMylor, pp. 290-316. Edward Elgar Publishing Ltd, Aldershot. A reprint of chapter 1 of **1989(u)**.
- 1994(g)** [Japanese translation of **1986(e)**]. Orion Literary Agency, Tokyo.
- 1994(h)** *Die beiden Grundprobleme der Erkenntnistheorie*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Second edition, revised, of **1979(z₂)**.
- 1994(i)** Emancipation Through Knowledge. In *Challenges to the Enlightenment: In Defense of Reason and Science*, edited by Paul Kurtz & Timothy J. Madigan for The Academy of Humanism, pp. 58-72. Prometheus Books, Buffalo, New York. A reprint of chapter 10 of **1992(t)**.
- 1994(j)** *Otkrytiia spetsialnogo jeji nepriatel'. Volume I: Ubranuti Platonem*. Oikymenhi, Prague. Czech translation by Milos Calda of volume I of **1992(i)**. Continued in **1994(z₉)**.
- 1994(k)** Karl Popper remembers Tomas Masaryk. *Open Society News*, Summer 1994. The Soros Foundation, New York. An excerpt from a speech delivered in Prague on 25th May 1994, on the occasion of the award of honorary doctorate in medical sciences by Charles University.
- 1994(l)** *El universo abierto. Un argumento en favor del indeterminismo. Post Scriptum a La lógica de la investigación científica*, volume II, edited by W.W. Bartley III, Editorial Tecnos, Madrid. Third impression of **1984(r)**.
- 1994(m)** *La lógica de la investigación científica*. Editorial Tecnos, Madrid. Tenth impression of **1962(r)**.
- 1994(n)** *Verso una teoria evoluzionistica della conoscenza*. Armando Editore, Rome. A reprint of **1991(z₃)**, with a new title.
- 1994(o)** *Ausgangspunkte: Meine Intellektuelle Entwicklung*. Hoffmann und Campe, Hamburg. First paperback edition of **1979(w)**, **1992(x)**.
- 1994(p)** Una patente per fare tv. In *Reset: Cattiva maestra televisione*, pp. 13-25. Reset S.r.l., Milan. Italian translation of an interview with Giancarlo Bosetti. Contains an Introduction by the interviewer, and essays by John Condry and by Charles S. Clark. English translations: **1995(g)**, **1995(z)**; German: **1994(t)**, **1995(g)**; French: **1994(z₆)**, **1995(g)**; Spanish: **1995(g)**; Russian: **1995(g)**; Dutch: **1996(z₁)**.

- 1994(q)** (with John C. Eccles) *Das Ich und sein Gehirn*. R. Piper & Co. Verlag, Munich. Eleventh edition (fourth edition in Serie Piper paperback) of **1982(q)**.
- 1994(r)** *The Myth of the Framework: In Defence of Science and Rationality*, edited by M.A. Notturmo. Routledge, London.
- Further impressions, editions, and translations of **1994(r)** into other languages are indicated in the year in which they are published.
- 1994(s)** *Knowledge and the Body-Mind Problem: In Defence of Interaction*, edited by M.A. Notturmo. Routledge, London.
- Further impressions, editions, and translations of **1994(s)** into other languages are indicated in the year in which they are published.
- 1994(t)** Eine Lizenz fürs Fernsehmachen. *Live: Hauszeitung des Schweizer Fernsehens DRS*, October 94/8, pp. 9-12. German translation by Antonio Antoniazzi & Rene Bardet of **1994(p)**.
- 1994(u)** *En busca de un mundo mejor*. Ediciones Paidós Iberica, S.A., Barcelona. Second impression of **1994(d)**.
- 1994(v)** *Conjeturas y refutaciones: El desarrollo del conocimiento científico*. Ediciones Paidós, Barcelona. Fifth edition of **1967(n)**.
- 1994(w)** Die Macht des Fernsehens: Gegen den Mißbrauch der Television. *Frankfurter Rundschau*(Zeit und Bild, Feuilleton), 5th November 1994, p. 3. A reprint of **1994(t)**.
- 1994(x)** (with David Miller) Contributions to the Formal Theory of Probability. In *Patrick Suppes: Scientific Philosopher*. Volume 1. *Probability and Probabilistic Causality*, edited by Paul Humphreys, pp. 3-23. Kluwer Academic Publishers, Dordrecht, Boston, & London.
- 1994(y)** *Alles Leben ist Problemlösen: Über Erkenntnis, Geschichte und Politik*. R. Piper & Co. Verlag, Munich.
- Further impressions, editions, and translations of **1994(y)** into other languages are indicated in the year in which they are published.
- 1994(z)** *Vermutungen und Widerlegungen*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. German translation by Gretl Albert, Melitta Mew, the author, & Georg Siebeck of Part I (Introduction and chapters 1-10) of **1991(z)**. Continued in **1997(p)**.
- 1994(z₁)** *Alles Leben ist Problemlösen: Über Erkenntnis, Geschichte und Politik*. Wissenschaftliche Buchgesellschaft, Darmstadt. Bookclub edition of **1994(y)**.
- 1994(z₂)** *Alles Leben ist Problemlösen: Über Erkenntnis, Geschichte und Politik*. R. Piper & Co. Verlag, Munich. Second edition of **1994(y)**.

- 1994(z₃)** *Objective Knowledge. An Evolutionary Approach*. Clarendon Press, Oxford. Sixth impression of **1979(a)**.
- 1994(z₄)** *Busqueda sin término: Una autobiografía intelectual*. Editorial Tecnos, Madrid. Third impression of **1985(z₈)**.
- 1994(z₅)** (with Konrad Lorenz) *Die Zukunft ist offen. Das Altenberger Gespräch. Mit den Texten des Wiener Popper Symposiums*, edited by Franz Kreuzer. Serie Piper. R. Piper GmbH & Co. KG, Munich. Fifth impression of **1985(g)**.
- 1994(z₆)** *La télévision. Un danger pour la démocratie*. Anatolia Éditions, Paris. Contains an Introduction by Giancarlo Bosetti, and an essay by Jean Baudouin. French translation of **1994(p)**.
- 1994(z₇)** *Alles Leben ist Problemlösen: Über Erkenntnis, Geschichte und Politik*. R. Piper & Co. Verlag, Munich. Third edition of **1994(y)**.
- 1994(z₈)** *Bida historicismu*. Oikoymenh, Prague. Czech translation by Jana Odehnalová of **1991(n)**.
- 1994(z₉)** *Olivovník společenství jeji nepřítel*. Volume II: Oikoymenh, Prague. Czech translation by Jana Odehnalová of volume II of **1992(i)**. Continuation of **1994(j)**.
- 1994(z₁₀)** Henimod en evolutionær erkendelsesteori. *Kritik* (Copenhagen) **113**, pp. 30-39. Danish translation by Frederik Stjernfelt of the second essay in **1990(f)**.
- 1994(z₁₁)** *Conjecturas e refutações (O progresso do conhecimento científico)*. Editora Universidade de Brasília, Brasília. Third edition of **1981(z₈)**.
- 1994(z₁₂)** The Prague Lecture. In *Karl Popper in Prague: In Memoriam*. 3rd Medical Faculty of Charles University, and Central European University Foundation, Prague, pp. 12-15. Czech translation by Ivan Chvatík, *ibid.*, pp. 46-49. A reprint of **1994(k)**.
- 1994(z₁₃)** Contributions to a seminar held at the Prague College of the Central European University, 26th May, 1994. *Ibid.*, pp. 25-31. With contributions by Ernest Gellner and George Soros. Czech translation by Ivan Chvatík, *ibid.*, pp. 59-65.
- 1994(z₁₄)** Words of Thanks on Receiving The Open Society Prize. *Ibid.*, p. 34. Czech translation by Ivan Chvatík, *ibid.*, p. 69.
- 1994(z₁₅)** Przyszłość jest otwarta: Sir Karlem Popperem Rozmawia Adam Chmielewski. *Odra* (Wrocław) **11**, No. 396. Polish translation of an interview with Adam Chmielewski on 29th July 1994. English version: **1995(z₂)**; Italian translation: **1996(w)**.
- 1994(z₁₆)** *Congetture e confutazioni: Lo sviluppo della conoscenza scientifica*. Universale Paperbacks, Il Mulino, Bologna. Fourth impression of **1976(x)**.

- 1994(z₁₇)** [Korean translation by Woosuk Park of **1968(a)**, without *Appendices*].
Koreaone, Seoul. Contains an epilogue by the translator.
- 1995(a)** *Auf der Suche nach einer besseren Welt*. R. Piper & Co. Verlag, Munich.
Eighth revised edition of **1984(b)**.
- 1995(b)** *Alles Leben ist Problemlösen: Über Erkenntnis, Geschichte und Politik*.
R. Piper & Co. Verlag, Munich. Fourth edition of **1994(y)**.
- 1995(c)** *Alles Leben ist Problemlösen: Über Erkenntnis, Geschichte und Politik*.
R. Piper & Co. Verlag, Munich. Fifth edition of **1994(y)**.
- 1995(d)** *Alles Leben ist Problemlösen: Über Erkenntnis, Geschichte und Politik*.
Lizenzausgabe der Buchgemeinschaft Donauland Kremayr & Scheriau, Vienna.
A reprint of **1994(y)**.
- 1995(e)** (with Konrad Lorenz) *L'avenir est ouvert*. Flammarion, Paris. A reprint of
1990(a).
- 1995(f)** *Logica della scoperta scientifica: Il carattere autocorrettivo della scienza*. Giulio
Einaudi editore, Turin. Fourth impression of **1970(e)**.
- 1995(g)** Against Misuse of Television. *Kulturchronik*, No. 1. VG Bild-Kunst, Bonn,
pp. 19-24. English, German, French, Spanish, and Russian translations of
1994(p).
- 1995(h)** *Alles Leben ist Problemlösen: Über Erkenntnis, Geschichte und Politik*.
R. Piper & Co. Verlag, Munich. Sixth edition of **1994(y)**.
- 1995(i)** *En busca de un mundo mejor*. Ediciones Paidós Iberica, S.A., Barcelona.
Second impression of **1994(d)**.
- 1995(j)** *Objective Knowledge. An Evolutionary Approach*. Seventh impression of
1979(a).
- 1995(k)** *Vivlíá kai Sképsis*. Hoi aparchés tou evrópaikou vivliou stén Athína tón
klasikón chrónon. *Utopia*, A Bimonthly Review of Theory and Culture, No. 14,
1995, pp. 119-129. Greek translation by Irene Papadaki of **1983(e)**.
- 1995(l)** *Karl Popper Lesebuch*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen.
German translation by Eva Schiffer et al. of **1983(i)**.
- 1995(m)** *Alles Leben ist Problemlösen: Über Erkenntnis, Geschichte und Politik*.
R. Piper & Co. Verlag, Munich. Seventh edition of **1994(y)**.
- 1995(n)** *Arvauksia ja Kumoamisia: Tieteellisen tiedon kasvu*. Gaudeamus, Helsinki.
Finnish translation by Eero Eerola of **1991(z)**.

- 1995(o)** *Eine Welt der Propensitäten*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. German translation by Jürgen August Alt & Desirée Irrgang of **1990(f)**.
- 1995(p)** *Kakutseru no Sekai*. Shinzan Books, Tokyo. Japanese translation by Yutaka Tajima of **1990(f)**.
- 1995(q)** *Věno Hobaně: Intelektuální autobiografie*. PROSTOR Publishing House, Prague. Czech translation by Jana Odehnalová of **1992(p)**.
- 1995(r)** *Auf der Suche nach einer besseren Welt*. R. Piper & Co. Verlag, Munich. Ninth revised edition of **1984(b)**.
- 1995(s)** *Čto takové dialektika?*. *Voprosi filosofii*, No. 1, 1995, pp. 118-138. Russian translation by G.A. Novichkova of chapter 15 of **1991(z)**.
- 1995(t)** *Il mito della cornice: Difesa della razionalità e della scienza*. Società editrice Il Mulino, Bologna. Italian translation by Paola Palminiello of **1994(r)**.
- 1995(u)** *La responsabilidad de vivir*. Ediciones Paidós Iberica, S.A., Barcelona. Spanish translation by Concha Roldán of **1994(y)**.
- 1995(v)** Grundprobleme der Erkenntnislogik. In *Das goldene Zeitaler der Österreichischen Philosophie*, edited by Kurt Rudolf Fischer. WUV Universitätsverlag, Vienna, pp. 225-242. A reprint of chapters 1 and 2 of **1994(e)**.
- 1995(w)** *Yoryoku sekai wo motomete*. Mirai-sha, Tokyo. Japanese translation by Makoto Kogawara & Ysanyuki Kageyama of **1995(a)**.
- 1995(x)** *Popper: Escritos Selectos*. Fondo de Cultura Económica, Mexico City. Spanish translation by Sergio René Madero Báez of **1985(e)**.
- 1995(y)** Against the Misuse of Television. *Network 1*, No. 4, 1995. Association of Christian Broadcasters, Wapadrand, South Africa. English translation of **1994(w)**.
- 1995(z)** *Hateshinaki tankyu chiteki jiden*. Iwanami Shoten, Tokyo. Volume I of a paperback edition of **1978(w)**. Continued in **1996(m)**.
- 1995(z₁)** *Karl Popper Lesbuch*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Second impression of **1995(l)**.
- 1995(z₂)** The Future Is Open. Conversation with Karl Popper. *Dialogue and Universalism* (Warsaw) 8/9, 1995, pp. 21-30. English version of **1994(z₁₅)**.
- 1995(z₃)** *Conjectures and Refutations*. Routledge, London. Third impression of **1989(u)**.

- 1995(z₄)** *The Open Society and Its Enemies*. Routledge, London. Golden Jubilee edition in one volume of **1966(a)**.
- 1996(a)** *In Search of a Better World*. Routledge, London. Second impression and first paperback edition of **1992(t)**.
- 1996(b)** *The Myth of the Framework: In Defence of Science and Rationality*, edited by M.A. Notturmo. Routledge, London. First paperback edition of **1994(r)**.
- 1996(c)** *Knowledge and the Body-Mind Problem: In Defence of Interaction*, edited by M.A. Notturmo. Routledge, London. First paperback edition of **1994(s)**.
- 1996(d)** (with John C. Eccles) *Das Ich und sein Gehirn*. R. Piper & Co. Verlag, Munich. Twelfth edition (fifth edition in Serie Piper paperback) of **1982(q)**.
- 1996(e)** *Surian Skłonności*. Wydawnictwo Znak, Cracow. Polish translation by Adam Chmielewski of **1990(f)**.
- 1996(f)** *En busca de un mundo mejor*. Ediciones Paidós Iberica, S.A., Barcelona. Third impression of **1994(d)**.
- 1996(g)** *Verso una teoria evoluzionistica della conoscenza*. Armando Editore, Rome. Second impression of **1994(n)**.
- 1996(h)** The Propensity Interpretation of the Calculus of Probability, and the Quantum Theory. In *Science and Philosophy in the Twentieth Century: Logic, Probability, and Epistemology*. Garland Publishing Inc., New York & London, pp. 135-140. A reprint of **1957(e)**.
- 1996(i)** *La lógica de la investigación científica*. Circulo de Lectores, Barcelona. First book-club edition of **1994(m)**.
- 1996(j)** Science: Conjectures and Refutations. In *Introduction to the Philosophy of Science*, edited by Arthur Zucker. Prentice-Hall Inc., Englewood Cliffs, N.J. A reprint of chapter 1 of **1963(a)**.
- 1996(k)** *Kritisk rationalisme: Udvalgte essays om videnskab og samfund*. Nyt Nordisk Forlag / Arnold Busck, Copenhagen. Second impression of **1973(k)**.
- 1996(l)** *Tutta la Vita è Risolvere Problemi: Scritti sulla Conoscenza, la Storia e la Politica*. Rusconi Libri S.r.l., Milan. Italian translation by Dario Antiseri of **1995(m)**.
- 1996(m)** *Hateshinaki tankyu: chiteki - jiden*. Iwanami Shoten, Tokyo. Volume II of a paperback edition of **1978(w)**. Continuation of **1995(z₁)**.
- 1996(n)** *En Värld av Benägenheter*. Brutus Östlings Bokförlag Symposium, Stockholm/Stehag. Swedish translation by Ola Lindberg of **1990(f)**.

- 1996(o)** *La théorie quantique et le schisme en physique.* Hermann éditeurs des sciences et des arts, Paris. French translation by Emmanuel Malolo Dissakè of **1982(b)**.
- 1996(p)** *Alles Leben ist Problemlösen: Über Erkenntnis, Geschichte und Politik.* R. Piper & Co. Verlag, Munich. Eighth edition of **1994(y)**.
- 1996(q)** *Un mundo de propensiones.* Editorial Tecnos S.A. Madrid. Second edition of **1992(c)**.
- 1996(r)** *Wszechswiat Otwarty: Argument na rzecz indeterminizmu.* SIW Znak, Cracow. Polish translation by Adam Chmielewski of **1991(z₁)**.
- 1996(s)** *Alles Leben ist Problemlösen: Über Erkenntnis, Geschichte und Politik.* R. Piper & Co. Verlag, Munich. First paperback edition of **1994(y)**.
- 1996(t)** *Historicismus elendighed.* Gyldendal Bokhandel, Nordisk Forlag A.S., Copenhagen. Danish translation by Gitte Lyngs of **1991(n)**.
- 1996(u)** *El universo abierto. Un argumento en favor del indeterminismo. Post Scriptum a La lógica de la investigación científica,* volume II, edited by W.W. Bartley III. Editorial Tecnos, Madrid. Fourth impression of **1984(r)**.
- 1996(v)** *Teoría cuántica y el esima en física. Post Scriptum a La lógica de la investigación científica,* volume III, edited by W.W. Bartley III. Editorial Tecnos, Madrid. Third impression of **1985(v)**.
- 1996(w)** 'Il nostro dovere è di essere ottimisti'. *Ideazione*, No. 3, May/June 1996. Italian translation by Dario Antiseri of **1994(z₁₅)**.
- 1996(x)** *Come controllare chi comanda.* Ideazione Editrice S.r.l., Rome. Italian translation by Dario Antiseri of a selection of Karl Popper's writings, containing an Introduction by the translator. Unauthorized.
- 1996(y)** *Otvorenoto obshtestvo i negovite vragove,* volume II. Soros Foundation, Sofia. Bulgarian translation by Kamen Lozev of volume II of **1996(a)**. Continuation of **1993(z)**.
- 1996(z)** *La conoscenza e il problema corpo-mente.* Editrice Il Mulino, Bologna. Italian translation by Federico Laudisa of **1994(s)**.
- 1996(z₁)** Een wet inzake de televisie. *Nexus*, No. 15, 1996, pp. 158-166. Nexus Institute, Tilburg. Dutch translation by Jeanne Holierhoek of **1994(p)**.
- 1996(z₂)** *The Lesson of This Century.* Routledge, London. English translation by Patrick Camiller of **1992(z₃)**.
- 1996(z₃)** *Congetture e confutazioni. La sviluppo della conoscenza scientifica.* RCS Libri

- & Grandi Opere, Milan. A reprint of **1972(e)** in the series 'I Classici del Pensiero'.
- 1996(z₄)** *Tuuhchleh Uchun Gurlanchlal*. CEU Press, Budapest. Mongolian translation by Battumur Doljinsuren of **1991(n)**.
- 1996(z₅)** *Alles Leben ist Problemlösen: Über Erkenntnis, Geschichte und Politik*. R. Piper & Co. Verlag, Munich. Second paperback edition of **1994(y)**.
- 1996(z₆)** *Conjectures and Refutations*. Routledge, London. Fourth impression of **1989(u)**.
- 1996(z₇)** *Congetture e confutazioni. Lo sviluppo della conoscenza scientifica*. Universale Paperbacks Il Mulino, Bologna. Fifth impression of **1976(x)**.
- 1996(z₈)** *Tutta la Vita è Risolvere Problemi: Scritti sulla Conoscenza, la Storia e la Politica*. Edizione CDE S.p.A., Milan. First book club edition of **1996(l)**.
- 1996(z₉)** *Bijeda Historicismma*. Kruzak, Zagreb. Croat translation by Danko Ivanec of **1991(n)**.
- 1996(z₁₀)** Una patente per fare tv. In *Reset: Cattiva maestra televisione*, pp. 33-45. Reset S.r.l., Milan. A reprint of **1994(p)**, in an enlarged edition, which contains also an essay by Karol Wojtyła.
- 1996(z₁₁)** L'informazione violenta. In *L'informazione violenta*, pp. 15-34. Società Aperta Edizioni S.r.l., Milan. Italian translation of an interview with Maria Teresa De Vito on 13th April 1993. Contains an introduction by Stefano Folli, and four appendices.
- 1997(a)** (with John C. Eccles) *Das Ich und sein Gehirn*. R. Piper & Co. Verlag, Munich. Thirteenth edition (sixth edition in Serie Piper paperback) of **1982(q)**.
- 1997(b)** La scienza: congetture e confutazioni. In *La ricerca contemporanea*, edited by Mario Trombino. Casa Editrice Poseidonia S.r.l., Bologna. Italian translation by Maurizio Villani of chapter 1 of **1963(a)**.
- 1997(c)** *Auf der Suche nach einer besseren Welt*. R. Piper & Co. Verlag, Munich. Tenth revised edition of **1984(b)**.
- 1997(d)** *Megismerés. Történelem. Politika*. AduPrint, Budapest. Hungarian translation of **1996(z₅)** by Darai Lajos Mihály.
- 1997(e)** *El mito del marco común: En defensa de la ciencia y la racionalidad*. Ediciones Paidós Ibérica, S.A., Barcelona. Spanish translation by Marco Aurelio Galmarini of **1996(b)**.
- 1997(f)** *Budoucnost je otevřená*. Nakladatelství Vysehrad, Prague. Czech translation by Jaroslav Kohout & Eva Stuchlíková of **1994(z₅)**.

- 1997(g)** *Historicismens elendighed*. Samlerens Bogklub, Byldendalske Boghandel, Nordisk Forlag A.S., Copenhagen. Danish translation by Gitte Lyngs of **1991(n)**.
- 1997(h)** *Alles Leben ist Problemlösen: Über Erkenntnis, Geschichte und Politik*. R. Piper & Co. Verlag, Munich. Third paperback edition of **1994(y)**.
- 1997(i)** *Conștienterea și problema raportului corp-minte: O pledoarie pentru interacționism*. Editura Trei, Bucharest. Romanian translation by Florin Lobont & Mihail Radu Solcan of **1996(c)**.
- 1997(j)** *Etat paternaliste ou Etat minimal: Remarques théoriques et pratiques sur la gestion de l'Etat démocratique*. Éditions de l'Aire, Vevey. French translation by Corinne Verdun-Moser of **1988(w)**.
- 1997(k)** *Yonllin Sahocwa Ke Chokdei*. Volume I: *Platzonkwa Utopia*; volume II: *Hegelkwa Marr*. Mineumsa Publishing Co., Seoul. Korean translation by Han ku Lee of **1991(l)** and **1992(z7)**.
- 1997(l)** *Tutta la Vita è Risolvere Problemi: Scritti sulla Conoscenza, la Storia e la Politica*. Rusconi Libri S.r.l., Milan. Second edition of **1996(i)**.
- 1997(m)** *Das Elend des Historizismus*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. Seventh edition of **1965(g)**.
- 1997(n)** *W Poszukiwaniu Lepszego Świata*. Książka i Wiedza, Warsaw. Polish translation by Antoni Malinowski of **1984(b)**.
- 1997(o)** *La lógica de la investigación científica*. Editorial Tecnos, Madrid. Eleventh impression of **1962(r)**.
- 1997(p)** *Vermutungen und Widerlegungen*. J.C.B. Mohr (Paul Siebeck) Verlag, Tübingen. German translation by Gretl Albert, Melitta Mew, the author, & Georg Siebeck of Part II (Chapters 11-10 and *Addenda*) of **1991(z)**. A continuation of **1994(z)**.
- 1997(q)** *El cuerpo y la mente*. Ediciones Paidós Ibérica, S.A., Barcelona. Spanish translation by Olga Domínguez Scheidereiter of **1994(s)**. Contains an introduction by José Antonio Marina.
- 1998(a)** *The World of Parmenides. Essays on the Presocratic Enlightenment*, edited by A.F. Petersen. Routledge, London. *In preparation*.

Note This is a somewhat simplified list, but contains all Popper's recorded publications. It is known that there are some omissions, especially amongst unauthorized translations and interviews.